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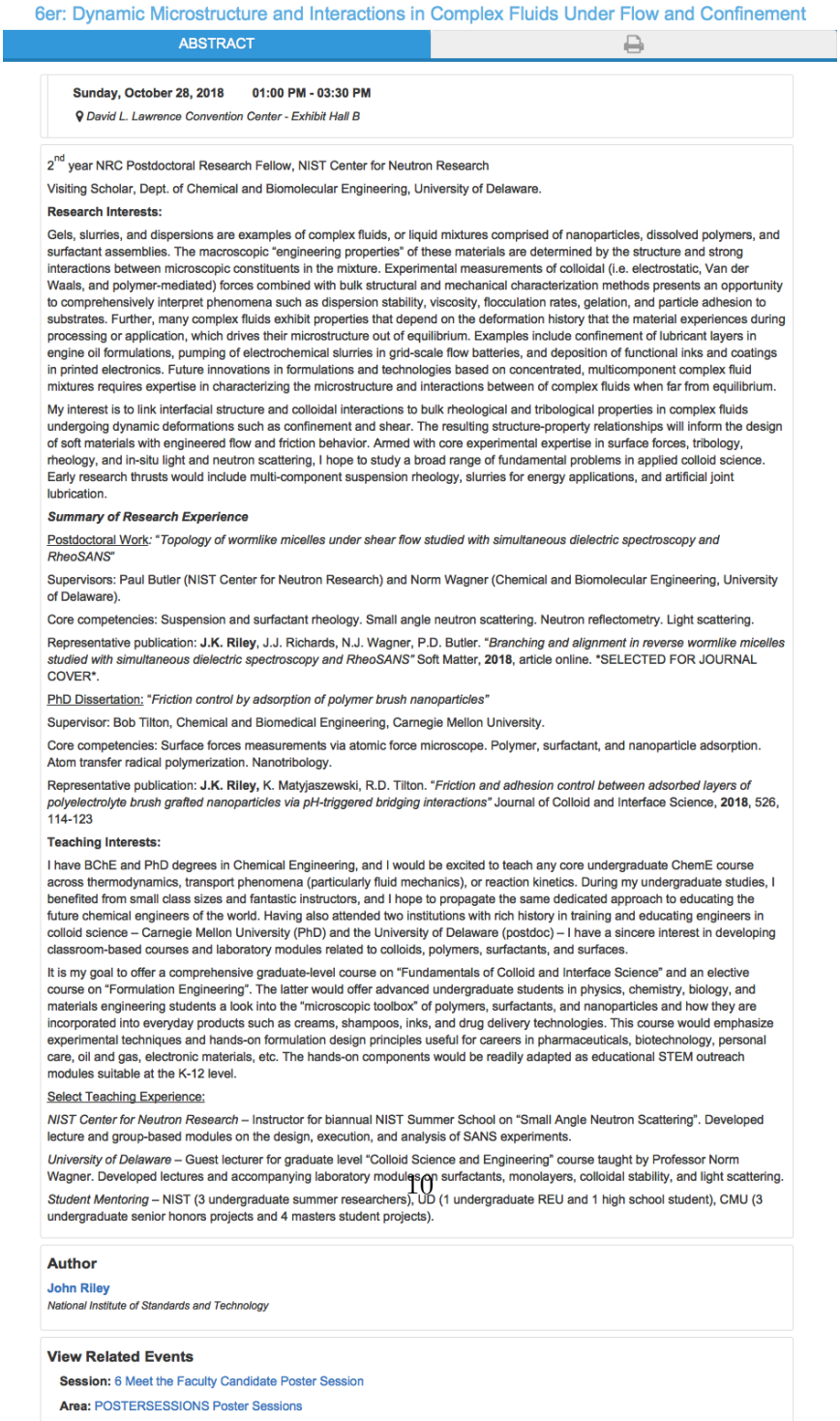
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# 1 #AIChEAnnual Here are a few talks from @CMU_-Cheme on Sunday @ChEnected tweet:thread

## 1.1 6er: Dynamic Microstructure and Interactions in Complex Fluids Under Flow and Confinement https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/542197 #AIChEAnnual #PhdAlum



6er: Dynamic Microstructure and Interactions in Complex Fluids Under Flow and Confinement

ABSTRACT

**Sunday, October 28, 2018 01:00 PM - 03:30 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

2nd year NRC Postdoctoral Research Fellow, NIST Center for Neutron Research

Visiting Scholar, Dept. of Chemical and Biomolecular Engineering, University of Delaware.

**Research Interests:**

Gels, slurries, and dispersions are examples of complex fluids, or liquid mixtures comprised of nanoparticles, dissolved polymers, and surfactant assemblies. The macroscopic "engineering properties" of these materials are determined by the structure and strong interactions between microscopic constituents in the mixture. Experimental measurements of colloidal (i.e. electrostatic, Van der Waals, and polymer-mediated) forces combined with bulk structural and mechanical characterization methods presents an opportunity to comprehensively interpret phenomena such as dispersion stability, viscosity, flocculation rates, gelation, and particle adhesion to substrates. Further, many complex fluids exhibit properties that depend on the deformation history that the material experiences during processing or application, which drives their microstructure out of equilibrium. Examples include confinement of lubricant layers in engine oil formulations, pumping of electrochemical slurries in grid-scale flow batteries, and deposition of functional inks and coatings in printed electronics. Future innovations in formulations and technologies based on concentrated, multicomponent complex fluid mixtures requires expertise in characterizing the microstructure and interactions between of complex fluids when far from equilibrium.

My interest is to link interfacial structure and colloidal interactions to bulk rheological and tribological properties in complex fluids undergoing dynamic deformations such as confinement and shear. The resulting structure-property relationships will inform the design of soft materials with engineered flow and friction behavior. Armed with core experimental expertise in surface forces, tribology, rheology, and in-situ light and neutron scattering, I hope to study a broad range of fundamental problems in applied colloid science. Early research thrusts would include multi-component suspension rheology, slurries for energy applications, and artificial joint lubrication.

***Summary of Research Experience***

Postdoctoral Work: "*Topology of wormlike micelles under shear flow studied with simultaneous dielectric spectroscopy and RheoSANS*"

Supervisors: Paul Butler (NIST Center for Neutron Research) and Norm Wagner (Chemical and Biomolecular Engineering, University of Delaware).

Core competencies: Suspension and surfactant rheology. Small angle neutron scattering. Neutron reflectometry. Light scattering.

Representative publication: **J.K. Riley**, J.J. Richards, N.J. Wagner, P.D. Butler. "*Branching and alignment in reverse wormlike micelles studied with simultaneous dielectric spectroscopy and RheoSANS*" Soft Matter, **2018**, article online. *SELECTED FOR JOURNAL COVER*.

PhD Dissertation: "*Friction control by adsorption of polymer brush nanoparticles*"

Supervisor: Bob Tilton, Chemical and Biomedical Engineering, Carnegie Mellon University.

Core competencies: Surface forces measurements via atomic force microscope. Polymer, surfactant, and nanoparticle adsorption. Atom transfer radical polymerization. Nanotribology.

Representative publication: **J.K. Riley,** K. Matyjaszewski, R.D. Tilton. "*Friction and adhesion control between adsorbed layers of polyelectrolyte brush grafted nanoparticles via pH-triggered bridging interactions*" Journal of Colloid and Interface Science, **2018**, 526, 114-123

**Teaching Interests:**

I have BChE and PhD degrees in Chemical Engineering, and I would be excited to teach any core undergraduate ChemE course across thermodynamics, transport phenomena (particularly fluid mechanics), or reaction kinetics. During my undergraduate studies, I benefited from small class sizes and fantastic instructors, and I hope to propagate the same dedicated approach to educating the future chemical engineers of the world. Having also attended two institutions with rich history in training and educating engineers in colloid science – Carnegie Mellon University (PhD) and the University of Delaware (postdoc) – I have a sincere interest in developing classroom-based courses and laboratory modules related to colloids, polymers, surfactants, and surfaces.

It is my goal to offer a comprehensive graduate-level course on "Fundamentals of Colloid and Interface Science" and an elective course on "Formulation Engineering". The latter would offer advanced undergraduate students in physics, chemistry, biology, and materials engineering students a look into the "microscopic toolbox" of polymers, surfactants, and nanoparticles and how they are incorporated into everyday products such as creams, shampoos, inks, and drug delivery technologies. This course would emphasize experimental techniques and hands-on formulation design principles useful for careers in pharmaceuticals, biotechnology, personal care, oil and gas, electronic materials, etc. The hands-on components would be readily adapted as educational STEM outreach modules suitable at the K-12 level.

Select Teaching Experience:

*NIST Center for Neutron Research* – Instructor for biannual NIST Summer School on "Small Angle Neutron Scattering". Developed lecture and group-based modules on the design, execution, and analysis of SANS experiments.

*University of Delaware* – Guest lecturer for graduate level "Colloid Science and Engineering" course taught by Professor Norm Wagner. Developed lectures and accompanying laboratory modules on surfactants, monolayers, colloidal stability, and light scattering.

*Student Mentoring* – NIST (3 undergraduate summer researchers), UD (1 undergraduate REU and 1 high school student), CMU (3 undergraduate senior honors projects and 4 masters student projects).

**Author**

**John Riley**

*National Institute of Standards and Technology*

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## 1.2 6bg: Catalysis Informatics: Accelerating Search and Discovery of New Catalysts https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/542295 @jacob_boes @SUNCAT_Center #FacultyCandidate #PhdAlum of @CMU_ChemE

6bg: Catalysis Informatics: Accelerating Search and Discovery of New Catalysts

ABSTRACT

**Sunday, October 28, 2018 01:00 PM - 03:30 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

**Research Interests:**

I am interested in the development of tools for catalysis informatics. More specifically, this includes methods for improving the rate at which we search through possible catalytic systems of interest using database mining, machine learning, and high-throughput screening techniques.

During my PhD, I studied computational catalysis at Carnegie Mellon University. I worked with Professor John Kitchin to develop a new kind of atomic potential constructed from neural networks. These potentials have become increasingly popular in the last decade with the advent of fingerprinting schemes, proposed by Behler and Parrinello, which allow them to be trained to a wide variety of atomic input structures. The methods are now capable of reproducing potential energy surfaces (PES) of a multitude of chemical systems with levels of accuracy unrivaled by their physical potential counterparts[1,2]. They are also significantly faster to evaluate than a full quantum chemistry calculation resulting in the ability to study much larger unit cells and more complex chemistry.

For the last year, I have been working as a post-doc at Stanford working with Thomas Bligaard and the catalytic methods development team at SUNCAT. Here I am developing the tools needed to realize high-throughput techniques for the field of computational catalysis. To accomplish this goal, there is a need to enumerate the space of all possible catalytic surfaces of interest. In cheminformatics this is performed through using graph theory tools which allows molecules to be considered as component vertices (the atoms) connected by bonds (edges). With this basic precept, a great number of molecular structures of interest can be represented computationally; a critical first step to any computational high-throughput implementation.

For catalysis, there is a bulk with and surface to be considered along with the adsorbed molecule and producing a unique descriptor is not trivial. In the field of materials informatics, systems are represented in 3D periodically bound boxes, so graphical methodologies are not commonly used. My recent work has focused on beginning from bulk structures and generating graphs from these periodic systems. From a bulk structure, any slab of interest in heterogeneous catalysis can be generated. Finally, unique adsorption sites can be identified through symmetries and used to add molecules to the surface in a systematic way, resulting in a method for producing a unique connectivity matrix for slabs with adsorbates; the basis for a complete catalysis informatics approach. These methodologies are available through the open source Python package, CatKit, for which I am the lead developer.

As available computational resources continue to increase, the rate of data generation in the field of catalysis is quickly beginning to outstrip the ability of those in the field to interpret it all. To improve the ability of all those in the field of catalysis to continue making rapid scientific progress, there is a need to develop better databases which are capable of storing and searching the information; similar to the way AFLOW, Materials Project, OQMD, and others have done for the field of material science. For catalysts, this is significantly more challenging due to the even larger number of possible surfaces and adsorption configurations of interest. Even with a robust database structure capable of storing a breadth of relevant information, tools capable of intelligent and sophisticated queries to the database are also needed. Machine-learning not only benefits from such large repositories of information, it can also essential in making more intelligent selections from a large number of interesting structures to find new and interesting results. These tools will prove to be fundamental for high-throughput research.

**Teaching Interests:**

As an instructor, I teach: thermodynamics, kinetics, molecular simulation (and underlying quantum theory), reaction engineering, functional coding languages, and general chemical engineering. Chemical engineers fulfill a diverse range of demand in industry positions today. As modern tools become increasingly computationally intensive, there is an ever growing need to shift the emphasis of education to computational tools and techniques. As such, I strongly support the integration of coding into the classroom. I am also interested in perusing educational research to do with code based instruction which I describe in greater detail below.

I have completed instruction on educational techniques for engineering in an evidence-based teaching course specific to STEM fields. I am a strong proponent of the flipped classroom as a pedagogical teaching model. This model requires students to study lecture material before the class and implement it on assignments given during class. Among other benefits, such a system allows the instructor to provide immediate feedback to students which is shown to improve student retention rates and longevity of student knowledge. Further details of my pedagogical teaching strategies and achievements can be found on my website.

I would like to study the potential benefits of teaching code-based chemical engineering tools in an immersive environment. The concept is similar to that of language immersion, in which various subject material is taught through the medium of a second language. Language immersion is known to dramatically improve student comprehension of the second language in technical communication with no discernible negative impact on primary language skills. Teaching students to solve chemical engineering problems can similarly be framed in the medium of a coding language, such as Python or MatLab.

**Relevant Work**

1. Jacob R. Boes, Mitchell C. Groenenboom, John A. Keith, and John R. Kitchin. Neural network and ReaxFF comparison for Au properties. International Journal of Quantum Chemistry, 116(13):979-987, 2016. DOI

2. Jacob R. Boes and John R. Kitchin. Modeling Segregation on AuPd(111) Surfaces with Density Functional Theory and Monte Carlo Simulations. The Journal of Physical Chemistry C, 121(6):3479-3487, 2017. DOI

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**Session:** 6 Meet the Faculty Candidate Poster Session

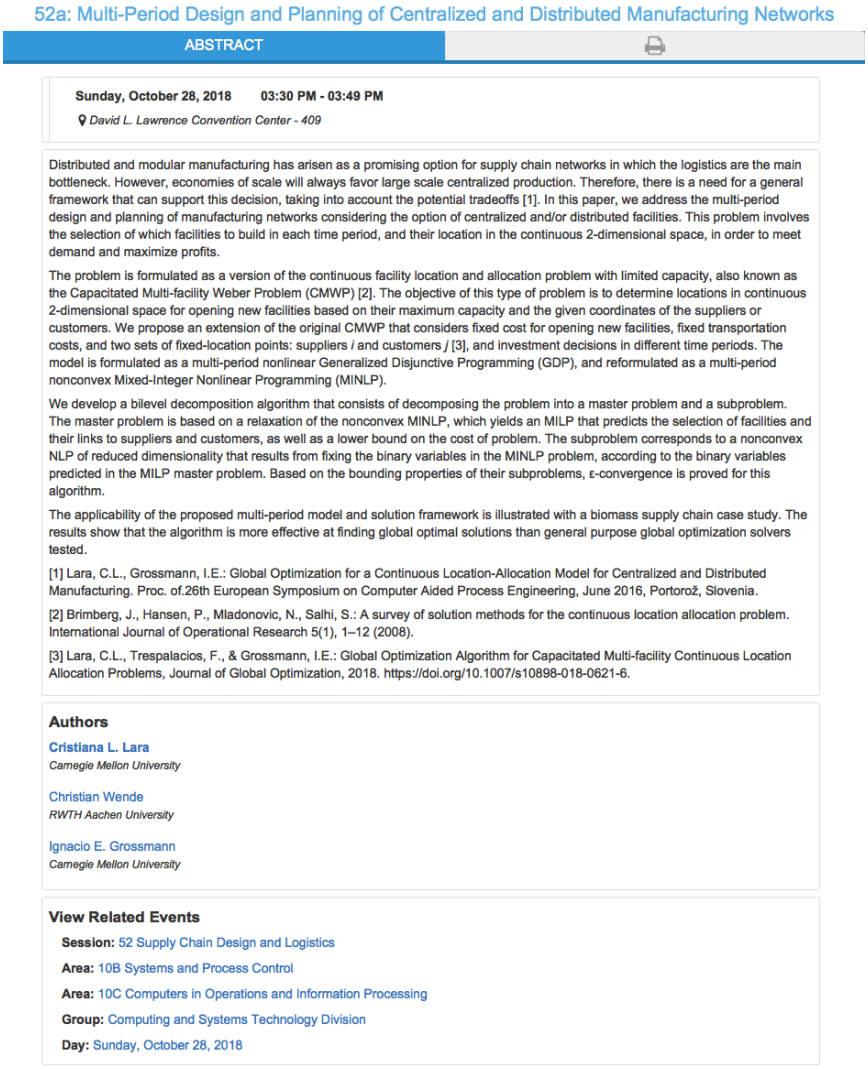
**Area:** POSTERSESSIONS Poster Sessions

**Area:** 04 Education

**Area:** T1 Meet the Faculty Candidate Poster Session – Sponsored by the Education Division

**Group:** Meet the Faculty Candidate Poster Session – Sponsored by the Education Division

## 1.3 52a: Multi-Period Design and Planning of Centralized and Distributed Manufacturing Networks https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/535345 @crislopes-lara @RWTH @AIChECAST10



52a: Multi-Period Design and Planning of Centralized and Distributed Manufacturing Networks

ABSTRACT

**Sunday, October 28, 2018** **03:30 PM - 03:49 PM**

*David L. Lawrence Convention Center - 409*

Distributed and modular manufacturing has arisen as a promising option for supply chain networks in which the logistics are the main bottleneck. However, economies of scale will always favor large scale centralized production. Therefore, there is a need for a general framework that can support this decision, taking into account the potential tradeoffs [1]. In this paper, we address the multi-period design and planning of manufacturing networks considering the option of centralized and/or distributed facilities. This problem involves the selection of which facilities to build in each time period, and their location in the continuous 2-dimensional space, in order to meet demand and maximize profits.

The problem is formulated as a version of the continuous facility location and allocation problem with limited capacity, also known as the Capacitated Multi-facility Weber Problem (CMWP) [2]. The objective of this type of problem is to determine locations in continuous 2-dimensional space for opening new facilities based on their maximum capacity and the given coordinates of the suppliers or customers. We propose an extension of the original CMWP that considers fixed cost for opening new facilities, fixed transportation costs, and two sets of fixed-location points: suppliers *i* and customers *j* [3], and investment decisions in different time periods. The model is formulated as a multi-period nonlinear Generalized Disjunctive Programming (GDP), and reformulated as a multi-period nonconvex Mixed-Integer Nonlinear Programming (MINLP).

We develop a bilevel decomposition algorithm that consists of decomposing the problem into a master problem and a subproblem. The master problem is based on a relaxation of the nonconvex MINLP, which yields an MILP that predicts the selection of facilities and their links to suppliers and customers, as well as a lower bound on the cost of problem. The subproblem corresponds to a nonconvex NLP of reduced dimensionality that results from fixing the binary variables in the MINLP problem, according to the binary variables predicted in the MILP master problem. Based on the bounding properties of their subproblems, ε-convergence is proved for this algorithm.

The applicability of the proposed multi-period model and solution framework is illustrated with a biomass supply chain case study. The results show that the algorithm is more effective at finding global optimal solutions than general purpose global optimization solvers tested.

[1] Lara, C.L., Grossmann, I.E.: Global Optimization for a Continuous Location-Allocation Model for Centralized and Distributed Manufacturing. Proc. of.26th European Symposium on Computer Aided Process Engineering, June 2016, Portorož, Slovenia.

[2] Brimberg, J., Hansen, P., Mladonovic, N., Salhi, S.: A survey of solution methods for the continuous location allocation problem. International Journal of Operational Research 5(1), 1–12 (2008).

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**Session:** 52 Supply Chain Design and Logistics

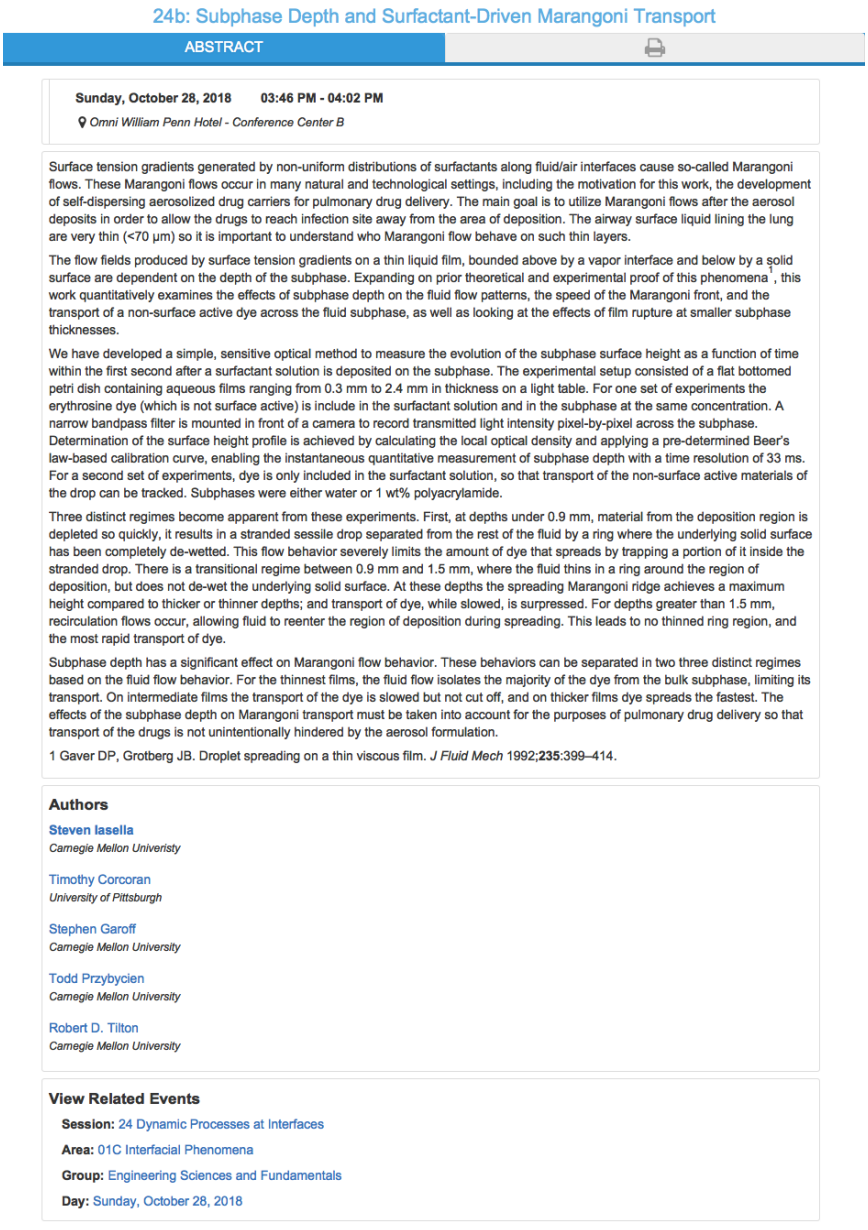
**Area:** 10B Systems and Process Control

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Sunday, October 28, 2018

## 1.4 24b: Subphase Depth and Surfactant-Driven Marangoni Transport https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528141



24b: Subphase Depth and Surfactant-Driven Marangoni Transport

ABSTRACT

**Sunday, October 28, 2018** **03:46 PM - 04:02 PM**

*Omni William Penn Hotel - Conference Center B*

Surface tension gradients generated by non-uniform distributions of surfactants along fluid/air interfaces cause so-called Marangoni flows. These Marangoni flows occur in many natural and technological settings, including the motivation for this work, the development of self-dispersing aerosolized drug carriers for pulmonary drug delivery. The main goal is to utilize Marangoni flows after the aerosol deposits in order to allow the drugs to reach infection site away from the area of deposition. The airway surface liquid lining the lung are very thin (<70 μm) so it is important to understand who Marangoni flow behave on such thin layers.

The flow fields produced by surface tension gradients on a thin liquid film, bounded above by a vapor interface and below by a solid surface are dependent on the depth of the subphase. Expanding on prior theoretical and experimental proof of this phenomena[1], this work quantitatively examines the effects of subphase depth on the fluid flow patterns, the speed of the Marangoni front, and the transport of a non-surface active dye across the fluid subphase, as well as looking at the effects of film rupture at smaller subphase thicknesses.

We have developed a simple, sensitive optical method to measure the evolution of the subphase surface height as a function of time within the first second after a surfactant solution is deposited on the subphase. The experimental setup consisted of a flat bottomed petri dish containing aqueous films ranging from 0.3 mm to 2.4 mm in thickness on a light table. For one set of experiments the erythrosine dye (which is not surface active) is include in the surfactant solution and in the subphase at the same concentration. A narrow bandpass filter is mounted in front of a camera to record transmitted light intensity pixel-by-pixel across the subphase. Determination of the surface height profile is achieved by calculating the local optical density and applying a pre-determined Beer's law-based calibration curve, enabling the instantaneous quantitative measurement of subphase depth with a time resolution of 33 ms. For a second set of experiments, dye is only included in the surfactant solution, so that transport of the non-surface active materials of the drop can be tracked. Subphases were either water or 1 wt% polyacrylamide.

Three distinct regimes become apparent from these experiments. First, at depths under 0.9 mm, material from the deposition region is depleted so quickly, it results in a stranded sessile drop separated from the rest of the fluid by a ring where the underlying solid surface has been completely de-wetted. This flow behavior severely limits the amount of dye that spreads by trapping a portion of it inside the stranded drop. There is a transitional regime between 0.9 mm and 1.5 mm, where the fluid thins in a ring around the region of deposition, but does not de-wet the underlying solid surface. At these depths the spreading Marangoni ridge achieves a maximum height compared to thicker or thinner depths; and transport of dye, while slowed, is surpressed. For depths greater than 1.5 mm, recirculation flows occur, allowing fluid to reenter the region of deposition during spreading. This leads to no thinned ring region, and the most rapid transport of dye.

Subphase depth has a significant effect on Marangoni flow behavior. These behaviors can be separated in two three distinct regimes based on the fluid flow behavior. For the thinnest films, the fluid flow isolates the majority of the dye from the bulk subphase, limiting its transport. On intermediate films the transport of the dye is slowed but not cut off, and on thicker films dye spreads the fastest. The effects of the subphase depth on Marangoni transport must be taken into account for the purposes of pulmonary drug delivery so that transport of the drugs is not unintentionally hindered by the aerosol formulation.

1 Gaver DP, Grotberg JB. Droplet spreading on a thin viscous film. *J Fluid Mech* 1992;**235**:399–414.

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**View Related Events**

**Session:** 24 Dynamic Processes at Interfaces

**Area:** 01C Interfacial Phenomena

**Group:** Engineering Sciences and Fundamentals

**Day:** Sunday, October 28, 2018

## 1.5 17c: Structure-Function-Dynamics Relationships in Next Generation Protein-Polymer Conjugates @DrAlanRussell @MatyPolymerLab https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/532740

17c: Structure-Function-Dynamics Relationships in Next Generation Protein-Polymer Conjugates

ABSTRACT

**Sunday, October 28, 2018 04:06 PM - 04:24 PM**

*Westin Convention Center - Westmoreland West-Central*

Protein-polymer conjugates utilize synthetic polymers to modulate the biological activity of proteins. The physiochemical properties of the polymers are designed so that they impart a desired effect on the protein including stealth from the immune system, stability in harsh environments, or sensing in a response to an external stimuli. The synthesis of protein-polymer conjugates, specifically when using amino reactive initiators and a grafting-from approach, is not well controlled and therefore, it is not well understood how the polymers impart their effect on the protein depending on polymer charge, molecular weight, modification density, and attachment site. Clarifying these structure-function-dynamic type relationships would provide the foundation to create designer protein-polymer conjugates, using polymer-based protein engineering, for a myriad of applications ranging from therapeutics to agrochemicals to food biotechnology.

There are many hypotheses for how polymers stabilize proteins including, but not limited to, polymers creating a stabilizing hydration layer around the protein (similar to osmolytes), polymers providing protection from autolysis, or polymers increasing the structural stability of the protein. The covalently attached polymers are free to move around, can adopt different conformations, and can participate in various noncovalent interactions at the protein-polymer interface. These noncovalent interactions include van der Waals interactions, electrostatic interactions, hydrogen bonds, or a combination of them all, and can lead to bioconjugate stabilization. Here, we sought to develop a mechanistic understanding of bioconjugate structure-function-dynamic relationships through a series of experimental and theoretical studies to determine the driving force for conjugate activity and stability. It is difficult to determine specific interactions in protein-polymer conjugates experimentally at the macroscale level, so we also used molecular dynamic (MD) simulations. MD simulations can be used to study the dynamic motions of biomolecules over time while visualizing specific interactions. MD simulations have conventionally been performed on native proteins, but recent efforts have extended into studying protein-polymer conjugates.

In this presentation, we will first describe how we were able to modulate the activities and pH dependencies of fully modified grafted-from bioconjugates by engineering the polymer charge and chain length. We will also propose a mechanism for how covalently attached polymers stabilize proteins under acidic conditions, depending on the physiochemical properties of the polymer, that challenges the commonly accepted hypotheses. We will further validate the experimental studies with theoretical molecular dynamics simulations. This work will provide a guide for the synthesis and application of next generation protein-polymer conjugates.

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**Session:** 17 Biobased Materials: Design and Application

**Area:** 08B Biomaterials

**Area:** 15C Bioengineering

**Group:** Food, Pharmaceutical & Bioengineering Division

**Day:** Sunday, October 28, 2018

## 1.6 51c: Plasmoalgorithms, a Collection of Decomposition Algorithms for Graph-Based Problem Representations @bbrunaud https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528251 @AIChECAST10

51c: Plasmoalgorithms, a Collection of Decomposition Algorithms for Graph-Based Problem Representations

ABSTRACT

**Sunday, October 28, 2018 04:08 PM - 04:27 PM**

*David L. Lawrence Convention Center - 410*

Mathematical programming has been successfully applied to solve decision-making problems in a variety of applications and industries, such as petroleum, chemical, pharmaceutical, and consumer products[1]. The success of the first experiences in these applications has continuously motivated the need for solving larger, and more complex problems. Furthermore, the recent focus on optimization under uncertainty has increased the problem sizes by at least one order of magnitude. At the same time solvers have improved significantly over the years[2], while computing hardware has not increased in speed but in number of parallel cores available. To keep improving and take advantage of this scenario the advancement of decomposition algorithms is necessary.

Currently, implementing and testing decomposition algorithms is a common and time consuming task in Process Systems Engineering (PSE) research. Lagrangean decomposition[3], Benders decomposition[4], and derived algorithms have been the choice for solving large-scale problems. The implementations of these algorithms are usually tailored to the application, there is an absence of general purpose implementations. A few attempts in this direction include the Benders decomposition implementation in CPLEX since version 12.7[5], and DSP[6], an implementation of dual decomposition for solving stochastic programming problems with integer recourse. The first one is a commercial code, while the second one is a complex code written in C++, making them hard to extend and improve.

We present PlasmoAlgorithms, a collection of general decomposition algorithms implemented in Julia programming language, using Plasmo[7] graphs as an input. Plasmo is a Julia package that allows the generation of graphs of models, where each node represents an independent model, and each edge represents a linking constraint. Since the Julia programming language is easy to code, and efficient, PlasmoAlgorithms is a good candidate to become a meeting place for the PSE and Operations Research communities to implement state-of-the-art algorithms, and for testing of novel ideas.

PlasmoAlgorithms currently includes implementations of multilevel Benders decomposition, Lagrangean decomposition, and Cross decomposition. The Benders implementation uses a multicuts, in which an independent cut is generated by each subproblem; and also includes cuts suitable for mixed-integer subproblems. The Lagrangean decomposition implementation includes several options for updating the multipliers, including a probing subgradient method, in which the bound of the candidate step is first probed to decide the length of the step taken. The Cross decomposition implementation follows the algorithm proposed by Mitra et al[8]. All the code is open source, and the implementations are modular enough to allow for easy extension of the algorithms.

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**View Related Events**

**Session:** 51 Software Tools and Implementations for Process Systems Engineering

**Area:** 10 Computing Systems and Technology Division

**Group:** Computing and Systems Technology Division

**Day:** Sunday, October 28, 2018

## 1.7 52d: Solving Robust Vehicle Routing Via a Branch-Price-and-Cut Approach https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528612 @AIChECAST10

52d: Solving Robust Vehicle Routing Via a Branch-Price-and-Cut Approach

ABSTRACT

**Sunday, October 28, 2018 04:27 PM - 04:46 PM**

*David L. Lawrence Convention Center - 409*


The Vehicle Routing Problem (VRP) is one of the most highly studied combinatorial optimization problems in the area of Supply Chains and Logistics [1]. It aims to determine a set of optimal routes for vehicles to serve customers such that all known demands are satisfied and in a way such that all vehicle capacities are respected. The Branch-Price-and-Cut (BP&C) method has been gradually accepted as the state-of-the-art algorithm to tackle deterministic VRP and its variants. The BP&C algorithm is sophisticated in a way that the linear relaxation at every node in the branch-and-bound tree is solved via Column Generation (CG) and cutting planes are added to strengthen the relaxation. In our VRP context, columns denote feasible routes, and they are generated by solving a Shortest Path Problem with Resource Constraints (SPPRC) via a labeling algorithm [2]. It has been shown that Rounded Capacity Inequalities (RCI) [3] and limited-memory subset row cuts [4] can effectively tighten the linear model in the deterministic setting.

Recently, VRP under demand uncertainty [5][6] has attracted considerable attention. It is motivated by the fact that customer demands are often not readily available at the time the distributor needs to commit to some routing plan. In this work, we combine Robust Optimization (RO) [7] and BP&C in order to develop an effective approach for robust VRP under demand uncertainty. When extending BP&C for a robust VRP, the optimal routing design has to be robust feasible. There are two different perspectives for how to achieve this. The first and most intuitive way is to generate deterministically feasible routes via solving a standard SPPRC and enforce robust feasibility via robust RCI [6]. This approach is very generic and can be ideally applied under any uncertainty set selection. Another approach is to directly generate robust feasible columns. This necessitates an efficient algorithm to exactly solve a robust SPPRC, which can be a tough proposition, since this problem is in general strongly NP-hard [8]. Recently, Pessoa *et al.* [9] showed that for some special structure uncertainty sets (e.g., Γ uncertainty set [10]), robust columns can be generated via solving a sequence of polynomially many deterministic SPPRC.

In this work, we first evaluate the two approaches and compare their performances using several well-known uncertainty sets, including the Γ set and budget set [6] among others. The comparison studies are mainly focused on the computational performance and theoretical limitations to different uncertainty sets. In addition, we consider combining the two techniques together and generating robust feasible columns in the CG stage as well as separating robust cuts in the cut generation stage to further tighten the linear relaxation.


References:


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**View Related Events**

**Session:** 52 Supply Chain Design and Logistics

**Area:** 10B Systems and Process Control

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Sunday, October 28, 2018

## 1.8 47d: Microkinetic Analysis of Ethylene Hydrogenation on Pd-Based Catalysts: Effect of Subsurface Hydrogen on Mechanism and Rate Control https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/527751 @aiche_cre

47d: Microkinetic Analysis of Ethylene Hydrogenation on Pd-Based Catalysts: Effect of Subsurface Hydrogen on Mechanism and Rate Control

ABSTRACT

**Sunday, October 28, 2018 04:36 PM - 04:58 PM**

*David L. Lawrence Convention Center - 404*


Ethylene, $C_2H_4$, hydrogenation to ethane, $C_2H_6$, has been widely studied to gain insight into more complex hydrogenation reactions. Yet, hydrogenation of $C_2H_4$ is a complex reaction itself, as it occurs through a series of elementary steps. Reaction kinetics depend sensitively on temperature, reactant partial pressures and the catalyst. Hence, it is difficult to analyze the reaction mechanism and elucidate the kinetic parameters of each elementary step, experimentally. Moreover, studies in recent years show that subsurface hydrogen plays a critical role in hydrogenation reactions. However, presence of subsurface hydrogen further complicates the reaction mechanism and kinetics.

In this study, activity of $C_2H_4$ hydrogenation on Pd, $Pd_{85}Cu_{15}$, and $Pd_{70}Cu_{30}$ foil catalysts were measured experimentally to elucidate a reaction mechanism that accounts for subsurface hydrogen. Experiments were performed over a temperature range of 300 – 450 K at 1 bar, with various $H_2$ and $C_2H_4$ inlet partial pressures. Experimental $C_2H_6$ conversion data was fit to models derived from two mechanisms; (i) classical Horiuti-Polayni (HP) mechanism, and (ii) mechanism that includes subsurface hydrogen in elementary steps. For each catalyst, suitable model for the reaction was selected through statistical analysis. Degree of rate control (DRC) analysis for each intermediate (IS) and transition (TS) state was also performed based on the selected model. The experimental data and the analysis results show that:

- The activity of catalysts for $C_2H_4$ hydrogenation increase in the order: $Pd_{70}Cu_{30} < Pd_{85}Cu_{15} < Pd$.
- Mechanism that includes the effect of subsurface hydrogen defines the reaction better compared to classical HP mechanism on all Pd-based catalysts used in this study.
- $C_2H_5$ - - H complex, with H supplied from surface for Pd, from subsurface for $Pd_{70}Cu_{30}$ and $Pd_{85}Cu_{15}$, is the significant TS in DRC.
- DRC analysis for IS showed that H occupying a surface site is the important species on Pd and $Pd_{85}Cu_{15}$, whereas it is the subsurface H on $Pd_{70}Cu_{30}$.

To our knowledge, this is the first experimental study on Pd-based foil catalysts that elucidates the detailed reaction mechanism, kinetics and DRC of species/states in ethylene hydrogenation, including the effect of subsurface hydrogen. Results of this study emphasize (i) the importance of subsurface hydrogen by proposing a new mechanism for ethylene hydrogenation on Pd-based foil catalysts, and (ii) the fact that only one state/species or elementary reaction cannot be considered as rate determining, as overall reaction is a combination off all these steps and states.



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**View Related Events**

**Session:** 47 Reaction Path Analysis

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Sunday, October 28, 2018

## 1.9 51f: Data Driven Modeling in Alamo: Feature Selection and Non-Parametric Modeling Applications https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/526364 @AIChECAST10

51f: Data Driven Modeling in Alamo: Feature Selection and Non-Parametric Modeling Applications

ABSTRACT

**Sunday, October 28, 2018** **05:05 PM - 05:24 PM**

*David L. Lawrence Convention Center - 410*

ALAMO is a computational methodology designed to learn algebraic models from data. The models produce via this approach constitute a linear combination of parametric transformations of input variables. Recent developments comparing variants of the lasso and best subset selection [1] have motivated the inclusion of additional heuristics to identify optimally predictive models. The linear model selection methodology utilized by ALAMO is explained and demonstrated across a diverse set of benchmark problems. The problem of feature engineering, or the identification of promising transformations of input variables for inclusion in linear models, is directly addressed by the ALAMO methodology. Diverse sets of algebraic features are applied to optimization benchmark problems in order to identify accurate models that are tailored for optimization with equation oriented solvers.

When physical interpretability of a model is not prioritized, it is common to use non-parametric regression techniques to provide smooth interpolations of available data. ALAMO is capable of using non-parametric transformations of points in the domain of interest as features for linear model selection. Examples utilizing Gaussian radial basis functions are explored in order to identify optimal interpolative models. Constrained regression techniques used by ALAMO can force these non-parametric models to obey insights enforced by the modeler, enhancing their ability to model accurately in extrapolative domains.

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**Session:** 51 Software Tools and Implementations for Process Systems Engineering

**Area:** 10 Computing Systems and Technology Division

**Group:** Computing and Systems Technology Division

**Day:** Sunday, October 28, 2018

## 1.10 Come join us at our reception!



**Please Join Us in Pittsburgh for the 2018 AIChE Carnegie Mellon Reception**

**Sunday, October 28, 2018**
*Singleton Room, Roberts Hall*

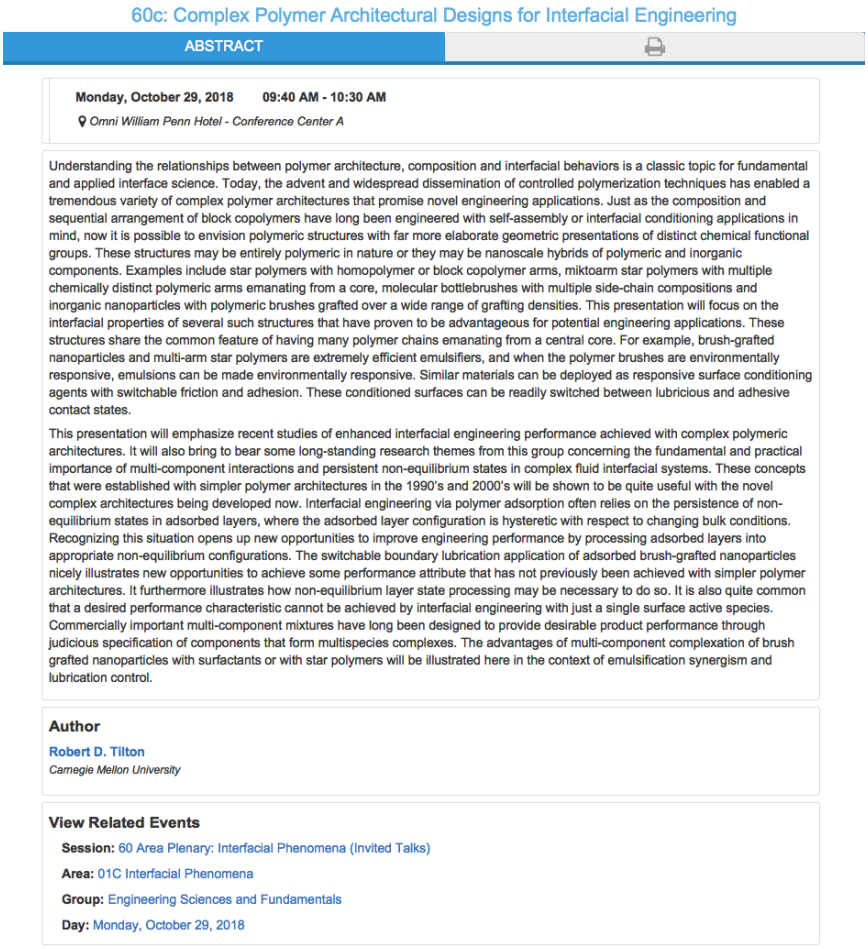
**Cocktail Reception:** 6:00–7:30pm

**Buffet Dinner:** 7:30–9:30pm

Tour the TechSpark makerspace, Hamerschlag Hall C105

# 2 #AIChEAnnual Here are a few talks from @CMU_-Cheme on Monday @ChEnected :tweet:thread:w

## 2.1 60c: Complex Polymer Architectural Designs for Interfacial Engineering #invited https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/540836



60c: Complex Polymer Architectural Designs for Interfacial Engineering

ABSTRACT

**Monday, October 29, 2018** **09:40 AM - 10:30 AM**

*Omni William Penn Hotel - Conference Center A*

Understanding the relationships between polymer architecture, composition and interfacial behaviors is a classic topic for fundamental and applied interface science. Today, the advent and widespread dissemination of controlled polymerization techniques has enabled a tremendous variety of complex polymer architectures that promise novel engineering applications. Just as the composition and sequential arrangement of block copolymers have long been engineered with self-assembly or interfacial conditioning applications in mind, now it is possible to envision polymeric structures with far more elaborate geometric presentations of distinct chemical functional groups. These structures may be entirely polymeric in nature or they may be nanoscale hybrids of polymeric and inorganic components. Examples include star polymers with homopolymer or block copolymer arms, miktoarm star polymers with multiple chemically distinct polymeric arms emanating from a core, molecular bottlebrushes with multiple side-chain compositions and inorganic nanoparticles with polymeric brushes grafted over a wide range of grafting densities. This presentation will focus on the interfacial properties of several such structures that have proven to be advantageous for potential engineering applications. These structures share the common feature of having many polymer chains emanating from a central core. For example, brush-grafted nanoparticles and multi-arm star polymers are extremely efficient emulsifiers, and when the polymer brushes are environmentally responsive, emulsions can be made environmentally responsive. Similar materials can be deployed as responsive surface conditioning agents with switchable friction and adhesion. These conditioned surfaces can be readily switched between lubricious and adhesive contact states.

This presentation will emphasize recent studies of enhanced interfacial engineering performance achieved with complex polymeric architectures. It will also bring to bear some long-standing research themes from this group concerning the fundamental and practical importance of multi-component interactions and persistent non-equilibrium states in complex fluid interfacial systems. These concepts that were established with simpler polymer architectures in the 1990's and 2000's will be shown to be quite useful with the novel complex architectures being developed now. Interfacial engineering via polymer adsorption often relies on the persistence of non-equilibrium states in adsorbed layers, where the adsorbed layer configuration is hysteretic with respect to changing bulk conditions. Recognizing this situation opens up new opportunities to improve engineering performance by processing adsorbed layers into appropriate non-equilibrium configurations. The switchable boundary lubrication application of adsorbed brush-grafted nanoparticles nicely illustrates new opportunities to achieve some performance attribute that has not previously been achieved with simpler polymer architectures. It furthermore illustrates how non-equilibrium layer state processing may be necessary to do so. It is also quite common that a desired performance characteristic cannot be achieved by interfacial engineering with just a single surface active species. Commercially important multi-component mixtures have long been designed to provide desirable product performance through judicious specification of components that form multispecies complexes. The advantages of multi-component complexation of brush grafted nanoparticles with surfactants or with star polymers will be illustrated here in the context of emulsification synergism and lubrication control.

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**View Related Events**

**Session:** 60 Area Plenary: Interfacial Phenomena (Invited Talks)

**Area:** 01C Interfacial Phenomena

**Group:** Engineering Sciences and Fundamentals

**Day:** Monday, October 29, 2018

## 2.2 76e: Forty Years of Computers and Chemical Engineering (1977-2017): Analysis of the Field Via Natural Language Processing Techniques https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/531257 @DowChemical @AIChECAST10

76e: Forty Years of Computers and Chemical Engineering (1977-2017): Analysis of the Field Via Natural Language Processing Techniques

ABSTRACT

**Monday, October 29, 2018** **10:03 AM - 10:30 AM**

*David L. Lawrence Convention Center - 407*

Recent advances in machine learning and natural language processing have produced a variety of tools for computerized mining of textual databases. However, while the mining of numerical data has enjoyed considerable attention, relatively little work has been done by chemical engineers in the analysis of textual information. In this work, we analyzed a large body of literature in chemical engineering using state-of-the-art text mining techniques. Particularly, the focus of attention is in process systems engineering and computer applications in the field of chemical engineering. Given its prominence and relevance to the field, the forty year old journal of Computers and Chemical Engineering (CACE) was investigated for our purposes.

In this work we are interested in answering following two questions. What are the main topics that have appealed to the CACE community since the journal was launched? How has interest in these topics evolved over time? These questions were addressed through topic modeling techniques, including non-negative matrix factorization and the structural topic model [1-4]. The results show that CACE covers diverse topics from process synthesis to artificial neural networks. A total of 18 research topics were identified when we maximized topic coherence [5]. Two dominant research topics that draw a great attention to the community are mathematical programming and process control. The topic prevalence analysis shows that the use of machine learning techniques can be traced back to 1990s. Additionally, we discovered that there is a close relationship between topics "artificial neural network" and "fault diagnosis." These and other findings from our analysis will be discussed in detail.

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**View Related Events**

**Session:** 76 Division Plenary: CAST (Invited Talks)

**Area:** 10 Computing Systems and Technology Division

**Group:** Computing and Systems Technology Division

**Day:** Monday, October 29, 2018

## 2.3 136a: Novel Approaches for the Integration of Supply Chain Planning and Scheduling @bbrunaud @DowChemical @AIChECAST10 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528569

136a: Novel Approaches for the Integration of Supply Chain Planning and Scheduling

ABSTRACT

**Monday, October 29, 2018** **12:30 PM - 12:49 PM**

*David L. Lawrence Convention Center - 408*

Supply chain decisions are hierarchically organized in strategic, tactical, and operational decisions. In practice, these planning processes are frequently conducted independently with minimal exchange of information between them, thereby leading to suboptimal solutions. Achieving a better coordination between these processes allows companies to capture benefits that are currently out of their reach and improve the communication among their functional areas[1]. Optimization methods for the integration of strategic and tactical decisions are well established[2]. Nevertheless, the integration of tactical and operational decisions has not been solved[3],[4]. There are at least three shortcomings that need to be addressed to create integrated planning and scheduling systems that can be used in real-world applications: 1) the models proposed in previous work are too simplistic, assuming a mid-term planning level setting inventory targets for the scheduling; 2) they aim to solve the scheduling model in the horizon of the planning model, and 3) there are no efficient algorithms capable of handling industrial-sized problems in reasonable amount of time.

In this work, we present an integrated formulation of short term multisite planning with batch scheduling using the UOPSS modeling framework[5]. The planning level considers an (s,S) inventory policy for all inventories in the system[6], and distinguishes between make-to-order and make-to-stock products[7]. The scheduling level receives planned orders, corresponding to a mix of inventory replenishment orders and customer orders, to determine the optimal production schedule. The scheduling objective includes minimization of priority-dependent backlog and changeovers. To address the difference in timescales we present an objective scaling approach. In the proposed scaling method the objective function of the model with the largest horizon (planning) is downscaled to make it comparable to the objective of the model of the shorter horizon (scheduling). The method is evaluated through simulation, and compared with traditional integration methods: full-space solution, and representative day approach. Results on industrially sized problems clearly indicate significantly improved solutions with the proposed approach. To address the computational complexity added by the integration of planning and scheduling, Benders decomposition with mixed-integer subproblems inspired by the stochastic programming literature[8],[9], and cross decomposition[10] methods, have also been developed to effectively address the integration problem.

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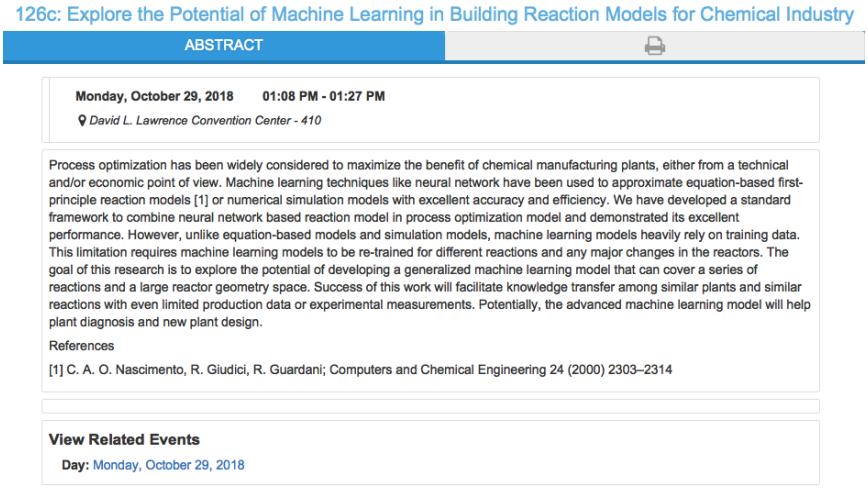
**Session:** 136 CAST Director's Student Presentation Award Finalists

**Area:** 10 Computing Systems and Technology Division

**Group:** Computing and Systems Technology Division

**Day:** Monday, October 29, 2018

## 2.4 126c: Explore the Potential of Machine Learning in Building Reaction Models for Chemical Industry https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/522672 @Dow-Chemical @AIChECAST10



126c: Explore the Potential of Machine Learning in Building Reaction Models for Chemical Industry

ABSTRACT

**Monday, October 29, 2018** **01:08 PM - 01:27 PM**

*David L. Lawrence Convention Center - 410*

Process optimization has been widely considered to maximize the benefit of chemical manufacturing plants, either from a technical and/or economic point of view. Machine learning techniques like neural network have been used to approximate equation-based first-principle reaction models [1] or numerical simulation models with excellent accuracy and efficiency. We have developed a standard framework to combine neural network based reaction model in process optimization model and demonstrated its excellent performance. However, unlike equation-based models and simulation models, machine learning models heavily rely on training data. This limitation requires machine learning models to be re-trained for different reactions and any major changes in the reactors. The goal of this research is to explore the potential of developing a generalized machine learning model that can cover a series of reactions and a large reactor geometry space. Success of this work will facilitate knowledge transfer among similar plants and similar reactions with even limited production data or experimental measurements. Potentially, the advanced machine learning model will help plant diagnosis and new plant design.

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**Day:** Monday, October 29, 2018

## 2.5 136e: Identification of Optimal Dopant Patterns in a Doped Perovskite Oxygen Carrier https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/535033 @NETL_DOE @AIChECAST10

136e: Identification of Optimal Dopant Patterns in a Doped Perovskite Oxygen Carrier

ABSTRACT

**Monday, October 29, 2018 01:46 PM - 02:05 PM**

*David L. Lawrence Convention Center - 408*


Perovskites have been shown to exhibit fast and tunable oxygen release, making them good candidates for oxygen carriers in chemical looping combustion (CLC) [1,2,3,4]. The introduction of dopant atoms in the lattice can have a significant impact on both the lattice structure and the reducibility of the perovskite. As our system of interest, we consider a $BaFeO_3$ perovskite doped with indium that has been shown to exhibit oxygen release at lower temperatures than the undoped counterpart, potentially enabling lower temperature operation of the CLC process. The presence of indium in the lattice is predicted to increase the strength of bonds to neighboring oxygen atoms while bonds to oxygen further away are weakened. Due to the combinatorial number of ways that dopant could be placed in the lattice, it is not obvious which arrangement of dopant would lead to the best release of oxygen.

To that end, this work aims to formulate and solve a mathematical optimization model to identify doped perovskite lattices with optimal properties. To translate the search for a highly reducible perovskite into a mathematical optimization formulation, we first quantify the impact of dopant on the reducibility of the perovskite via a suitable metric, namely the oxygen excess energy, which is defined as the energy difference between the original lattice and the lattice with a particular oxygen removed. This quantity can be calculated accurately by Hubbard corrected density functional theory given a supercell with a particular arrangement of dopant around an oxygen atom in question [5,6]. In order to explore the impact that dopant has on the oxygen excess energy, a motif of relevant locations was defined and the set of possible configurations of dopant in a motif were enumerated. This resulted in 74 rotationally-unique ways to place dopant in the ten nearest sites near a given oxygen. The oxygen excess energy for each of these motifs was evaluated at T=0K and then additional free energy excess energy calculations were performed at process conditions.

Given a complete set of evaluations, we selected a subset of motifs with the lowest oxygen excess energy to consider as "target" motifs to pack in an optimal material. A mathematical optimization model was formulated to relate the placement of dopant atoms in a supercell of the perovskite lattice to the number of oxygen sites that achieved one of the target motifs. Using binary variables and indicator constraints, we were able to model the problem as a mixed integer linear program. This modelling approach is generic in the sense that additional constraints, such as bounds on dopant concentration, fabricability, and stability of the material can be enforced to further restrict the search to materials of interest. In addition to the basic formulation, we derive several approaches to symmetry breaking that are able to improve the tractability of the model. A key advantage of casting our problem as a mathematical optimization model is the ability to rigorously guarantee the exploration of the entire search space and the ability to collect a list of the top designs for evaluation offline against secondary criteria.

Our optimal solutions demonstrate that a significant fraction of oxygen sites can display a target motif. By comparing the quality of our optimal designs against the baseline qualities of designs formed by random dopant placement, we show that nanostructured designs can significantly outperform unstructured materials. Furthermore, we can use our model of randomly doped perovskites to explain trends in the experimentally observed temperature of oxygen release. While the precise placement of dopant in the lattice may be currently impractical, the identification of ideal dopant patterns can serve to guide future development of synthesis methods and to establish rigorous upper bounds on the performance of doped perovskite materials.


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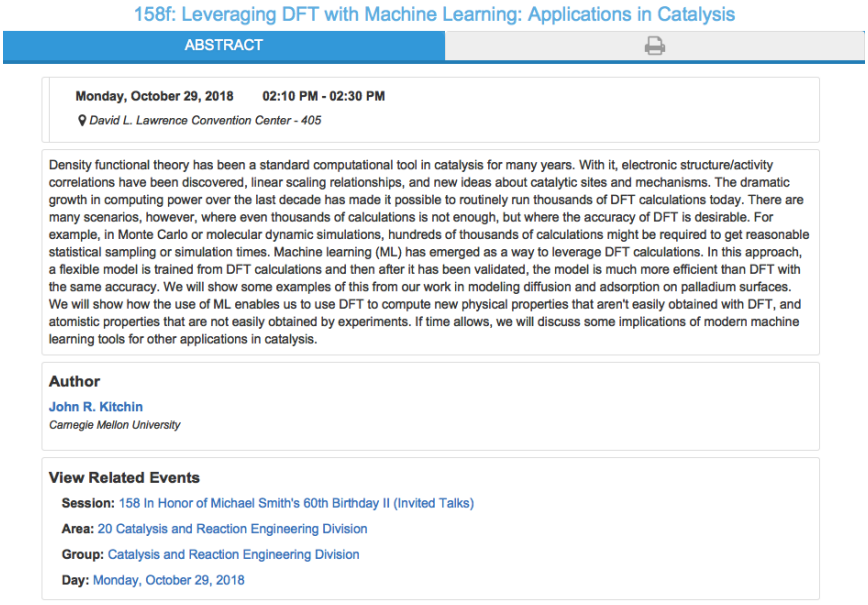
**Session:** 136 CAST Director's Student Presentation Award Finalists

**Area:** 10 Computing Systems and Technology Division

**Group:** Computing and Systems Technology Division

**Day:** Monday, October 29, 2018

## 2.6 158f: Leveraging DFT with Machine Learning: Applications in Catalysis https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/529563 @johnkitchin @aiche_cre



158f: Leveraging DFT with Machine Learning: Applications in Catalysis

ABSTRACT

**Monday, October 29, 2018** **02:10 PM - 02:30 PM**

*David L. Lawrence Convention Center - 405*

Density functional theory has been a standard computational tool in catalysis for many years. With it, electronic structure/activity correlations have been discovered, linear scaling relationships, and new ideas about catalytic sites and mechanisms. The dramatic growth in computing power over the last decade has made it possible to routinely run thousands of DFT calculations today. There are many scenarios, however, where even thousands of calculations is not enough, but where the accuracy of DFT is desirable. For example, in Monte Carlo or molecular dynamic simulations, hundreds of thousands of calculations might be required to get reasonable statistical sampling or simulation times. Machine learning (ML) has emerged as a way to leverage DFT calculations. In this approach, a flexible model is trained from DFT calculations and then after it has been validated, the model is much more efficient than DFT with the same accuracy. We will show some examples of this from our work in modeling diffusion and adsorption on palladium surfaces. We will show how the use of ML enables us to use DFT to compute new physical properties that aren't easily obtained with DFT, and atomistic properties that are not easily obtained by experiments. If time allows, we will discuss some implications of modern machine learning tools for other applications in catalysis.

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**Session:** 158 In Honor of Michael Smith's 60th Birthday II (Invited Talks)

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Monday, October 29, 2018

## 2.7 126g: A Comparison of Mathematical Optimization and Deep Reinforcement Learning for Supply Chain Materials Planning @christiandhubbs @DowChemical https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530734 @AIChECAST10

126g: A Comparison of Mathematical Optimization and Deep Reinforcement Learning for Supply Chain Materials Planning

ABSTRACT

**Monday, October 29, 2018** **02:24 PM - 02:43 PM**

*David L. Lawrence Convention Center - 410*

This paper presents a novel approach to supply chain planning optimization through the utilization of deep reinforcement learning and compares it to modern, mathematical optimization techniques. The supply chain is modeled as a factory with multiple products and stochastic demand for each product. The deep reinforcement learning agent interacts with this environment to learn by experience, as captured in neural networks, to determine which product to produce and when to produce it to satisfy customer demand. The neural networks approximate the value of each state with respect to a value function that accounts for on time delivery, the level of inventory on hand, and the customer service level. The reinforcement learning agent learns through simulations of the manufacturing environment and demand patterns in order to maximize the reward received through the value function. In parallel, we formulate a mixed-integer program that has to make identical decisions under the same constraints. The mixed-integer program can see orders out to a certain horizon, and does not use any information from historical order patterns. We compare and contrast the performance between the two methodologies on several scheduling test examples, and suggest how to choose between them, as well as how they may be used in concert.

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**Session:** 126 Advances in Machine Learning and Intelligent Systems

**Area:** 10E Data and Information Systems

**Group:** Computing and Systems Technology Division

**Day:** Monday, October 29, 2018

## 2.8 182h: Autotuning with Derivative-Free Optimization @bsauk2 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530288 @AIChECAST10

182h: Autotuning with Derivative-Free Optimization

ABSTRACT

**Monday, October 29, 2018 03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

As the landscape of high performance computing evolves, there is a growing need to design software that is optimal on a variety of system architectures. To address this need, algorithms are designed with tunable parameters to allow for performance portability. However, identifying optimal tuning parameters requires solving a large global optimization problem without an explicit algebraic function. Past research has been conducted to automatically determine good tuning parameters through the use of hill climbing algorithms [1], Nelder-mead simplex algorithms [5], genetic algorithms [2], and simulated annealing [4].

In this work we investigate the benefits of autotuning with derivative-free optimization (DFO). While most of the search techniques implemented in autotuners utilize local DFO solvers or heuristic based methods, by employing model-based and global DFO solvers [6], we are able to quickly find high quality tuning parameters. We compare our proposed methodology against state-of-the-art autotuning techniques such as OpenTuner [1], Active Harmony [7], and Starchart [3]. We carry out experiments on dense linear algebra kernels that are essential to the performance of numerous algorithms. In this work, we tune GPU algorithms which have a more complex parameter space than traditional CPU algorithms. We demonstrate the ability of our approach to tune any system by tuning a collection of algorithms on different GPU architectures.

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**Session:** 182 Interactive Session: Applied Mathematics and Numerical Analysis

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**Day:** Monday, October 29, 2018

## 2.9 189i Capturing Non-Ideal Surfactant/Nanoparticle Interfacial Structure with Variable Coverage Molecular Simulations https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/526829 @UlissiGroup @aichecomsef

189i: Capturing Non-Ideal Surfactant/Nanoparticle Interfacial Structure with Variable Coverage Molecular Simulations

ABSTRACT

**Monday, October 29, 2018** **03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

Surfactants adsorbed to hard nanoparticle interfaces are poorly understood despite their importance in nanotechnologies because of their complex non-ideal behavior, the non-deformable nanoparticle interface, and the similar length scales of surfactant molecules and surfaces. We develop a molecular thermodynamic approach to directly simulate isotherms of surfactants at interfaces with arbitrary coverages and to describe surfactants that are not well described with simple molecular thermodynamic theories. All-atom molecular simulations are used to estimate the chemical potential of non-ideal surfactant structures by calculating the free energy required to concentrate a fixed number of adsorbed surfactants at low density and under ideal conditions onto a small subsection of the interface with high local density. Standard adsorption calculations are used to reference the low density state to the bulk liquid surfactant potential, thus providing equilibrium coverage estimates. This approach captures non-ideal effects such as specific surfactant-surfactant interactions, amphiphilic surfactant structure, and more than one monolayer of adsorbates. We show that this new approach reproduces analytical results for hard spheres at the interface and can resolve qualitative changes in surfactant structure at between low and high loadings.

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**Session:** 189 Poster Session: Computational Molecular Science and Engineering Forum (CoMSEF)

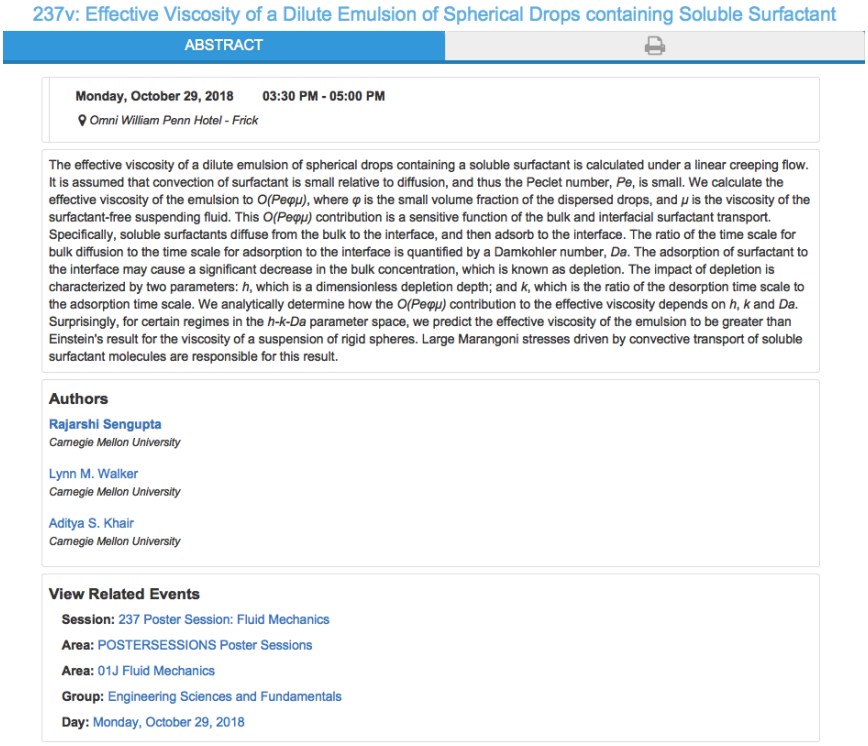
**Area:** POSTERSESSIONS Poster Sessions

**Area:** 21 Computational Molecular Science and Engineering Forum

**Group:** Computational Molecular Science and Engineering Forum

**Day:** Monday, October 29, 2018

## 2.10 237v: Effective Viscosity of a Dilute Emulsion of Spherical Drops containing Soluble Surfactant @AdityaSKhair1 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/543311



237v: Effective Viscosity of a Dilute Emulsion of Spherical Drops containing Soluble Surfactant

ABSTRACT

**Monday, October 29, 2018** **03:30 PM - 05:00 PM**

*Omni William Penn Hotel - Frick*

The effective viscosity of a dilute emulsion of spherical drops containing a soluble surfactant is calculated under a linear creeping flow. It is assumed that convection of surfactant is small relative to diffusion, and thus the Peclet number, *Pe*, is small. We calculate the effective viscosity of the emulsion to $O(Pe\varphi\mu)$, where $\varphi$ is the small volume fraction of the dispersed drops, and $\mu$ is the viscosity of the surfactant-free suspending fluid. This $O(Pe\varphi\mu)$ contribution is a sensitive function of the bulk and interfacial surfactant transport. Specifically, soluble surfactants diffuse from the bulk to the interface, and then adsorb to the interface. The ratio of the time scale for bulk diffusion to the time scale for adsorption to the interface is quantified by a Damkohler number, *Da*. The adsorption of surfactant to the interface may cause a significant decrease in the bulk concentration, which is known as depletion. The impact of depletion is characterized by two parameters: *h*, which is a dimensionless depletion depth; and *k*, which is the ratio of the desorption time scale to the adsorption time scale. We analytically determine how the $O(Pe\varphi\mu)$ contribution to the effective viscosity depends on *h*, *k* and *Da*. Surprisingly, for certain regimes in the *h*-*k*-*Da* parameter space, we predict the effective viscosity of the emulsion to be greater than Einstein's result for the viscosity of a suspension of rigid spheres. Large Marangoni stresses driven by convective transport of soluble surfactant molecules are responsible for this result.

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**Session:** 237 Poster Session: Fluid Mechanics

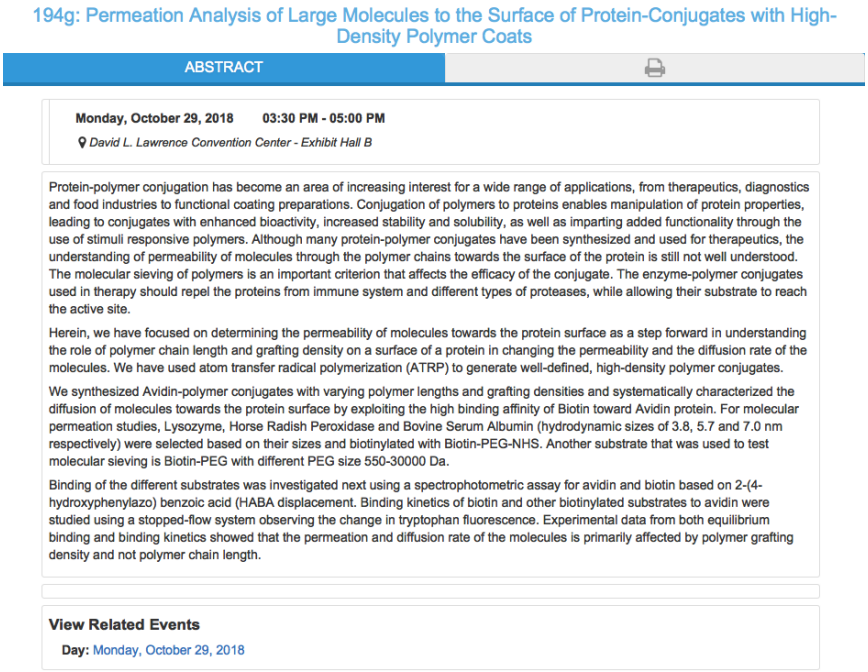
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**Area:** 01J Fluid Mechanics

**Group:** Engineering Sciences and Fundamentals

**Day:** Monday, October 29, 2018

## 2.11 194g: Permeation Analysis of Large Molecules to the Surface of Protein-Conjugates with High-Density Polymer Coats @DrAlanRussell https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/520868



194g: Permeation Analysis of Large Molecules to the Surface of Protein-Conjugates with High-Density Polymer Coats

ABSTRACT

**Monday, October 29, 2018** **03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

Protein-polymer conjugation has become an area of increasing interest for a wide range of applications, from therapeutics, diagnostics and food industries to functional coating preparations. Conjugation of polymers to proteins enables manipulation of protein properties, leading to conjugates with enhanced bioactivity, increased stability and solubility, as well as imparting added functionality through the use of stimuli responsive polymers. Although many protein-polymer conjugates have been synthesized and used for therapeutics, the understanding of permeability of molecules through the polymer chains towards the surface of the protein is still not well understood. The molecular sieving of polymers is an important criterion that affects the efficacy of the conjugate. The enzyme-polymer conjugates used in therapy should repel the proteins from immune system and different types of proteases, while allowing their substrate to reach the active site.

Herein, we have focused on determining the permeability of molecules towards the protein surface as a step forward in understanding the role of polymer chain length and grafting density on a surface of a protein in changing the permeability and the diffusion rate of the molecules. We have used atom transfer radical polymerization (ATRP) to generate well-defined, high-density polymer conjugates.

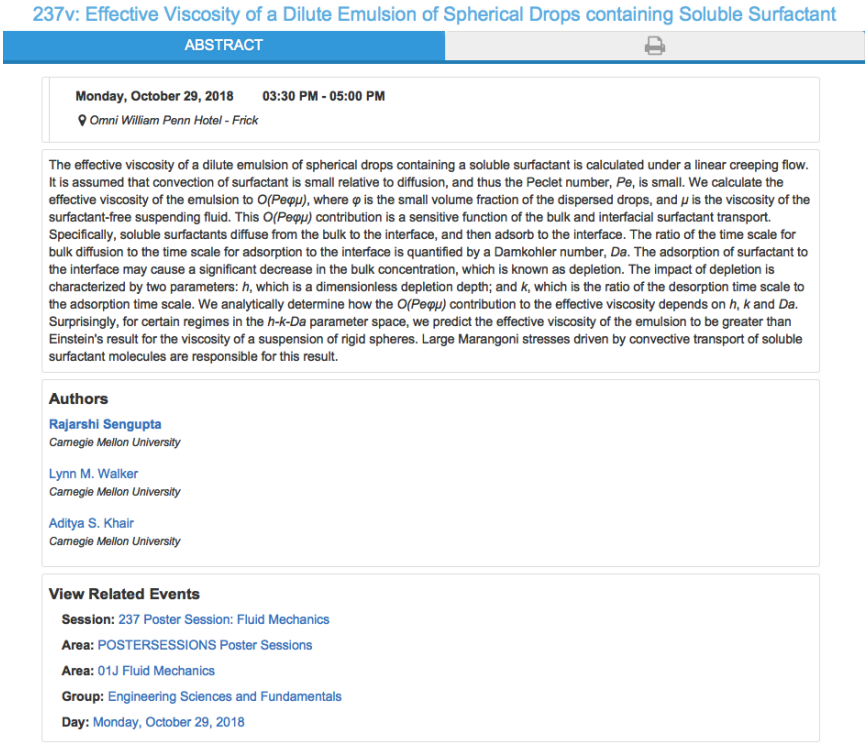
We synthesized Avidin-polymer conjugates with varying polymer lengths and grafting densities and systematically characterized the diffusion of molecules towards the protein surface by exploiting the high binding affinity of Biotin toward Avidin protein. For molecular permeation studies, Lysozyme, Horse Radish Peroxidase and Bovine Serum Albumin (hydrodynamic sizes of 3.8, 5.7 and 7.0 nm respectively) were selected based on their sizes and biotinylated with Biotin-PEG-NHS. Another substrate that was used to test molecular sieving is Biotin-PEG with different PEG size 550-30000 Da.

Binding of the different substrates was investigated next using a spectrophotometric assay for avidin and biotin based on 2-(4-hydroxyphenylazo) benzoic acid (HABA displacement. Binding kinetics of biotin and other biotinylated substrates to avidin were studied using a stopped-flow system observing the change in tryptophan fluorescence. Experimental data from both equilibrium binding and binding kinetics showed that the permeation and diffusion rate of the molecules is primarily affected by polymer grafting density and not polymer chain length.

**View Related Events**

**Day:** Monday, October 29, 2018

## 2.12 237v: Effective Viscosity of a Dilute Emulsion of Spherical Drops containing Soluble Surfactant https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/543311 @AdityaSKhair1 @LwalkerCMU



237v: Effective Viscosity of a Dilute Emulsion of Spherical Drops containing Soluble Surfactant

ABSTRACT

**Monday, October 29, 2018** **03:30 PM - 05:00 PM**

*Omni William Penn Hotel - Frick*

The effective viscosity of a dilute emulsion of spherical drops containing a soluble surfactant is calculated under a linear creeping flow. It is assumed that convection of surfactant is small relative to diffusion, and thus the Peclet number, *Pe*, is small. We calculate the effective viscosity of the emulsion to $O(Pe\varphi\mu)$, where $\varphi$ is the small volume fraction of the dispersed drops, and $\mu$ is the viscosity of the surfactant-free suspending fluid. This $O(Pe\varphi\mu)$ contribution is a sensitive function of the bulk and interfacial surfactant transport. Specifically, soluble surfactants diffuse from the bulk to the interface, and then adsorb to the interface. The ratio of the time scale for bulk diffusion to the time scale for adsorption to the interface is quantified by a Damkohler number, *Da*. The adsorption of surfactant to the interface may cause a significant decrease in the bulk concentration, which is known as depletion. The impact of depletion is characterized by two parameters: *h*, which is a dimensionless depletion depth; and *k*, which is the ratio of the desorption time scale to the adsorption time scale. We analytically determine how the $O(Pe\varphi\mu)$ contribution to the effective viscosity depends on *h*, *k* and *Da*. Surprisingly, for certain regimes in the *h*-*k*-*Da* parameter space, we predict the effective viscosity of the emulsion to be greater than Einstein's result for the viscosity of a suspension of rigid spheres. Large Marangoni stresses driven by convective transport of soluble surfactant molecules are responsible for this result.

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**Session:** 237 Poster Session: Fluid Mechanics

**Area:** POSTERSESSIONS Poster Sessions

**Area:** 01J Fluid Mechanics

**Group:** Engineering Sciences and Fundamentals

**Day:** Monday, October 29, 2018

## 2.13 190o: In Situ Growth of Acetylcholinesterase-Oxime Polymer Conjugate Scavengers of Organophosphate Nerve Agent Toxicity @DrAlanRussell @MatyPolymerLab https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/534015

190o: In Situ Growth of Acetylcholinesterase-Oxime Polymer Conjugate Scavengers of Organophosphate Nerve Agent Toxicity

ABSTRACT

**Monday, October 29, 2018 03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

**In Situ Growth of Acetylcholinesterase-Oxime Polymer Conjugate Scavengers of Organophosphate Nerve Agent Toxicity**

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**Abstract**

Acetylcholinesterase (AChE) is a serine hydrolase which rapidly catalyzes hydrolysis of the neurotransmitter, acetylcholine (ACh, Figure 1A). In keeping with its activity, AChE serves to terminate synaptic transmission in neuromuscular junctions and maintain a tolerable level of diffusing substrate, ACh. Organo-phosphorus (OP) compounds, nerve agents, inhibit AChE causing ACh accumulation in cholinergic synapses. The toxic effects of accumulated ACh range from excessive salivation through muscular paralysis to respiratory failure. Nerve agent toxicity can be treated by intravenous injection of the OP scavenger, butyrylcholinesterase (BChE).[1] Strong nucleophiles, such as oximes, are used for reactivation of OP-AChE conjugates by displacing the OP moiety from the enzyme's active site (Figure 1B) and deactivating circulating Ops. Quaternary pyridinium oximes, for example, are used for nucleophilic reactivation of AChE that is covalently inhibited by OPs.[2] Treatment by pyridinium oximes, however, requires rapid administration after exposure to OPs and is limited by a short half-life in the blood. Attachment of synthetic polymers to form protein-polymer conjugates combines properties derived from both biologic and synthetic materials. Such conjugates have become central components of the biologic drug and food industries. Atom transfer radical polymerization (ATRP), that was developed in the early 1990s, has proven to be a powerful technique with control of polymer chain length and molecular weight, producing a narrow size distribution and a large library of available architectures.[3] We have previously generated a series of protein-polymer conjugates using ATRP to increase protein pH and temperature stability, enable dissolution of protein in organic solvent and enhance trans-epithelial protein transport.[4-6] We will describe the design, synthesis and initial functional characterization of AChE-polymer conjugates, with oxime pendant on the polymer chain, to study self-reactivation following inhibition.

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**Session:** 190 Poster Session: Engineering Fundamentals in Life Science

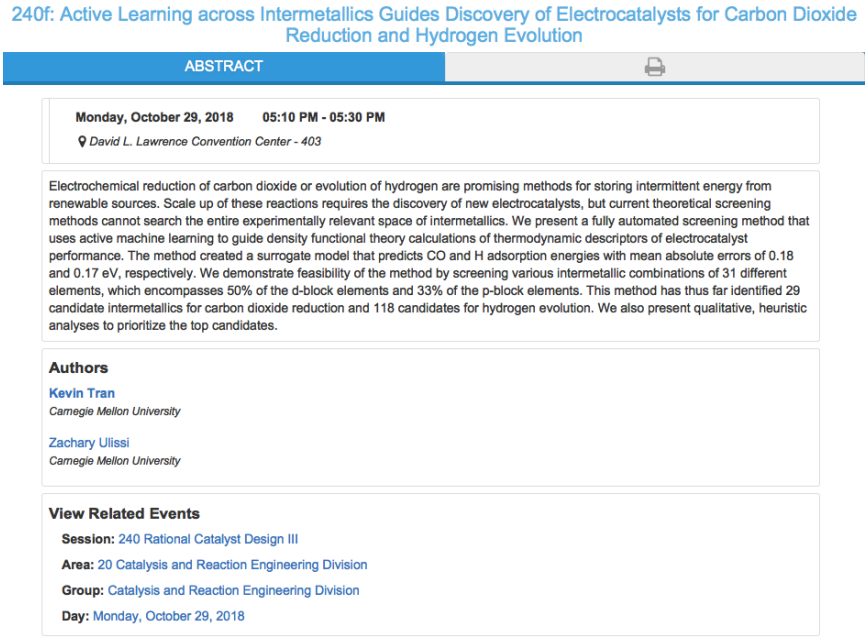
**Area:** POSTERSESSIONS Poster Sessions

**Area:** 15D Engineering Fundamentals in Life Science

**Group:** Food, Pharmaceutical & Bioengineering Division

**Day:** Monday, October 29, 2018

## 2.14 240f Active Learning across Intermetallics Guides Discovery of Electrocatalysts for Carbon Dioxide Reduction and Hydrogen Evolution https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/523436 @UlissiGroup @aiche_cre



240f: Active Learning across Intermetallics Guides Discovery of Electrocatalysts for Carbon Dioxide Reduction and Hydrogen Evolution

ABSTRACT

**Monday, October 29, 2018** **05:10 PM - 05:30 PM**

*David L. Lawrence Convention Center - 403*

Electrochemical reduction of carbon dioxide or evolution of hydrogen are promising methods for storing intermittent energy from renewable sources. Scale up of these reactions requires the discovery of new electrocatalysts, but current theoretical screening methods cannot search the entire experimentally relevant space of intermetallics. We present a fully automated screening method that uses active machine learning to guide density functional theory calculations of thermodynamic descriptors of electrocatalyst performance. The method created a surrogate model that predicts CO and H adsorption energies with mean absolute errors of 0.18 and 0.17 eV, respectively. We demonstrate feasibility of the method by screening various intermetallic combinations of 31 different elements, which encompasses 50% of the d-block elements and 33% of the p-block elements. This method has thus far identified 29 candidate intermetallics for carbon dioxide reduction and 118 candidates for hydrogen evolution. We also present qualitative, heuristic analyses to prioritize the top candidates.

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**View Related Events**

**Session:** 240 Rational Catalyst Design III

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Monday, October 29, 2018

## 2.15 240g Design of Optimal Metallic Surface Reconstructions for Heterogeneous Catalysis https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/534923 @UlissiGroup @aiche_cre

240g: Design of Optimal Metallic Surface Reconstructions for Heterogeneous Catalysis

ABSTRACT

**Monday, October 29, 2018 05:30 PM - 05:50 PM**

*David L. Lawrence Convention Center - 403*

Transition metal nanoparticles have been used to enhance the reactivity of catalysts by introducing significantly more surface area per amount of precious metal. Furthermore, some reactions show additional increase in reactivity on a per area basis due to the relative abundance of under-coordinated surface sties on nanoclusters [1]. Recently, synthesis methods such as electrochemical deposition, galvanic displacement, and the formation of near surface alloys have enabled the formation of nanostructured surfaces with pits, grooves, and multimetallic sites designed to introduce unique controls over the reactivity of surface sites [2,3]. The formation of these structures results in a combinatorial number of surface designs to be proposed, and leads to the challenging task of identifying the most reactive surface to target for catalyst design. To support this effort, we propose the use of mathematical optimization models to help guide this materials design process [4].

As a design space, we consider the choices of whether to place an atom or not in a periodic tile representing several layers around an infinite catalyst surface. We can generically model the characteristics of reaction sites with suitable descriptors, such as the coordination or generalized coordination number, and indicate if a site constitutes an ideal location for a desired chemical reaction [5,6,7]. Symmetry-breaking constraints are introduced to eliminate the significant number of equivalent representations of a design and are necessary to solve the model for reasonable tile sizes. The resulting model can be solved to identify the pattern of modifications to the base crystallographic plane that results in a maximal packing of ideal reaction sites per unit area.

While the developed framework can be applied with some basic approximations of stability, it is desirable to develop additional simplified structure function models to predict if designs are likely to be stable under reactive conditions. In order to determine which nanostructured designs are stable, we first enumerated a set of over one hundred Pt surfaces with modifications from the face centered cubic (FCC) {111} plane. The surface energy of these reconstructions and several reference Pt crystal configurations was calculated via density functional theory (DFT) using the RPBE exchange correlation functional and a linear extrapolation method [8].

This analysis yielded a number of reconstructions whose excess from the FCC {111} surface energy was low enough to make them plausible under normal conditions. In order to incorporate this information into our mathematical optimization model, we regressed a simplified structure-function relationship between the coordination of surface atoms and the surface energy evaluated by DFT. This allows us to predict the surface energy of nanostructured surfaces to within a small error by simply counting up the number of sites at a particular coordination. Furthermore, the contribution of different coordination numbers to surface energy was found to be roughly linear, suggesting that a simple bond counting model was sufficient to capture the trend in the data.

Given such a bond-counting model, we were able to explicitly incorporate constraints in our optimization model that required the predicted surface energy of designs to fall below a threshold of surface energy. We show that the choice of threshold can be chosen to correspond to the surface energy of the least stable simple crystal plane that is exhibited in experimental conditions. Alternatively, the optimization model can be iteratively solved while the threshold is scanned through a range of values to generate a Pareto-optimal frontier representing the tradeoff of stability against reactivity of nanostructured surfaces. The resulting surfaces were further evaluated via DFT to more accurately determine their actual stability and reactivity. The outcome of this framework is a systematic way to identify patterns of nanostructure that can be of interest for further study.

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**Session:** 240 Rational Catalyst Design III

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Monday, October 29, 2018

# 3 #AICHeAnnual Here are a few talks from @CMU_Cheme on Tuesday @ChEnected :tweet:thread

## 3.1 253a: Enhancing Relaxations for Nonconvex Mixed-Integer Quadratically-Constrained Quadratic Programs https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/532505 @AIChECAST10

253a: Enhancing Relaxations for Nonconvex Mixed-Integer Quadratically-Constrained Quadratic Programs

ABSTRACT

**Tuesday, October 30, 2018 08:00 AM - 08:19 AM**

*David L. Lawrence Convention Center - 409*

Enhancing Relaxations for Nonconvex Mixed-Integer Quadratically Constrained Quadratic Programs

Mixed-integer quadratically-constrained quadratic programs (MIQCQPs), and related classes of problems such as quadratically-constrained quadratic programs (QCQPs) and quadratic programs (QPs) arise in a wide variety of engineering and scientific applications including pooling and blending in process networks [12], facility location [23], production planning [8], quadratic assignment [9] and portfolio optimization [15]. Due to their practical importance, these classes of problems have been extensively studied in the mathematical programming literature. They are typically solved using spatial branch-and-bound algorithms, whose efficiency primarily depends on the quality of the relaxations constructed to bound the problem.

Commonly used relaxations for these type of problems can be classified into four groups. The first group involves relaxations obtained via standard factorable programming techniques whereby each quadratic function is decomposed iteratively by introducing intermediate variables and constraints for each nonconvex term, until each intermediate expression can be outer-approximated by its convex and/or concave envelopes [10, 20, 22]. While these techniques are relatively simple to implement, they typically result in large relaxation gaps. The second group is given by semi-definite programming (SDP) relaxations. This approach has received significant attention in recent years [1, 5, 6, 21]. While SDP relaxations often provide strong bounds, they are computationally expensive to solve, which limits their use within general purpose branch-and-bound global solvers. The third group relies on the reformulation-linearization techniques (RLT) [17–19]. Though these techniques typically yield tight bounds, the resulting relaxations grow quickly in size, and thus can become expensive to solve. The fourth group consists of factorable relaxations augmented by adding various classes of valid inequalities such as SDP-based cuts [7, 16], RTL-based cuts [24, 25], facets of the envelopes of edge-concave and multilinear subexpressions [2, 3, 11, 13], and mixed-integer cuts [4]. These augmented relaxations provide stronger bounds than standard factorable relaxations while remaining computationally inexpensive to solve. Due to its practicality, this class of relaxations is employed in most general-purpose global optimization packages.

In this work, we show how certain facets of the Boolean Quadric Polytope [14] can be utilized in order to enhance factorable relaxations of MIQCQPs constructed by global optimization solvers. To illustrate the benefits of this approach, we propose a computational implementation which integrates these facets as cutting planes into the branch-and-reduce global solver BARON. Our implementation consists of several algorithmic developments including tools for the identification of substructures in a graph representation of the problem, as well as efficient cut generation and separation algorithms which are embedded at every node of the branch-and-bound tree. We demonstrate the impact of the proposed implementation by conducting an extensive computational study on a large collection of problems selected from publicly available test sets.

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## 3.2 300b: Integration of Crude-Oil Scheduling and Refinery Planning By Lagrangean Decomposition Approach https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/527210 @bernalde

300b: Integration of Crude-Oil Scheduling and Refinery Planning By Lagrangean Decomposition Approach

ABSTRACT

**Tuesday, October 30, 2018 08:00 AM - 08:25 AM**

*David L. Lawrence Convention Center - 323*

**Integration of crude-oil scheduling and refinery planning by Lagrangean decomposition approach**

Haokun Yang, David E. Bernal, Ignacio E. Grossmann

Chemical Engineering Department, Carnegie Mellon University, Pittsburgh PA.

Problems of crude-oil scheduling and refinery planning are usually considered hard to solve to optimality at an enterprise level. The former deals with the crude oil inventory considering several types of crude oil which are delivered by different vessels and used as feedstock for the Crude Distillation Unit (CDU). The latter deals with the oil production starting at the CDU until final products involving refining and blending operations. Mixed-integer Nonlinear Programming (MINLP) models have been developed with both discrete-time and continuous-time formulation for the scheduling problem and solved with linearization based algorithms, e.g. [1] and [2]. On the other hand, the refinery planning often involves the pooling problem and usually consists of optimizing feedstocks, unit settings, as well as final product blending and shipping. The CDU modeling has used traditionally used approaches like fixed yield or swing cuts Linear Programming (LP) models. Recently, Alattas et al. [3] developed a nonlinear model of the CDU based on fractionation indices and integrated it in an MINLP model involving production planning, also considering the changeover of crudes [4]. Since both problems aim at an economic objective and are linked by the feedstock of the CDU, it becomes interesting to solve an integrated problem which may result in a better solution than the one obtained by solving each one separately. The challenge of this approach is the time scale difference between the two problems which may result in a large-scale MINLP model.

Lagrangean decomposition is a solution method which decomposes an optimization problem with complicating constraints into more easily solvable subproblems which solved separately and iteratively converge to the solution of the original problem by dualizing those complicating constraints. This method has shown a significant advantage in solving scheduling and planning integrated problems [5]. Mouret et al. [6] integrated a crude oil scheduling model with a continuous-time formulation and a simplified refinery planning model using an Artificial Neural Network (ANN) for the CDU modeling [7]. In Mouret's work, the advantages of considering an integrated approach of the crude oil scheduling and refinery planning problem were showed by achieving a time efficient framework for these problems using Lagranean decomposition.

The integrated refinery problem formulation and solution method using a Lagrangean decomposition algorithm are described in this work. We have studied the effect of having a continuous-time and a discrete-time formulation for the crude oil scheduling model and used several modeling approaches for the CDU. Two different kinds of comparisons have been made using the results of this work. Our model, which includes both the scheduling and planning simultaneously using Lagrangean decomposition with linking constraints for the CDU, is compared against a non-integrated approach to the crude oil scheduling and followed by the refinery planning. The results show an improvement of the economic objective function with respect to the sequential approach. Furthermore, we present a comparison of the solution time against solving the monolithically integrated model using state-of-the-art MINLP solvers, showing the advantage of using a Lagrangian Decomposition algorithm for this problem.

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**Area:** 16 Fuels and Petrochemicals Division

**Group:** Fuels and Petrochemicals Division

**Day:** Tuesday, October 30, 2018

## 3.3 285a: Synergistic Impact of Polymer/Surfactant Complexation on the Colloidal Depletion Force https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/526072

285a: Synergistic Impact of Polymer/Surfactant Complexation on the Colloidal Depletion Force

ABSTRACT

**Tuesday, October 30, 2018 08:00 AM - 08:16 AM**

*Omni William Penn Hotel - Conference Center A*


Colloidal suspensions prepared in solutions of surfactants or macromolecules ("depletants") may experience attractive depletion forces. Depletants could also be nanoparticles co-dispersed in the colloidal suspension. Depletion forces can be prominent in multicomponent formulated suspension products, such as paints or pharmaceutical suspensions, and they are frequently exploited in wastewater treatment processes to de-stabilize suspended colloidal solids. The depletion force arises due to exclusion of depletants from the gap between two interacting particles upon sufficiently close approach. A higher osmotic pressure is created in the bulk than in the gap, which leads to a net attractive force between the interacting particles. Osmotic pressure is a colligative property. In the case of charged depletants (polyelectrolytes) it depends not only on the depletant concentration but also on its charge and the concentration of background electrolytes. The depletion attraction is in fact the primary minimum in an oscillatory structural force. At higher depletant concentrations, or in the presence of more highly charged depletants, ordering and structuring transitions of depletants in the gap produces forces that oscillate between attraction and repulsion as a function of gap thickness. The advent of sensitive force measurement instruments such as the surface forces apparatus, total internal reflection microscopy and colloidal probe atomic force microscopy has enabled direct measurement of depletion and oscillatory structural forces. Most prior studies of depletion forces considered solutions of only one type of depletant. Commercially important colloidal systems are mostly multi-component. Therefore, it is necessary to understand the combined effect of multiple types of depletants on the depletion force. When ionic surfactants are present in nonionic polymer solutions, they can bind to the polymers and form a charged complex, essentially endowing a nonionic polymer with polyelectrolyte character. This charged complex formation has a profound impact on the depletion force experienced by the suspended particles in a mixed system as compared to that in surfactant-only or polymer-only solutions. We observe that anionic surfactant complexation with nonionic polymers produces a synergistic effect on depletion forces, whereby significant forces are measured at concentrations where neither single-component system would have yielded detectable depletion forces. This is primarily due to the enhanced osmotic pressure of polyelectrolytes relative to nonionic polymers.

In this project we focused on understanding the combined impact of a commercially significant nonionic triblock copolymer, poly (ethylene oxide-b-propylene oxide-b-ethylene oxide) (Pluronic F108) and a sodium dodecyl sulfate (SDS) anionic surfactants on the depletion force between a silica sphere and silica flat disc. Colloid probe atomic force microscopy was used to obtain force vs. distance profiles in the presence of varying concentrations of NaCl as background electrolyte. Sodium ion selective electrode and dynamic light scattering (DLS) experiments were used to monitor the concentration-dependent complexation of SDS with F108. Modeling of the ion-selective electrode analyses provided a measure of the charge of the polymer/surfactant complexes for varying surfactant and polymer concentrations. The combination of ion-selective electrode and DLS meaurements indicated complexation commences above a threshold surfactant concentration, as has been well demonstrated in past complexation studies, and that not all the SDS added beyond that point binds to the polymer. Free SDS present in the solution may form micelles before saturation of F108 complexation, especially at high F108 concentration. These free SDS micelles also contribute to the depletion interaction. The magnitude of depletion (and oscillatory) forces in solutions containing F108/SDS complexes was compared to that in SDS-only and F108-only solutions for a wide range of SDS concentrations and for two F108 concentrations spanning one decade in concentration.

The magnitude of depletion force was observed to increase with increasing SDS concentration with and without F108. Depletion force was not detected when the polymer solutions contained no SDS. The presence of a depletion force was detected at a much lower SDS concentration with polymer as compared to SDS-only solutions. Similar findings were obtained with respect to the first repulsive maximum in the oscillatory structural force. The synergistic enhancement of depletion forces was observed only in a finite range of SDS concentrations, corresponding to approximately 0.5 to 2.5 times the critical micelle concentration of SDS for solutions containing 10000 ppm F108. It was observed in the finite range of 1.5 to 2 times the CMC of SDS for solutions containing 1000 ppm F108. At higher SDS concentrations, forces measured in the mixtures resembled those measured in the SDS-only solutions. This was because at low SDS concentrations the F108/SDS complexes were the dominant depletants. At higher SDS concentrations, the concentration of free unbound SDS micelles increased to the point where they were the dominant depletant. Synergism was progressively diminished with increasing NaCl concentration, as expected based on its suppression of the polyelectrolyte osmotic pressure effect.



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**Session:** 285 Fundamentals of Interfacial Phenomena I

**Area:** 01C Interfacial Phenomena

**Group:** Engineering Sciences and Fundamentals

**Day:** Tuesday, October 30, 2018

## 3.4 300b: Integration of Crude-Oil Scheduling and Refinery Planning By Lagrangean Decomposition Approach https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/527210 @bernalde

300b: Integration of Crude-Oil Scheduling and Refinery Planning By Lagrangean Decomposition Approach

ABSTRACT

**Tuesday, October 30, 2018 08:00 AM - 08:25 AM**

*David L. Lawrence Convention Center - 323*

**Integration of crude-oil scheduling and refinery planning by Lagrangean decomposition approach**

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Problems of crude-oil scheduling and refinery planning are usually considered hard to solve to optimality at an enterprise level. The former deals with the crude oil inventory considering several types of crude oil which are delivered by different vessels and used as feedstock for the Crude Distillation Unit (CDU). The latter deals with the oil production starting at the CDU until final products involving refining and blending operations. Mixed-integer Nonlinear Programming (MINLP) models have been developed with both discrete-time and continuous-time formulation for the scheduling problem and solved with linearization based algorithms, e.g. [1] and [2]. On the other hand, the refinery planning often involves the pooling problem and usually consists of optimizing feedstocks, unit settings, as well as final product blending and shipping. The CDU modeling has used traditionally used approaches like fixed yield or swing cuts Linear Programming (LP) models. Recently, Alattas et al. [3] developed a nonlinear model of the CDU based on fractionation indices and integrated it in an MINLP model involving production planning, also considering the changeover of crudes [4]. Since both problems aim at an economic objective and are linked by the feedstock of the CDU, it becomes interesting to solve an integrated problem which may result in a better solution than the one obtained by solving each one separately. The challenge of this approach is the time scale difference between the two problems which may result in a large-scale MINLP model.

Lagrangean decomposition is a solution method which decomposes an optimization problem with complicating constraints into more easily solvable subproblems which solved separately and iteratively converge to the solution of the original problem by dualizing those complicating constraints. This method has shown a significant advantage in solving scheduling and planning integrated problems [5]. Mouret et al. [6] integrated a crude oil scheduling model with a continuous-time formulation and a simplified refinery planning model using an Artificial Neural Network (ANN) for the CDU modeling [7]. In Mouret's work, the advantages of considering an integrated approach of the crude oil scheduling and refinery planning problem were showed by achieving a time efficient framework for these problems using Lagranean decomposition.

The integrated refinery problem formulation and solution method using a Lagrangean decomposition algorithm are described in this work. We have studied the effect of having a continuous-time and a discrete-time formulation for the crude oil scheduling model and used several modeling approaches for the CDU. Two different kinds of comparisons have been made using the results of this work. Our model, which includes both the scheduling and planning simultaneously using Lagrangean decomposition with linking constraints for the CDU, is compared against a non-integrated approach to the crude oil scheduling and followed by the refinery planning. The results show an improvement of the economic objective function with respect to the sequential approach. Furthermore, we present a comparison of the solution time against solving the monolithically integrated model using state-of-the-art MINLP solvers, showing the advantage of using a Lagrangian Decomposition algorithm for this problem.

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[5] B. Brunaud and I. E. Grossmann, "Perspectives in Multilevel Decision-making in the Process Industry," pp. 1–34, 2017.

[6] S. Mouret, I. E. Grossmann, and P. Pestiaux, "A new Lagrangian decomposition approach applied to the integration of refinery planning and crude-oil scheduling," *Comput. Chem. Eng.*, vol. 35, no. 12, pp. 2750–2766, 2011.

[7] T. Gueddar and V. Dua, "Novel model reduction techniques for refinery-wide energy optimisation," *Appl. Energy*, vol. 89, no. 1, pp. 117–126, 2012.

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**Session:** 300 Refining and Petrochemical Plant Modelling and Operations Improvements I

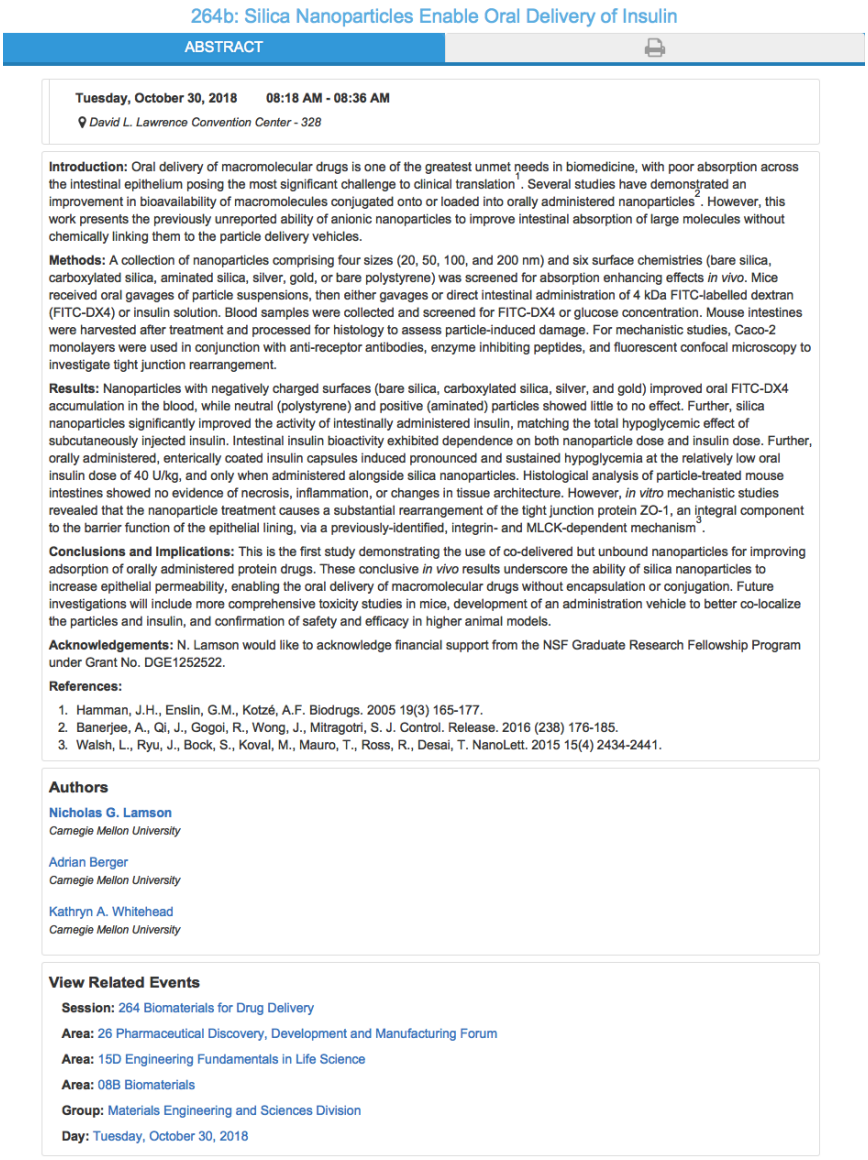
**Area:** 02A Distillation and Absorption

**Area:** 16 Fuels and Petrochemicals Division

**Group:** Fuels and Petrochemicals Division

**Day:** Tuesday, October 30, 2018

## 3.5 264b: Silica Nanoparticles Enable Oral Delivery of Insulin https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/522263 @Nick_Lamson @KWhiteheadLab



264b: Silica Nanoparticles Enable Oral Delivery of Insulin

ABSTRACT

**Tuesday, October 30, 2018 08:18 AM - 08:36 AM**

*David L. Lawrence Convention Center - 328*

**Introduction:** Oral delivery of macromolecular drugs is one of the greatest unmet needs in biomedicine, with poor absorption across the intestinal epithelium posing the most significant challenge to clinical translation[1]. Several studies have demonstrated an improvement in bioavailability of macromolecules conjugated onto or loaded into orally administered nanoparticles[2]. However, this work presents the previously unreported ability of anionic nanoparticles to improve intestinal absorption of large molecules without chemically linking them to the particle delivery vehicles.

**Methods:** A collection of nanoparticles comprising four sizes (20, 50, 100, and 200 nm) and six surface chemistries (bare silica, carboxylated silica, aminated silica, silver, gold, or bare polystyrene) was screened for absorption enhancing effects *in vivo*. Mice received oral gavages of particle suspensions, then either gavages or direct intestinal administration of 4 kDa FITC-labelled dextran (FITC-DX4) or insulin solution. Blood samples were collected and screened for FITC-DX4 or glucose concentration. Mouse intestines were harvested after treatment and processed for histology to assess particle-induced damage. For mechanistic studies, Caco-2 monolayers were used in conjunction with anti-receptor antibodies, enzyme inhibiting peptides, and fluorescent confocal microscopy to investigate tight junction rearrangement.

**Results:** Nanoparticles with negatively charged surfaces (bare silica, carboxylated silica, silver, and gold) improved oral FITC-DX4 accumulation in the blood, while neutral (polystyrene) and positive (aminated) particles showed little to no effect. Further, silica nanoparticles significantly improved the activity of intestinally administered insulin, matching the total hypoglycemic effect of subcutaneously injected insulin. Intestinal insulin bioactivity exhibited dependence on both nanoparticle dose and insulin dose. Further, orally administered, enterically coated insulin capsules induced pronounced and sustained hypoglycemia at the relatively low oral insulin dose of 40 U/kg, and only when administered alongside silica nanoparticles. Histological analysis of particle-treated mouse intestines showed no evidence of necrosis, inflammation, or changes in tissue architecture. However, *in vitro* mechanistic studies revealed that the nanoparticle treatment causes a substantial rearrangement of the tight junction protein ZO-1, an integral component to the barrier function of the epithelial lining, via a previously-identified, integrin- and MLCK-dependent mechanism[3].

**Conclusions and Implications:** This is the first study demonstrating the use of co-delivered but unbound nanoparticles for improving adsorption of orally administered protein drugs. These conclusive *in vivo* results underscore the ability of silica nanoparticles to increase epithelial permeability, enabling the oral delivery of macromolecular drugs without encapsulation or conjugation. Future investigations will include more comprehensive toxicity studies in mice, development of an administration vehicle to better co-localize the particles and insulin, and confirmation of safety and efficacy in higher animal models.

**Acknowledgements:** N. Lamson would like to acknowledge financial support from the NSF Graduate Research Fellowship Program under Grant No. DGE1252522.

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**Session:** 264 Biomaterials for Drug Delivery

**Area:** 26 Pharmaceutical Discovery, Development and Manufacturing Forum

**Area:** 15D Engineering Fundamentals in Life Science

**Area:** 08B Biomaterials

**Group:** Materials Engineering and Sciences Division

**Day:** Tuesday, October 30, 2018

## 3.6 304b: Sustainable Optimal Strategic Planning for Shale Water Management https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/523808 @UA_Universidad

304b: Sustainable Optimal Strategic Planning for Shale Water Management

ABSTRACT

**Tuesday, October 30, 2018 08:20 AM - 08:40 AM**

*David L. Lawrence Convention Center - 317*

The development of new additives in the shale gas industry, which tolerate the use of high Total Dissolved Solids (TDS) base fluid, has allowed reusing the wastewater in the drilling of subsequent wells (U.S. Environmental Protection Agency, 2016). This practice is currently the most popular and cost-effective option for shale gas water management. Although it minimizes freshwater consumption, producers should take into consideration possible long-term issues and challenges. For instance, the TDS concentration will increase significantly, around 2-3 times, which can represent a cost barrier to reuse the water for fracturing operations. Besides, as the number of drilled wells decrease this practice becomes less attractive. Specifically, the volume of fracturing fluid required to fracture new wells may be less than the volume of water generated by producing wells in the area.

Currently, several works have been reported on the optimization of shale gas water management (Gao and You, 2015; Guerra et al., 2016; Yang et al., 2015). However, most of the works published in the literature consider that the water blending ratio is restricted or the return to pad operations are not allowed. Drouven and Grossmann (2017) assume that the water-blending ratio is unrestricted, they over-estimate the friction reducers expenses. The MILP model that they proposed does not account the salt concentration of impaired water. Moreover, they only distinguish between two types of water: impaired water and freshwater. Hence, the model cannot handle any water management option when drilling operation decrease.

In this work, we propose a mixed-integer non-linear programming (MINLP) model considering the TDS concentration of flowback and impaired water as a function of time and for different water treatment solutions. We estimate the friction reducers expenses as a function of TDS concentration to determine if the level of TDS in impaired water is an impediment to reusing it in fracturing operations. Moreover, the model distinguishes between four types of water: impaired water, flowback water, desalinated water and freshwater. The objective is to maximize the "sustainability profit" in order to obtain a compromise solution between economic, environmental and social aspects. Only a single objective function is necessary since all the indicators are expressed in monetary terms (Zore et al., 2017).

The water management system comprises wellpads, shale gas wells in each wellpad, centralized water treatment technologies (CWT), freshwater sources, fracturing crews, and disposals wells.

After hydraulic fracturing, a portion of water called flowback water returns to the wellhead. The flowback water is stored in fracturing tanks (FT) onsite before basic treatment (pre-treatment) in mobile units or transport to CWT facility or Class II disposal. Pre-treatment includes technologies to remove suspended solids, oil and grease, and bacteria, certain ions that can cause the scale to form on equipment and interfere with fracturing chemical additives. After pre-treatment, the water can be used as a fracturing fluid in the same wellpad or in the neighboring wellpad, or can be desalinated in the onsite TDS removal technologies. The flowback water reused for fracturing operations is called impaired water and is stored in impaired water tanks (IWT). Several desalination technologies can be selected such as multi-stage membrane distillation (MMD), multi-effect evaporation with mechanical vapor recompression (MEE-MVR) or forward-reverse osmosis (FO-RO) hybrid. We consider that the outflow brine salinity in the onsite treatment is close to salt saturation conditions to achieve ZLD operation. Cost and salinity levels restrict the type of desalination unit that can be used for TDS removal. The onsite desalinated water can also be used as a fracturing fluid in the same wellpad, transported to the next wellpad or discharge for other usages. The flowback water can also be transported and treated in CWT. Desalinated water from CWT can select the same routes as the desalinated water in onsite technologies. Freshwater is obtained from an uninterruptible fresh water source. Desalinated water and freshwater are stored in freshwater tanks (FWT) and/or water impoundment. Transportation of freshwater, impaired water, flowback water and desalinated water can only be through trucks. Storage tanks and mobile treatment are assumed to be leased.

The problem is to determine: (1) wellpad fracturing start date (fracturing schedule), (2) number of tanks leased at each time period, (3) number of trucks needed in each time period, (4) flowback destination, reuse (impaired water) , treatment (onsite or offsite) or disposal, and (5) quality of water used to fracture each well.

The optimization problem is formulated as an MINLP model that includes: assignment constraints, material balance in storage tanks, mixer and splitters, logic constraints and an objective function. Salt material balances are modeled using total flows and salt composition as variables (bilinear terms). An advantage of using this representation is that the variables involved in the bilinear terms are well bounded, allowing us to define tight under and over estimators.

Them objective function includes the economic profit, the eco-profit and the social-profit.

- *Economic profit* includes revenues from natural gas minus the sum of the following expenses: wastewater disposal cost, storage tank cost, freshwater cost, friction reducer cost, wastewater and freshwater transport cost and onsite and offsite treatment cost.
- *Eco-profit* distinguishes between eco-benefit (raw material and products that unburden the environment) and eco-cost (raw material al products that burden the environment). Both terms are calculated by using eco-cost coefficients (Delft University of Technology, 2017). In our problem, impaired water and desalinated water used to fracture a neighboring well exhibit unburdening effect on the environment. However, natural gas, freshwater withdrawal, disposal and transportation burden the environment.
- *Social profit* includes social security contributions paid for the employed people to fracture a well, plus the social transfer by hiring people, minus social cost. We only take into account the numbers of jobs on a fracturing crew and the time that they are working to fracture a specific well. Once the well is completed, the number of jobs generated by truck drivers or maintenance team are not considered.

The proposed model is applied to a case study in Marcellus Play with 3 wellpads and 20 wells, three years discretized at one week per time period, four interruptible sources of fresh water, three class II disposal wells, two CWT plants and one fracturing crew. The MINLP problem consists of 2808 binary variables, 14977 continues variables and 12010 constraints.

The solution of the model shows that the capabilities to reuse impaired water to fracture other wells is the best economic and environmental practice for shale gas water management. The model, which can be effectively solved with the proposed decomposition technique, reveals that the level of TDS in impaired water is not an obstacle to reusing it for fracturing purposes. Also, it has been shown that onsite desalination treatment can be cost-effective for operators once no more wells to fracture are available.

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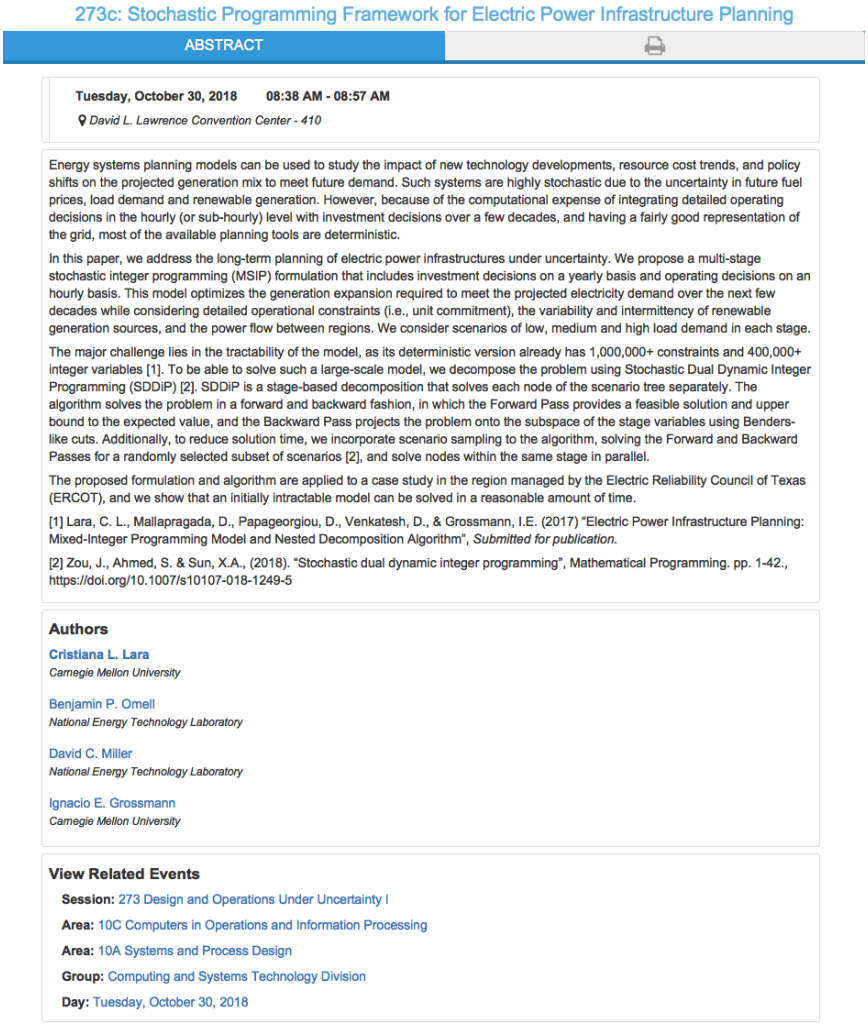
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### 3.7 273c: Stochastic Programming Framework for Electric Power Infrastructure Planning https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/536208 @crislopeslara @NETL_DOE @AIChECAST10



273c: Stochastic Programming Framework for Electric Power Infrastructure Planning

ABSTRACT

**Tuesday, October 30, 2018 08:38 AM - 08:57 AM**

*David L. Lawrence Convention Center - 410*

Energy systems planning models can be used to study the impact of new technology developments, resource cost trends, and policy shifts on the projected generation mix to meet future demand. Such systems are highly stochastic due to the uncertainty in future fuel prices, load demand and renewable generation. However, because of the computational expense of integrating detailed operating decisions in the hourly (or sub-hourly) level with investment decisions over a few decades, and having a fairly good representation of the grid, most of the available planning tools are deterministic.

In this paper, we address the long-term planning of electric power infrastructures under uncertainty. We propose a multi-stage stochastic integer programming (MSIP) formulation that includes investment decisions on a yearly basis and operating decisions on an hourly basis. This model optimizes the generation expansion required to meet the projected electricity demand over the next few decades while considering detailed operational constraints (i.e., unit commitment), the variability and intermittency of renewable generation sources, and the power flow between regions. We consider scenarios of low, medium and high load demand in each stage.

The major challenge lies in the tractability of the model, as its deterministic version already has 1,000,000+ constraints and 400,000+ integer variables [1]. To be able to solve such a large-scale model, we decompose the problem using Stochastic Dual Dynamic Integer Programming (SDDiP) [2]. SDDiP is a stage-based decomposition that solves each node of the scenario tree separately. The algorithm solves the problem in a forward and backward fashion, in which the Forward Pass provides a feasible solution and upper bound to the expected value, and the Backward Pass projects the problem onto the subspace of the stage variables using Benders-like cuts. Additionally, to reduce solution time, we incorporate scenario sampling to the algorithm, solving the Forward and Backward Passes for a randomly selected subset of scenarios [2], and solve nodes within the same stage in parallel.

The proposed formulation and algorithm are applied to a case study in the region managed by the Electric Reliability Council of Texas (ERCOT), and we show that an initially intractable model can be solved in a reasonable amount of time.

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**Session:** 273 Design and Operations Under Uncertainty I

**Area:** 10C Computers in Operations and Information Processing

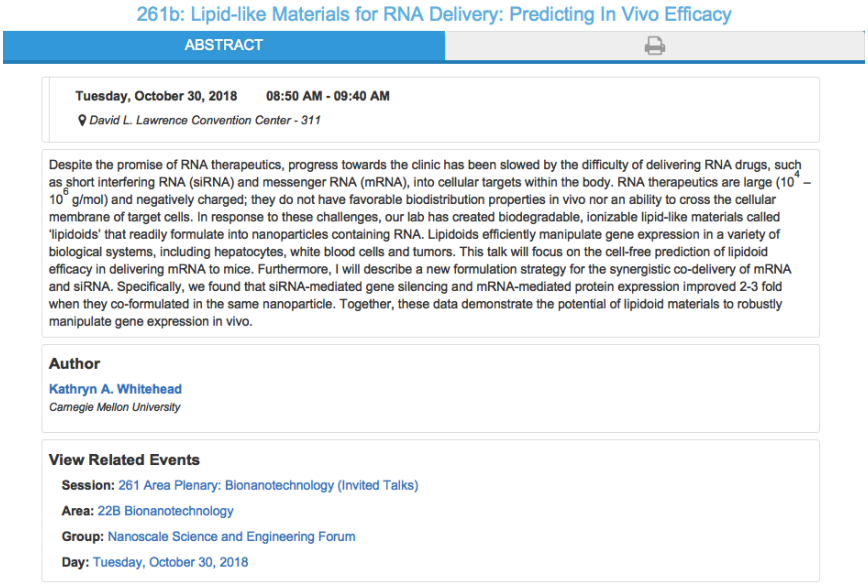
**Area:** 10A Systems and Process Design

**Group:** Computing and Systems Technology Division

**Day:** Tuesday, October 30, 2018

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## 3.8 261b: Lipid-like Materials for RNA Delivery: Predicting In Vivo Efficacy https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/525836 @KWhiteheadLab



261b: Lipid-like Materials for RNA Delivery: Predicting In Vivo Efficacy

ABSTRACT

**Tuesday, October 30, 2018 08:50 AM - 09:40 AM**

*David L. Lawrence Convention Center - 311*

Despite the promise of RNA therapeutics, progress towards the clinic has been slowed by the difficulty of delivering RNA drugs, such as short interfering RNA (siRNA) and messenger RNA (mRNA), into cellular targets within the body. RNA therapeutics are large ($10^4$ – $10^6$ g/mol) and negatively charged; they do not have favorable biodistribution properties in vivo nor an ability to cross the cellular membrane of target cells. In response to these challenges, our lab has created biodegradable, ionizable lipid-like materials called 'lipidoids' that readily formulate into nanoparticles containing RNA. Lipidoids efficiently manipulate gene expression in a variety of biological systems, including hepatocytes, white blood cells and tumors. This talk will focus on the cell-free prediction of lipidoid efficacy in delivering mRNA to mice. Furthermore, I will describe a new formulation strategy for the synergistic co-delivery of mRNA and siRNA. Specifically, we found that siRNA-mediated gene silencing and mRNA-mediated protein expression improved 2-3 fold when they co-formulated in the same nanoparticle. Together, these data demonstrate the potential of lipidoid materials to robustly manipulate gene expression in vivo.

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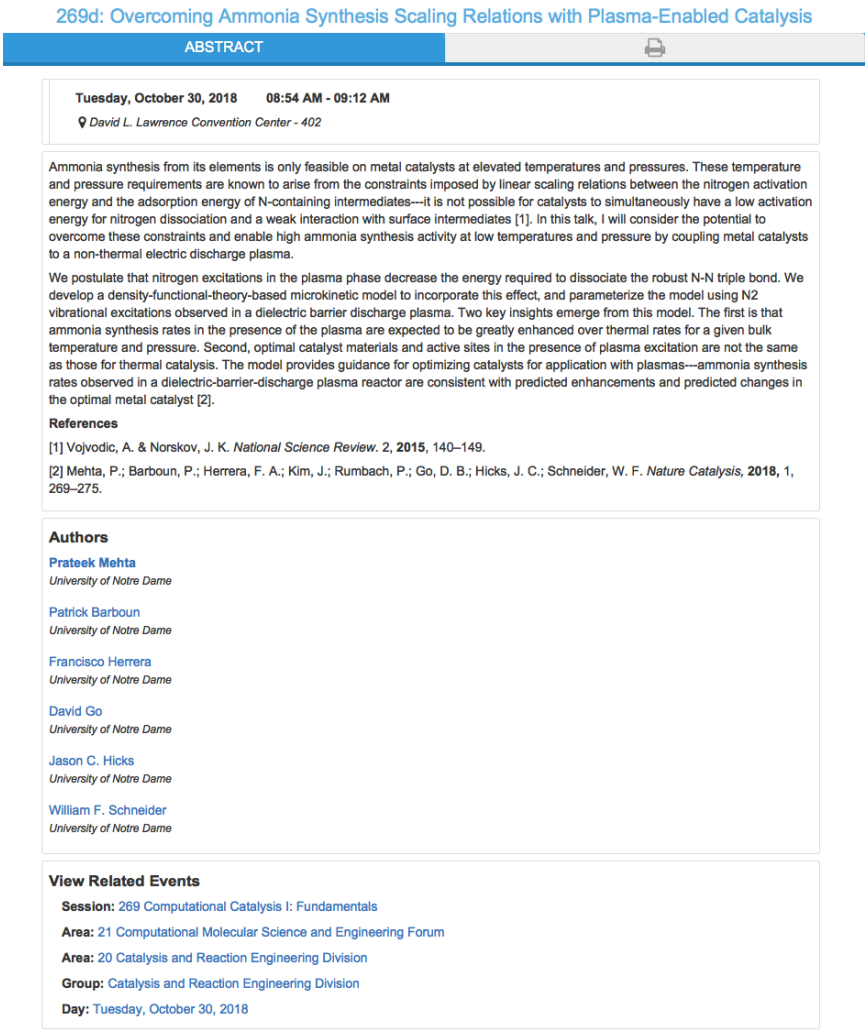
**Session:** 261 Area Plenary: Bionanotechnology (Invited Talks)

**Area:** 22B Bionanotechnology

**Group:** Nanoscale Science and Engineering Forum

**Day:** Tuesday, October 30, 2018

## 3.9 269d: Overcoming Ammonia Synthesis Scaling Relations with Plasma-Enabled Catalysis https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/532015 #MSAlum @prtk_-m @profwschneider @aiche_cre



269d: Overcoming Ammonia Synthesis Scaling Relations with Plasma-Enabled Catalysis

ABSTRACT

**Tuesday, October 30, 2018 08:54 AM - 09:12 AM**

*David L. Lawrence Convention Center - 402*

Ammonia synthesis from its elements is only feasible on metal catalysts at elevated temperatures and pressures. These temperature and pressure requirements are known to arise from the constraints imposed by linear scaling relations between the nitrogen activation energy and the adsorption energy of N-containing intermediates---it is not possible for catalysts to simultaneously have a low activation energy for nitrogen dissociation and a weak interaction with surface intermediates [1]. In this talk, I will consider the potential to overcome these constraints and enable high ammonia synthesis activity at low temperatures and pressure by coupling metal catalysts to a non-thermal electric discharge plasma.

We postulate that nitrogen excitations in the plasma phase decrease the energy required to dissociate the robust N-N triple bond. We develop a density-functional-theory-based microkinetic model to incorporate this effect, and parameterize the model using N2 vibrational excitations observed in a dielectric barrier discharge plasma. Two key insights emerge from this model. The first is that ammonia synthesis rates in the presence of the plasma are expected to be greatly enhanced over thermal rates for a given bulk temperature and pressure. Second, optimal catalyst materials and active sites in the presence of plasma excitation are not the same as those for thermal catalysis. The model provides guidance for optimizing catalysts for application with plasmas---ammonia synthesis rates observed in a dielectric-barrier-discharge plasma reactor are consistent with predicted enhancements and predicted changes in the optimal metal catalyst [2].

**References**

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[2] Mehta, P.; Barboun, P.; Herrera, F. A.; Kim, J.; Rumbach, P.; Go, D. B.; Hicks, J. C.; Schneider, W. F. *Nature Catalysis*, **2018,** 1, 269–275.

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**Session:** 269 Computational Catalysis I: Fundamentals

**Area:** 21 Computational Molecular Science and Engineering Forum

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Tuesday, October 30, 2018

## 3.10 285h: Effect of Surfactant Structure on Self-Assembly and Charging Processes in Anhydrous Nonpolar Liquids https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/536361

285h: Effect of Surfactant Structure on Self-Assembly and Charging Processes in Anhydrous Nonpolar Liquids.

ABSTRACT

**Tuesday, October 30, 2018 09:52 AM - 10:08 AM**

*Omni William Penn Hotel - Conference Center A*

Impedance spectroscopy has been used to determine the conductivity and concentration of charged inverse micelles of Span surfactants in dodecane.[JCIS, 449, 2-12, 2015] In this technique, an oscillating voltage is applied between two parallel electrode surfaces with surfactant solution sandwiched in between. The conductivity of the charged micelles is derived from the electrical resistance arising from the transport of charge carriers across the bulk solution towards the electrode surface. The concentration of charge carriers in solution is then obtained from the electrical capacitance of the diffuse clouds of counterions accumulating next to either electrode surface. Despite previous success with this technique, severe charge adsorption on the electrode surface has been shown to obscure measurement of the double layer capacitance and hence the charge carrier concentration.[JCIS, 449, 2-12, 2015] Charge adsorption also effectively limits the concentration range and catalog of surfactants available for impedance spectroscopy study. To expand the available concentration ranges of the impedance method, a passivating coating of plasma polymerized octafluorocyclobutane is fabricated and determined to successfully suppress charge adsorption. An equivalent circuit representing the surfactant solution and fluoropolymer coating is derived and used to model and interpret the impedance spectra. The circuit model captures most of the experimental charge transport processes with the exception of a small systematic deviation. To control and monitor the water content of solution during impedance experiments, a rigorous water elimination protocol involving nitrogen gas is developed and implemented. The conductivity and charge carrier trends with respect to surfactant concentration is then studied and rationalized with respect to surfactant structure and chemistry.

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**Session:** 285 Fundamentals of Interfacial Phenomena I

**Area:** 01C Interfacial Phenomena

**Group:** Engineering Sciences and Fundamentals

**Day:** Tuesday, October 30, 2018

## 3.11 253g: A Novel Branching Scheme for Problems with Reverse Convex Quadratic Constraints and Its Application to Packing Problems https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528773 @AIChECAST10

253g: A Novel Branching Scheme for Problems with Reverse Convex Quadratic Constraints and Its Application to Packing Problems

ABSTRACT

**Tuesday, October 30, 2018 09:54 AM - 10:13 AM**

*David L. Lawrence Convention Center - 409*

A quadratic constraint f(x) ≥ 0 is called reverse convex if f(x) is convex [1]. Reverse convex quadratic constraints are abundant in mathematical models for many industrial applications such as cutting and packing problems (CPP) [2] and wireless sensor network localization [3] among others. Though there have been impressive improvements in the general purpose global solvers for the past decades, solving problems with reverse convex quadratic constraints to ε-optimality is still a very challenging task. In this work, we propose a novel branching scheme in order to tackle this type of constraint and we compare our approach with global solvers via computational studies on cutting and packing instances.

Unlike a traditional, general purpose global solver, which must generically relax a nonconvex constraint and tighten that relaxation via branching on the domains of the native variables [4], our novel approach recognizes the physical meaning of the reverse convex quadratic constraint and proposes a direct branching on nonconvex constraints themselves. The inspiration emanates from two observations. First, branching on nonconvex constraints is enforcing their feasibility in a more direct way and therefore should be more effective in cutting off infeasible space. Second, in the context of specific applications such as CPP, due to a large number of these constraints and their close relationship, feasibility-based tightening techniques can be exploited in a more sophisticated fashion if the constraints feasible domain is explicitly split.

The CPP aims to find a packing of shapes that minimizes the amount of material needed to carve them out of a stock. This minimization of waste can be of utmost economic importance in certain large industrial sectors, wherein a minuscule reduction in the amount of stock material used would result in significant monetary savings across the whole industry. In this work, we focus on two types of CPP: circle packing problems (CP) [5] and irregular shape packing problems (ISP, also known as nesting problems) [2]. For CP and ISP, relevant studies in the CPP community have mostly focused on the development of heuristics to obtain large-scale packings. While heuristic methods are practically valuable for the generation of good solutions [6], they lack any guarantees of optimality and cannot rigorously quantify the quality of packings produced. The challenge of solving CP and ISP to global optimality results from the fact that enforcing no overlap between two shapes leads to certain nonconvexities that are still numerically very challenging for even the state-of-the-art global optimization solvers.

In this work, we focus on tackling reverse convex quadratic constraints and propose a novel branching scheme. In order to evaluate our approach's effectiveness and compare its performance with deterministic global solvers, we conduct comprehensive computational studies on a suite of circle packing instances and irregular shape packing instances [7]. Preliminary results indicate that our new branching method is competitive to global solvers including BARON [8], COUENNE [9] and GloMIQO [10], LINDOGlobal [11] and SCIP [12].

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**Session:** 253 Advances in Deterministic Global Optimization

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Tuesday, October 30, 2018

## 3.12 273g: Modeling for Reliability Optimization of System Design and Maintenance Based on Markov Chain Theory https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530733 @PraxairInc @AIChECAST10

273g: Modeling for Reliability Optimization of System Design and Maintenance Based on Markov Chain Theory

ABSTRACT

**Tuesday, October 30, 2018 09:54 AM - 10:13 AM**

*David L. Lawrence Convention Center - 410*

Plant availability has always been a critical consideration for the design and operation of chemical processes as it represents the expected fraction of normal operating time, which impacts directly the ability of meeting customer demands. The failure of a key unit is likely to cause the shut-down of the entire plant. Effective ways to reduce the impact of equipment breakdown on the entire plant include installing back-up units and carrying out regular inspections and maintenance. Currently, discrete event simulation tools are used to evaluate the reliability/availability of selected alternatives to simulate the behavior of every asset in a plant using historical maintenance data and statistical models (Sharda and Bury, 2008). However, this approach does not systematically consider all the alternatives as it would be the case in an optimization approach.

A number of works have been reported to optimize the reliability of chemical plants. Pistikopoulos et al. (2001) and Goel et al. (2003b) formulate an MILP model for the selection of units with different reliability and the corresponding production and maintenance planning for a fixed system configuration. Terrazas-Moreno et al. (2010) formulate an MILP model using Markov chains to optimize the expected stochastic flexibility of an integrated production site by the selection of pre-specified alternative plants and the design of intermediate storage. Lin et al. (2012) model a simple utility system using Markov chain and carry out RAM (reliability, availability & maintainability) analysis iteratively to decide the optimal reliability design.

While mixed-integer optimization models have been proposed to address the reliability problem from various aspects, there are virtually no systematic modeling approaches proposed to make decisions regarding the number and selection of parallel units, as well as maintenance policy for the optimal profitability of reliable chemical processes. In response to this gap, this work extends our recent mixed-integer framework (Ye et al., in press) and introduces a systematic approach to model the stochastic process of system failures and repairs as a continuous-time Markov chain.

Two decision levels are considered to increase the availability of the system. The first decision level is to install parallel units for certain processing stages, such that when the primary unit fails, the other units can fill in its place in order to reduce system downtime. The second decision level is to carry out periodic inspections, and conditional maintenance if the inspection result indicates that the equipment will fail shortly. By that strategy the system can avoid a number of repairs, which are more costly than maintenances in terms of both time and money. Thus, there are two types of trade-offs in the model. The first type is the trade-off between the costs incurred and the revenue gained from increasing system availability. The second type is the budget balance between the two decision levels and among the processing stages. A non-convex MINLP model based on Markov chain assumption is proposed accordingly. It is worth mentioning that when inspections and conditional maintenance are not considered, the model reduces to an MILP, and a small superstructure with two stages is solved directly with global optimization solvers to show that improved system availability and net profit can be obtained compared to the case when disregarding maintenance, although at the expense of longer computing time.

However, the size of the non-convex MINLP model grows geometrically with the number of processing stages in the system. In addition, the combination of the two decision levels creates a large number of bilinear and multilinear terms. In order to overcome these computational difficulties and make the model applicable to practical problems, a decomposition scheme is proposed to reduce the model complexity. Moreover, scenario reduction, i. e., preprocess the scenarios and eliminate those with consistently low probabilities, is also applied to reduce the size of the model. The non-convex MINLP model with the specialized solution methods is applied to the two-stage problem and reaches global optimum in significantly shorter time than the global solvers. It is then used to solve larger examples that are not directly solvable and is able to obtained solutions in acceptable time.

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**Area:** 10C Computers in Operations and Information Processing

**Area:** 10A Systems and Process Design

**Group:** Computing and Systems Technology Division

**Day:** Tuesday, October 30, 2018

## 3.13 343b: From Academia to Industry: Optimization Models for Shale Gas Development https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/524592 @EQT_Corporation @AIChECAST10

343b: From Academia to Industry: Optimization Models for Shale Gas Development

ABSTRACT

**Tuesday, October 30, 2018 12:49 PM - 01:08 PM**

*David L. Lawrence Convention Center - 409*

**From Academia to Industry: Optimization Models for Shale Gas Development**

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**Abstract**

For the past four years the Center for Advanced Process Decision-Making (CAPD) at Carnegie Mellon University (CMU) and EQT Corporation have been actively collaborating to develop optimization models for shale gas development. The CAPD is a research center at CMU that is engaged in Process Systems Engineering research for the process industries. The CAPD focuses on a number of different research areas including a) modeling and optimization algorithms, b) process synthesis and product design, c) enterprise-wide optimization and process operations, and d) molecular computing. EQT, on the other hand, is the leading natural gas producer in the United States with a strategic focus on shale gas development in the Marcellus Play. The company locates natural gas, develops acreage and transports produced gas to distribution systems.

In this presentation, we review how the shale gas development research project at CMU started and how it progressed over time. We show that a number of researchers associated with the CAPD made important contributions to this domain. In particular, Knudsen et al. (2014) developed a Lagrangean relaxation scheme for shut-in scheduling in shale-gas multi-pad systems. Thereafter, Cafaro et al. (2014) proposed a mixed-integer nonlinear programming (MINLP) model for the strategic planning of the shale gas supply chain. Next, Yang et al. (2014) developed a stochastic two-stage mixed-integer linear programming (MILP) scheduling models for shale plays water management. Most recently, Drouven et al. (2016) developed a multiperiod MINLP model for long-term, quality-sensitive shale gas development. Finally, Ondeck et al. (2018) developed a bi-criterion optimization approach for simultaneous planning and scheduling of shale gas production. In this talk we review the key findings of the aforementioned work, and we discuss their significance for the shale gas industry as a whole.

We also review how the research within the CAPD led to the formation of the Optimization Engineering team at EQT. We show how this team currently translates practical business and/or engineering problems within the company into mixed-integer programming models; and we demonstrate how these models are used to support faster and better decision-making across the organization.

Finally, we present a real-world case study to demonstrate how optimization models at EQT are currently being used to support safer and more economic shale gas development. The case study reveals how a rigorous optimization model for water management applied to an active shale gas development area can reduce the number of water hauling trucks on the road (a safety benefit), limit impaired water disposal volumes (an environmental benefit) and how it minimizes operational costs (an economic benefit).

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**Session:** 343 Industrial Applications in Design and Operations

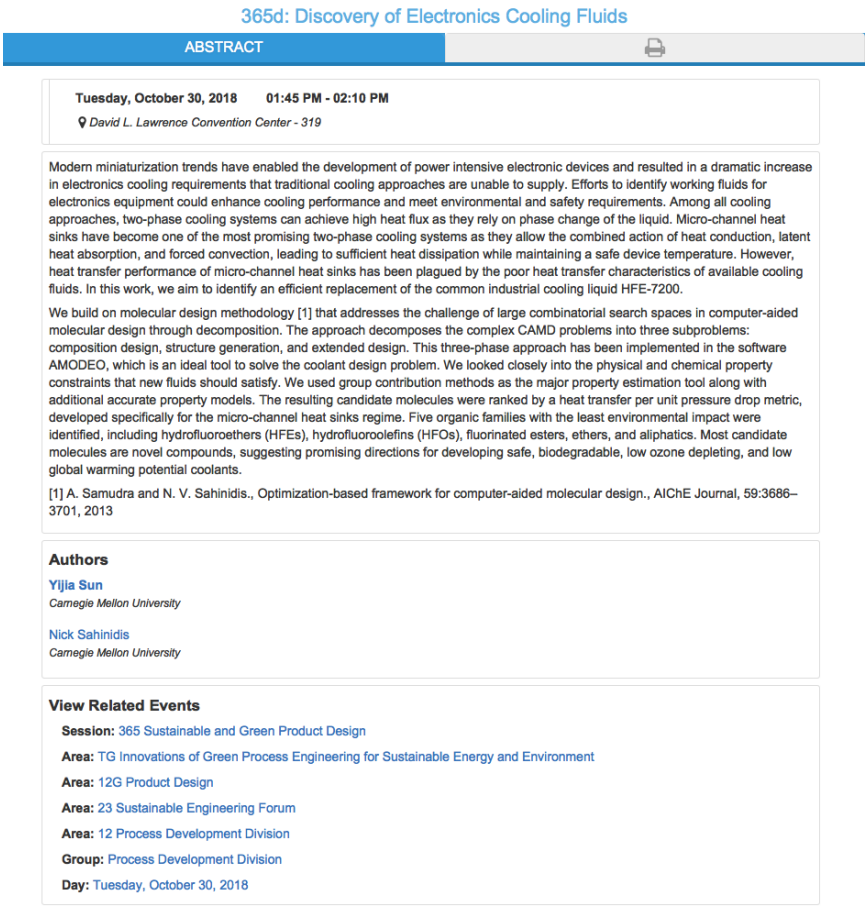
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**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Tuesday, October 30, 2018

## 3.14 365d: Discovery of Electronics Cooling Fluids https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528218



365d: Discovery of Electronics Cooling Fluids

ABSTRACT

**Tuesday, October 30, 2018 01:45 PM - 02:10 PM**

*David L. Lawrence Convention Center - 319*

Modern miniaturization trends have enabled the development of power intensive electronic devices and resulted in a dramatic increase in electronics cooling requirements that traditional cooling approaches are unable to supply. Efforts to identify working fluids for electronics equipment could enhance cooling performance and meet environmental and safety requirements. Among all cooling approaches, two-phase cooling systems can achieve high heat flux as they rely on phase change of the liquid. Micro-channel heat sinks have become one of the most promising two-phase cooling systems as they allow the combined action of heat conduction, latent heat absorption, and forced convection, leading to sufficient heat dissipation while maintaining a safe device temperature. However, heat transfer performance of micro-channel heat sinks has been plagued by the poor heat transfer characteristics of available cooling fluids. In this work, we aim to identify an efficient replacement of the common industrial cooling liquid HFE-7200.

We build on molecular design methodology [1] that addresses the challenge of large combinatorial search spaces in computer-aided molecular design through decomposition. The approach decomposes the complex CAMD problems into three subproblems: composition design, structure generation, and extended design. This three-phase approach has been implemented in the software AMODEO, which is an ideal tool to solve the coolant design problem. We looked closely into the physical and chemical property constraints that new fluids should satisfy. We used group contribution methods as the major property estimation tool along with additional accurate property models. The resulting candidate molecules were ranked by a heat transfer per unit pressure drop metric, developed specifically for the micro-channel heat sinks regime. Five organic families with the least environmental impact were identified, including hydrofluoroethers (HFEs), hydrofluoroolefins (HFOs), fluorinated esters, ethers, and aliphatics. Most candidate molecules are novel compounds, suggesting promising directions for developing safe, biodegradable, low ozone depleting, and low global warming potential coolants.

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**Session:** 365 Sustainable and Green Product Design

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**Group:** Process Development Division

**Day:** Tuesday, October 30, 2018

## 3.15 343e: Optimization of Circuitry Arrangements for Heat Exchangers https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/529957 @Mitsubishi_USA @AIChECAST10

343e: Optimization of Circuitry Arrangements for Heat Exchangers

ABSTRACT

**Tuesday, October 30, 2018 01:46 PM - 02:05 PM**

*David L. Lawrence Convention Center - 409*


Heat exchanger performance is important in various heating and air-conditioning systems that are widely used in residential and commercial applications. Although these systems are manufactured in various shapes and configurations [1], a common configuration is that of the crossflow fin-and-tube type, in which a refrigerant flows through a set of pipes and moist air flows across a possibly enhanced surface on the other side of the pipe, where thermal energy is transferred between the air and the refrigerant.

Heat exchanger performance optimization can be achieved by optimizing a number of different metrics including maximization of heating or cooling capacity, size reduction, component material reduction, manufacturing cost reduction, reduction of pumping power, or a combination of these metrics. Systematic optimization of heat exchanger design has been a long-standing research topic [2–4]. It is a particularly challenging problem mainly for the following reasons: (i) there is a highly discontinuous and nonlinear relationship between the circuitry design and the heat exchangers performances, and (ii) the decision space is extremely large making exhaustive search algorithms insufficient for searching the entirety of the solution space.

In this paper, we propose a new formulation for the refrigerant circuitry design problem. We model this problem as a constrained binary optimization problem. We incorporate only realistic manufacturing constraints to the optimization problem without requiring extensive domain knowledge. As a result, the proposed approach can be applied to different types of heat exchangers. In order to evaluate the heat exchanger performance using different refrigerant circuitry designs, we use CoilDesigner [5], a steady-state simulation and design tool for air to refrigerant heat exchangers, which acts as a black-box since the exact relationship of the objective functions with the decision variables is not explicit. We apply derivative-free optimization algorithms [6] to optimize heat exchanger performance. Although the derivative-free optimization literature has recently been attracting significant attention, it currently lacks systematic comparisons between mixed-integer constrained derivative-free optimization algorithms on industrially-relevant problems. Another contribution of this work is the systematic comparison of four different mixed-integer constrained derivative-free optimization algorithms (MIDACO [7], NOMAD [8], TOMLAB/glcDirect [9], and TOMLAB/glcSolve [9]) and a box-bounded derivative-free optimization algorithm (CMAES [10]) that are applied to optimize heat exchanger circuitry using two different thermal efficiency criteria.

Computational results on 17 different circuitry architectures show that TOMLAB/glcDirect and TOMLAB/glcSolve have the best performance among all derivative-free optimization solvers. Timewise, TOMLAB/glcSolve is faster than TOMLAB/glcDirect on smaller, but TOMLAB/glcDirect is much faster on larger instances and on average. We also formulated the circuitry optimization problem as a constraint satisfaction problem using Choco solver in order to automate the procedure of finding all possible feasible circuitry designs and verify the results of the derivative-free methods for small heat exchangers. The results show that the proposed method provides optimal refrigerant circuitries satisfying realistic manufacturing constraints.


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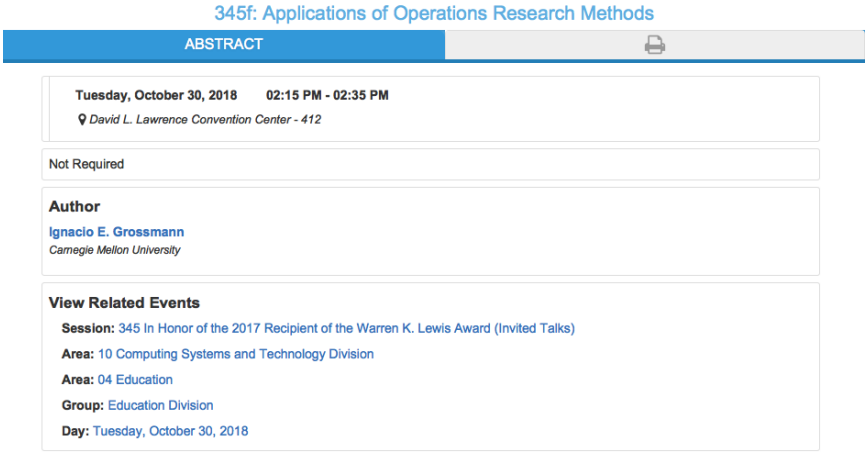
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**Area:** 10C Computers in Operations and Information Processing

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**Day:** Tuesday, October 30, 2018

## 3.16 345f: Applications of Operations Research Methods https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/540936 @AIChECAST10



## 3.17 367g: Constrained Subset Selection for the Regression of Multi-Component Helmholtz Energy Equations https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530932

367g: Constrained Subset Selection for the Regression of Multi-Component Helmholtz Energy Equations

ABSTRACT

**Tuesday, October 30, 2018 02:43 PM - 03:02 PM**

*David L. Lawrence Convention Center - 307*

Accurate thermodynamic properties are important for the development of new processes and technologies and simulating an optimized flowsheet. Unfortunately, equations of state typically used, such as Peng Robinson and Soave-Redlich Kwong, are inaccurate in critical regions of new technologies. A recent development in equations of state uses first principles to determine one unifying equation explicit in Helmholtz energy to overcome these inaccuracies. These equations have been developed for over 100 pure substances and have been expanded to mixtures [1, 2, 3, 4].

A major challenge in the development of Helmholtz energy equations is the fitting of varying datasets containing correlated data to the main unifying equation [3]. The current fitting procedures depend on cycling between linear and nonlinear regression techniques that are restricted to equality constraints and result in local solutions requiring multiple starts or an experienced user manually selecting an initial starting point [4]. Advancing these techniques to eliminate the need for multistart heuristics and provide the ability to enforce inequality constraints on the resulting models would allow us to control the thermodynamic behavior, improve extrapolation behavior, and maintain thermodynamically feasible limitations on the regressed equation while simultaneously fitting the data [7].

In order to address these challenges associated with fitting data to Helmholtz energy equations, we have developed a model-building procedure from thermodynamic data and have expanded its capabilities to include mixtures. The main idea is to apply best subset selection and allow for the controlling of thermodynamic slopes with inequality constraints while fitting all the data simultaneously. A bank of terms is chosen to best represent the different phase regions and thermodynamic behavior. In order to avoid overfitting, our procedure systematically selects a subset of these terms to optimally fit the multiple thermodynamic property data sets. We rely on a global optimization solver [7] to find an optimal solution that optimizes the fit as well as the number of terms in the model according to an information criterion. The proposed approach was used to fit a new Helmholtz energy equation for carbon dioxide binary mixtures. The resulting models are compared to established models such as cubic equations of state, EOS-CG [5], and the GERG-2008 [6].

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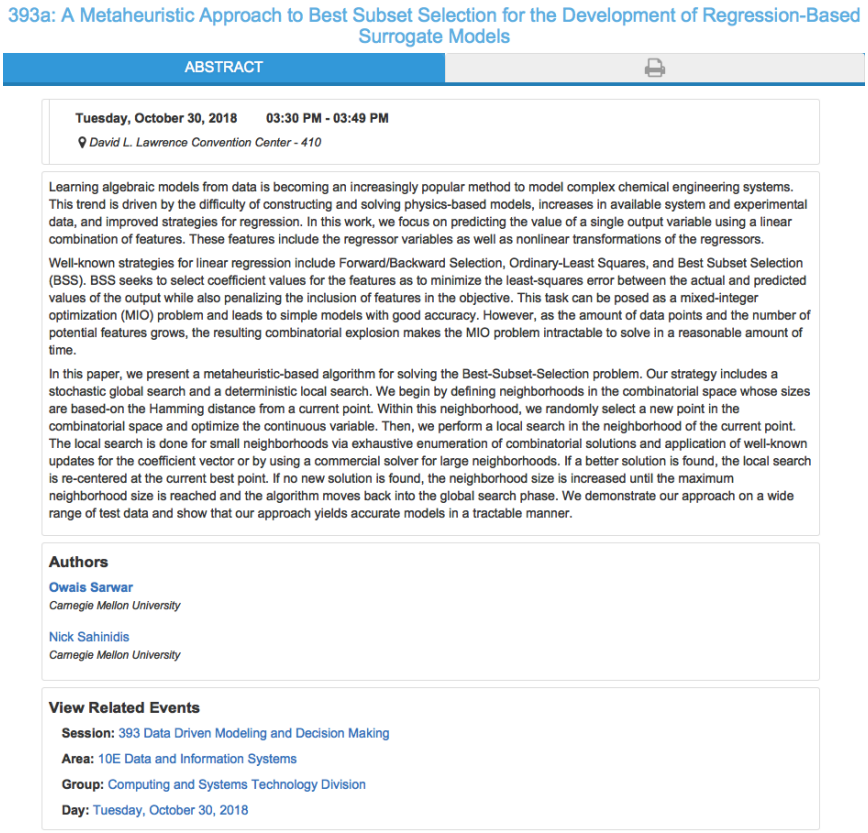
**Session:** 367 Thermophysical Properties and Phase Behavior

**Area:** 01A Thermodynamics and Transport Properties

**Group:** Engineering Sciences and Fundamentals

**Day:** Tuesday, October 30, 2018

## 3.18 393a: A Metaheuristic Approach to Best Subset Selection for the Development of Regression-Based Surrogate Models https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530253 @AIChECAST10



393a: A Metaheuristic Approach to Best Subset Selection for the Development of Regression-Based Surrogate Models

ABSTRACT

**Tuesday, October 30, 2018 03:30 PM - 03:49 PM**

*David L. Lawrence Convention Center - 410*

Learning algebraic models from data is becoming an increasingly popular method to model complex chemical engineering systems. This trend is driven by the difficulty of constructing and solving physics-based models, increases in available system and experimental data, and improved strategies for regression. In this work, we focus on predicting the value of a single output variable using a linear combination of features. These features include the regressor variables as well as nonlinear transformations of the regressors.

Well-known strategies for linear regression include Forward/Backward Selection, Ordinary-Least Squares, and Best Subset Selection (BSS). BSS seeks to select coefficient values for the features as to minimize the least-squares error between the actual and predicted values of the output while also penalizing the inclusion of features in the objective. This task can be posed as a mixed-integer optimization (MIO) problem and leads to simple models with good accuracy. However, as the amount of data points and the number of potential features grows, the resulting combinatorial explosion makes the MIO problem intractable to solve in a reasonable amount of time.

In this paper, we present a metaheuristic-based algorithm for solving the Best-Subset-Selection problem. Our strategy includes a stochastic global search and a deterministic local search. We begin by defining neighborhoods in the combinatorial space whose sizes are based-on the Hamming distance from a current point. Within this neighborhood, we randomly select a new point in the combinatorial space and optimize the continuous variable. Then, we perform a local search in the neighborhood of the current point. The local search is done for small neighborhoods via exhaustive enumeration of combinatorial solutions and application of well-known updates for the coefficient vector or by using a commercial solver for large neighborhoods. If a better solution is found, the local search is re-centered at the current best point. If no new solution is found, the neighborhood size is increased until the maximum neighborhood size is reached and the algorithm moves back into the global search phase. We demonstrate our approach on a wide range of test data and show that our approach yields accurate models in a tractable manner.

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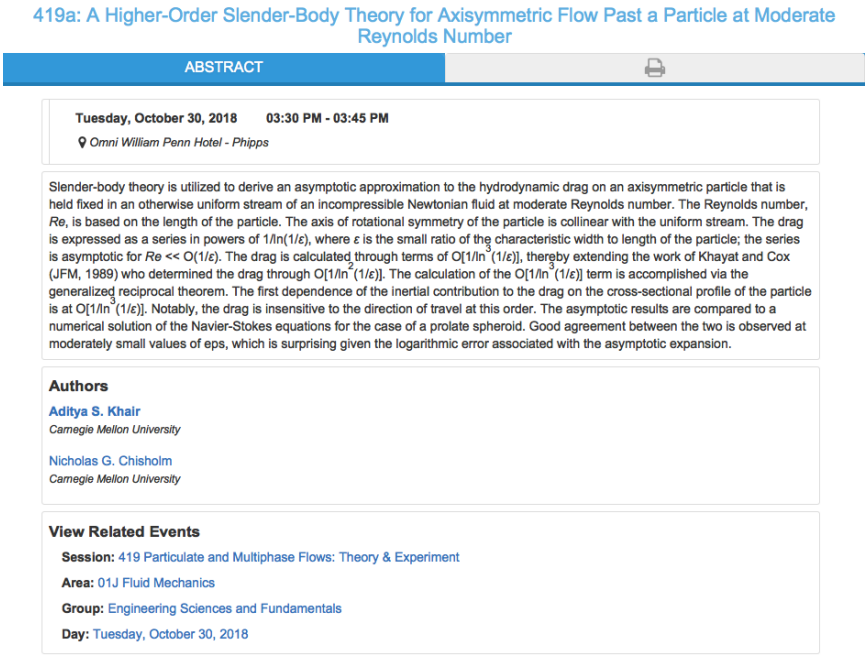
**Session:** 393 Data Driven Modeling and Decision Making

**Area:** 10E Data and Information Systems

**Group:** Computing and Systems Technology Division

**Day:** Tuesday, October 30, 2018

## 3.19 419a: A Higher-Order Slender-Body Theory for Axisymmetric Flow Past a Particle at Moderate Reynolds Number @AdityaSKhair1 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/529600



419a: A Higher-Order Slender-Body Theory for Axisymmetric Flow Past a Particle at Moderate Reynolds Number

ABSTRACT

**Tuesday, October 30, 2018** **03:30 PM - 03:45 PM**

*Omni William Penn Hotel - Phipps*

Slender-body theory is utilized to derive an asymptotic approximation to the hydrodynamic drag on an axisymmetric particle that is held fixed in an otherwise uniform stream of an incompressible Newtonian fluid at moderate Reynolds number. The Reynolds number, *Re*, is based on the length of the particle. The axis of rotational symmetry of the particle is collinear with the uniform stream. The drag is expressed as a series in powers of $1/\ln(1/\varepsilon)$, where $\varepsilon$ is the small ratio of the characteristic width to length of the particle; the series is asymptotic for $Re << O(1/\varepsilon)$. The drag is calculated through terms of $O[1/\ln^3(1/\varepsilon)]$, thereby extending the work of Khayat and Cox (JFM, 1989) who determined the drag through $O[1/\ln^2(1/\varepsilon)]$. The calculation of the $O[1/\ln^3(1/\varepsilon)]$ term is accomplished via the generalized reciprocal theorem. The first dependence of the inertial contribution to the drag on the cross-sectional profile of the particle is at $O[1/\ln^3(1/\varepsilon)]$. Notably, the drag is insensitive to the direction of travel at this order. The asymptotic results are compared to a numerical solution of the Navier-Stokes equations for the case of a prolate spheroid. Good agreement between the two is observed at moderately small values of eps, which is surprising given the logarithmic error associated with the asymptotic expansion.

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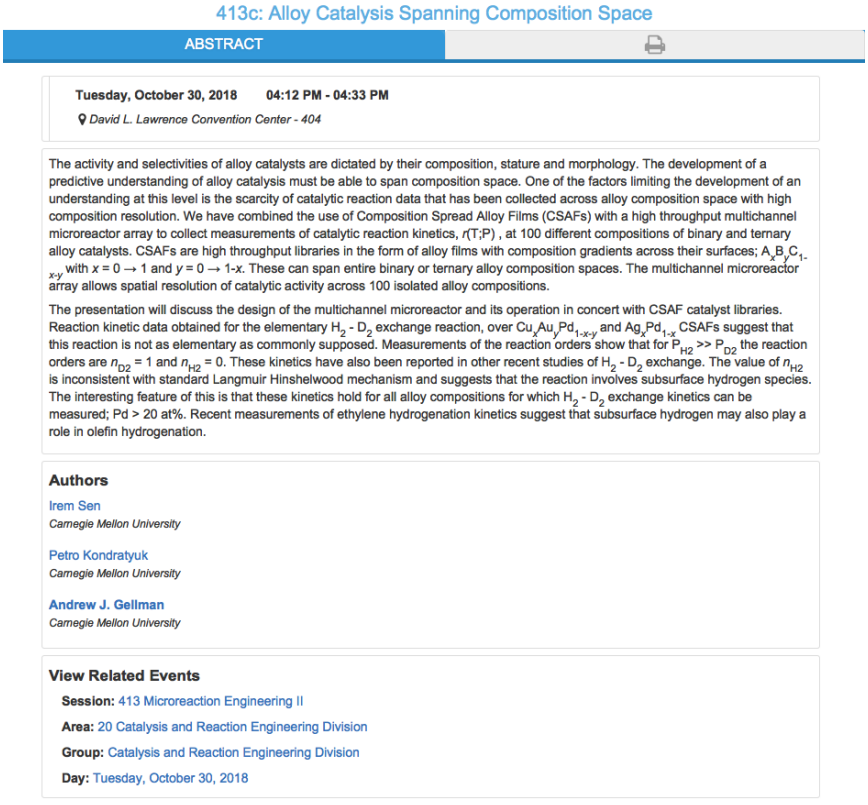
**Session:** 419 Particulate and Multiphase Flows: Theory & Experiment

**Area:** 01J Fluid Mechanics

**Group:** Engineering Sciences and Fundamentals

**Day:** Tuesday, October 30, 2018

## 3.20 413c: Alloy Catalysis Spanning Composition Space https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/531425 @iremsen @aiche_cre



413c: Alloy Catalysis Spanning Composition Space

ABSTRACT

**Tuesday, October 30, 2018 04:12 PM - 04:33 PM**

*David L. Lawrence Convention Center - 404*

The activity and selectivities of alloy catalysts are dictated by their composition, stature and morphology. The development of a predictive understanding of alloy catalysis must be able to span composition space. One of the factors limiting the development of an understanding at this level is the scarcity of catalytic reaction data that has been collected across alloy composition space with high composition resolution. We have combined the use of Composition Spread Alloy Films (CSAFs) with a high throughput multichannel microreactor array to collect measurements of catalytic reaction kinetics, $r(T;P)$ , at 100 different compositions of binary and ternary alloy catalysts. CSAFs are high throughput libraries in the form of alloy films with composition gradients across their surfaces; $A_xB_yC_{1-x-y}$ with $x = 0 \rightarrow 1$ and $y = 0 \rightarrow 1\text{-}x$. These can span entire binary or ternary alloy composition spaces. The multichannel microreactor array allows spatial resolution of catalytic activity across 100 isolated alloy compositions.

The presentation will discuss the design of the multichannel microreactor and its operation in concert with CSAF catalyst libraries. Reaction kinetic data obtained for the elementary $H_2$ - $D_2$ exchange reaction, over $Cu_xAu_yPd_{1-x-y}$ and $Ag_xPd_{1-x}$ CSAFs suggest that this reaction is not as elementary as commonly supposed. Measurements of the reaction orders show that for $P_{H2} >> P_{D2}$ the reaction orders are $n_{D2} = 1$ and $n_{H2} = 0$. These kinetics have also been reported in other recent studies of $H_2$ - $D_2$ exchange. The value of $n_{H2}$ is inconsistent with standard Langmuir Hinshelwood mechanism and suggests that the reaction involves subsurface hydrogen species. The interesting feature of this is that these kinetics hold for all alloy compositions for which $H_2$ - $D_2$ exchange kinetics can be measured; Pd > 20 at%. Recent measurements of ethylene hydrogenation kinetics suggest that subsurface hydrogen may also play a role in olefin hydrogenation.

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**View Related Events**

**Session:** 413 Microreaction Engineering II

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Tuesday, October 30, 2018

## 3.21 412d: Rapid Separation of DNA Digests in Entangled Micelle Networks https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/537649

412d: Rapid Separation of λDNA Digests in Entangled Micelle Networks

ABSTRACT

**Tuesday, October 30, 2018 04:30 PM - 04:45 PM**

*Omni William Penn Hotel - Frick*

The ability to separate nucleic acids quickly and accurately via electrophoretic means has been imperative in many medical and biological applications. In free solution, DNA has a length invariant electrophoretic mobility. As such, DNA electrophoresis is often carried out within a gel or gel-like sieving matrix. As an alternative method, micelle-ELFSE (end-labeled free-solution electrophoresis) achieves rapid, length-based separations in free solution by attaching non-ionic micelles to the end of 5'-alkylated DNA. Separation resolution is found to be highly sensitive to buffer thermal history and composition. To increase the read length of micelle-ELFSE, surfactant compositions and separation temperature were varied to systematically induce micellar growth. This formulation effort led to a read length of 1150 bases using a high concentration buffer (5x the overlap concentration) composed of 150 mM C12E5 and 3 M urea (6 wt%) at 33°C. Under these conditions, unalkylated DNA is separable by a sieving/reptation mechanism. Yet, end-alkylated DNA is separated by an ELSFE mechanism as shown by its reversed elution order with DNA length. As an example, the electropherogram of Figure 1 has two dsDNA ladders mixed; one that is end-alkylated and another that is not. The two ladders eluted independently: the unmodified DNA ladder separates via sieving with short DNA eluting first, whereas an end-alkylated ladder separates via ELFSE with long DNA eluting first.

Recent efforts were successful in separating unalkylated and alkylated kilobase (23kb) double stranded λDNA digests (Figure 2), demonstrating that, in principal, micelle-ELFSE permits the use of very large micellar drag-tag for kilobase DNA separation despite the presence of entanglement and sieving. One possible mechanistic explanation of the process is that alkylated DNA's alkyl group interact locally with nearby micelle crosslinks, causing them to dissociate at a greater rate and negating the sieving effects. In most cases, the mobility of alkylated DNA during micelle-ELFSE can be accurately predicted by the standard ELFSE equation assuming the drag-tag and DNA are hydrodynamically equivalent (non-segregated) during the separation. We do observe deviations from the non-segregated model for long kilobase DNA in high concentrated buffers in which the mobility is better predicted by the segregated model. We discuss our efforts to quantitatively model this behavior, and set upper limits for the lengths of single- and double-stranded DNA that can be resolved in entangled micelle networks. The ability to rapidly separate unalkylated and alkylated kilobase λDNA digests in free solution using standard benchtop CE instrument bypasses many complications associated with the microarray fabrication process. As such, we compare separation performance to other common kilobase DNA separation methods such as entropic trap, nano-arrays, microfluidic chip, etc.

We envision many applications for kilobase DNA separation via micelle-ELFSE. One such application involves using kilobase dsDNA saturated with intercalating fluorophore as high-affinity probes for sub-fM at-line detection of viral and bacterial contaminants in cell cultures. Micelle-ELFSE is an excellent candidate for PCR-less detection of nucleic acid directly from the culture media given that it's fast, highly selective and compatible with high levels of broad-spectrum lipophilic contaminants.

**Figure 1.** Simultaneous separation of unalkylated dsDNA PCR prodcuts and alkylated dsDNA with a C16 alkane modification to bind micelles in the running buffer (2 wt% C12E5). Unalkylated ladder separates via sieving with shorter DNA eluting first. Alkylated ladder separates via ELFSE with longer DNA eluting first.

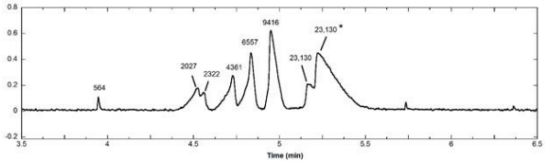



**Figure 2.** Simultaneous separation of unalkylated and alkylated HindIII digest of λDNA. DNA lengths labeled with (*) has a C18 alkane modification that binds to micelles in the running buffer.

**View Related Events**

**Day:** Tuesday, October 30, 2018

## 3.22 421e: Effective Generalized Disjunctive Programming (GDP) Models for Modular Plant Design https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/532195 @QtotheC @AIChECAST10

421e: Effective Generalized Disjunctive Programming (GDP) Models for Modular Plant Design

ABSTRACT

**Tuesday, October 30, 2018 04:46 PM - 05:05 PM**

*David L. Lawrence Convention Center - 409*

Evolving demands and increasing competition in the chemical processing industries drive a need for the most effective process design, at the right time. Engineers must identify the optimal configurations among many flowsheet alternatives in order to arrive at the desired design. One of the major approaches to synthesis is superstructure optimization, which involves postulating a superstructure representation of the alternatives, formulating a mathematical program to capture the decision logic, and applying a solution algorithm to determine the optimal configuration and operating conditions (Chen and Grossmann, 2017). Recently, there has been increased interest in modular designs due to their potential for improving safety, reliability, flexibility, and time-to-market (Baldea et al., 2017). While modularity on the product side has been addressed in depth (Rogers and Bottaci, 1997), new model extensions and methodologies for superstructure optimization must be developed to address modular process design challenges.

A hallmark of modular design is an emphasis shift from economies of scale towards economies of mass production, also referred to as "numbering-up" (Lier et al., 2015). This requires standardization of modules, a concept that has been explored in terms of aggregation of process units (Kampczyk et al., 2004) and cluster analysis to identify heat exchanger modules (Eilermann et al., 2017). This standardization implies that continuous design variables in the optimization formulation will now take values from some discrete set. While the introduction of discrete variables adds combinatorial complexity, it can also induce linearity in synthesis models, making them easier to solve (Voudouris and Grossmann, 1992).

In this work, motivated by Stephanopoulos and Westerberg (1980), we present a modification of the stage-wise heat exchanger network synthesis model from Yee and Grossmann (1990) that allows for incorporation of modular exchangers, selected from among a set of standard sizes. The modular design problem is formulated as a Generalized Disjunctive Programming (GDP) model Using this demonstration model, we show that discretization of the exchanger area allows for linear reformulations of the basic heat exchanger design equation, which is usually bilinear. Reformulation strengthens linear relaxations of the overall design problem, therefore aiding in the convergence of optimization solvers. Therefore, the consideration of modular process units and resulting discretization in design variables can even simplify solution of synthesis problems. We express the GDP model using the Pyomo optimization modeling library (Hart et al., 2017) in Python. Pyomo enables programmatic access to modify and manipulate modeling objects such as variables and constraints, allowing for automatic model interrogation and reformulation (Chen et al., 2018). Using Pyomo, we develop a plugin to automatically detect induced linearity from discrete modular design choices and reformulate the model to exploit the special structure.

Another central point of modular design is improved time-to-market due to expedited design, construction, commissioning, and startup of modular facilities. Several authors have examined the net present value (NPV) economic impact of switching to a modular design, including analysis on the sensitivity to the number of modules employed (Brodhagen et al., 2012; Lier and Grünewald, 2011; Mothes, 2015). We extend these studies using a model adapted from (Menezes et al., 2015) to examine the flowsheet design and construction scheduling of a process as a simultaneous GDP optimization problem, and demonstrate its value in balancing the trade-off between investment costs, operating costs, and time-to-market.

In this work, we demonstrate the value of effective GDP models both in capturing the logic and addressing the mathematics of modular chemical plant design. Through various case studies, we show how GDP enables investigation of key tenants of modular design.

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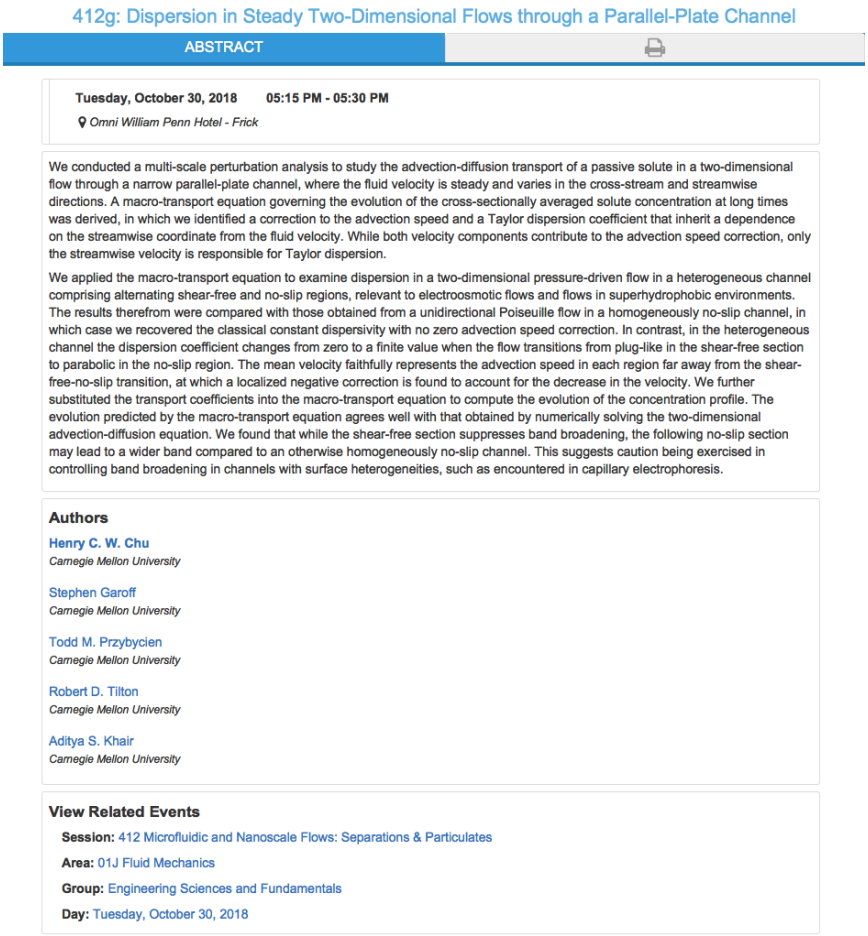
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**View Related Events**

**Session:** 421 Process Design: Conceptualization and Analysis of Chemical Processes I

**Area:** 10A Systems and Process Design

## 3.23 412g: Dispersion in Steady Two-Dimensional Flows through a Parallel-Plate Channel @AdityaSKhair1 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528937



412g: Dispersion in Steady Two-Dimensional Flows through a Parallel-Plate Channel

ABSTRACT

**Tuesday, October 30, 2018 05:15 PM - 05:30 PM**

*Omni William Penn Hotel - Frick*

We conducted a multi-scale perturbation analysis to study the advection-diffusion transport of a passive solute in a two-dimensional flow through a narrow parallel-plate channel, where the fluid velocity is steady and varies in the cross-stream and streamwise directions. A macro-transport equation governing the evolution of the cross-sectionally averaged solute concentration at long times was derived, in which we identified a correction to the advection speed and a Taylor dispersion coefficient that inherit a dependence on the streamwise coordinate from the fluid velocity. While both velocity components contribute to the advection speed correction, only the streamwise velocity is responsible for Taylor dispersion.

We applied the macro-transport equation to examine dispersion in a two-dimensional pressure-driven flow in a heterogeneous channel comprising alternating shear-free and no-slip regions, relevant to electroosmotic flows and flows in superhydrophobic environments. The results therefrom were compared with those obtained from a unidirectional Poiseuille flow in a homogeneously no-slip channel, in which case we recovered the classical constant dispersivity with no zero advection speed correction. In contrast, in the heterogeneous channel the dispersion coefficient changes from zero to a finite value when the flow transitions from plug-like in the shear-free section to parabolic in the no-slip region. The mean velocity faithfully represents the advection speed in each region far away from the shear-free-no-slip transition, at which a localized negative correction is found to account for the decrease in the velocity. We further substituted the transport coefficients into the macro-transport equation to compute the evolution of the concentration profile. The evolution predicted by the macro-transport equation agrees well with that obtained by numerically solving the two-dimensional advection-diffusion equation. We found that while the shear-free section suppresses band broadening, the following no-slip section may lead to a wider band compared to an otherwise homogeneously no-slip channel. This suggests caution being exercised in controlling band broadening in channels with surface heterogeneities, such as encountered in capillary electrophoresis.

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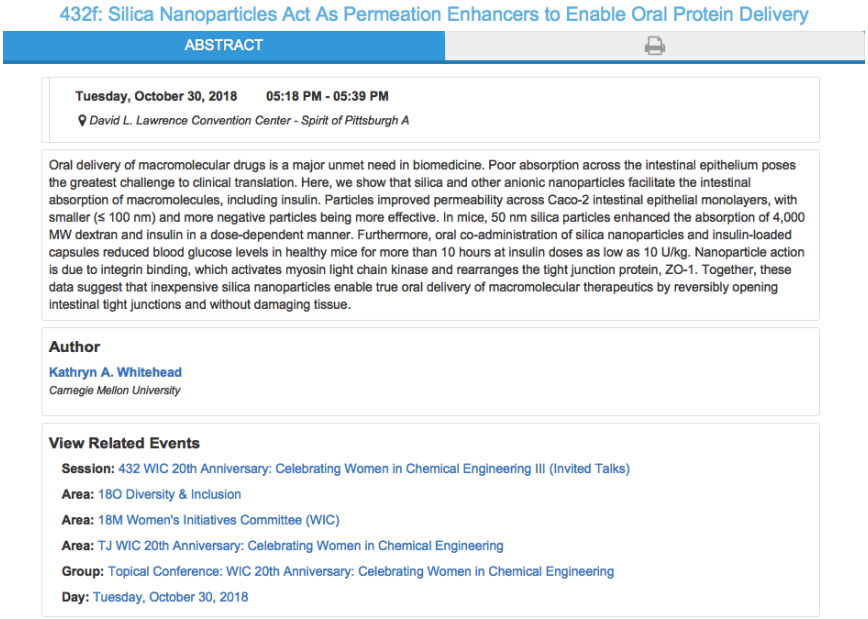
**Session:** 412 Microfluidic and Nanoscale Flows: Separations & Particulates

**Area:** 01J Fluid Mechanics

**Group:** Engineering Sciences and Fundamentals

**Day:** Tuesday, October 30, 2018

## 3.24 432f: Silica Nanoparticles Act As Permeation Enhancers to Enable Oral Protein Delivery https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/541751 @KWhiteheadLab @aichewic



432f: Silica Nanoparticles Act As Permeation Enhancers to Enable Oral Protein Delivery

ABSTRACT

**Tuesday, October 30, 2018 05:18 PM - 05:39 PM**

*David L. Lawrence Convention Center - Spirit of Pittsburgh A*

Oral delivery of macromolecular drugs is a major unmet need in biomedicine. Poor absorption across the intestinal epithelium poses the greatest challenge to clinical translation. Here, we show that silica and other anionic nanoparticles facilitate the intestinal absorption of macromolecules, including insulin. Particles improved permeability across Caco-2 intestinal epithelial monolayers, with smaller (≤ 100 nm) and more negative particles being more effective. In mice, 50 nm silica particles enhanced the absorption of 4,000 MW dextran and insulin in a dose-dependent manner. Furthermore, oral co-administration of silica nanoparticles and insulin-loaded capsules reduced blood glucose levels in healthy mice for more than 10 hours at insulin doses as low as 10 U/kg. Nanoparticle action is due to integrin binding, which activates myosin light chain kinase and rearranges the tight junction protein, ZO-1. Together, these data suggest that inexpensive silica nanoparticles enable true oral delivery of macromolecular therapeutics by reversibly opening intestinal tight junctions and without damaging tissue.

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**View Related Events**

**Session:** 432 WIC 20th Anniversary: Celebrating Women in Chemical Engineering III (Invited Talks)

**Area:** 18O Diversity & Inclusion

**Area:** 18M Women's Initiatives Committee (WIC)

**Area:** TJ WIC 20th Anniversary: Celebrating Women in Chemical Engineering

**Group:** Topical Conference: WIC 20th Anniversary: Celebrating Women in Chemical Engineering

**Day:** Tuesday, October 30, 2018

## 3.25 386i: Graduate Student Award Session: Silica Nanoparticles Enable Oral Delivery of Insulin https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/522312 @Nick__Lamson @KWhiteheadLab

386i: Graduate Student Award Session: Silica Nanoparticles Enable Oral Delivery of Insulin

ABSTRACT

**Tuesday, October 30, 2018 05:22 PM - 05:36 PM**

*David L. Lawrence Convention Center - 328*

**Introduction:** Oral delivery of macromolecular drugs is one of the greatest unmet needs in biomedicine, with poor absorption across the intestinal epithelium posing the most significant challenge to clinical translation[1]. Several studies have demonstrated an improvement in bioavailability of macromolecules conjugated onto or loaded into orally administered nanoparticles[2]. However, this work presents the previously unreported ability of anionic nanoparticles to improve intestinal absorption of large molecules without chemically linking them to the particle delivery vehicles.

**Methods:** A collection of nanoparticles comprising four sizes (20, 50, 100, and 200 nm) and six surface chemistries (bare silica, carboxylated silica, aminated silica, silver, gold, or bare polystyrene) was screened for absorption enhancing effects *in vivo*. Mice received oral gavages of particle suspensions, then either gavages or direct intestinal administration of 4 kDa FITC-labelled dextran (FITC-DX4) or insulin solution. Blood samples were collected and screened for FITC-DX4 or glucose concentration. Mouse intestines were harvested after treatment and processed for histology to assess particle-induced damage. For mechanistic studies, Caco-2 monolayers were used in conjunction with anti-receptor antibodies, enzyme inhibiting peptides, and fluorescent confocal microscopy to investigate tight junction rearrangement.

**Results:** Nanoparticles with negatively charged surfaces (bare silica, carboxylated silica, silver, and gold) improved oral FITC-DX4 accumulation in the blood, while neutral (polystyrene) and positive (aminated) particles showed little to no effect. Further, silica nanoparticles significantly improved the activity of intestinally administered insulin, matching the total hypoglycemic effect of subcutaneously injected insulin. Intestinal insulin bioactivity exhibited dependence on both nanoparticle dose and insulin dose. Further, orally administered, enterically coated insulin capsules induced pronounced and sustained hypoglycemia at the relatively low oral insulin dose of 40 U/kg, and only when administered alongside silica nanoparticles. Histological analysis of particle-treated mouse intestines showed no evidence of necrosis, inflammation, or changes in tissue architecture. However, *in vitro* mechanistic studies revealed that the nanoparticle treatment causes a substantial rearrangement of the tight junction protein ZO-1, an integral component to the barrier function of the epithelial lining, via a previously-identified, integrin- and MLCK-dependent mechanism[3].

**Conclusions and Implications:** This is the first study demonstrating the use of co-delivered but unbound nanoparticles for improving adsorption of orally administered protein drugs. These conclusive *in vivo* results underscore the ability of silica nanoparticles to increase epithelial permeability, enabling the oral delivery of macromolecular drugs without encapsulation or conjugation. Future investigations will include more comprehensive toxicity studies in mice, development of an administration vehicle to better co-localize the particles and insulin, and confirmation of safety and efficacy in higher animal models.

**Acknowledgements:** N. Lamson would like to acknowledge financial support from the NSF Graduate Research Fellowship Program under Grant No. DGE1252522.

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**View Related Events**

**Session:** 386 Biomaterials: Graduate Student Award Session

**Area:** 08B Biomaterials

**Group:** Materials Engineering and Sciences Division

**Day:** Tuesday, October 30, 2018

**3.26 You made it this far, don't forget our Dessert and Cocktail Reception!**




Tuesday, October 30
9:00 – 11:00pm

The Westin Convention Center, Allegheny Grand Ballroom

# 4 #AICHeAnnual Here are a few talks from @CMU_Cheme on Wednesday @ChEnected :tweet:thread

## 4.1 456a: Estimating Uncertain Atmospheric Aerosol Dynamics with an Input Observer https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/527346 @CAPS_CMU @AIChECAST10

456a: Estimating Uncertain Atmospheric Aerosol Dynamics with an Input Observer

ABSTRACT

**Wednesday, October 31, 2018 08:00 AM - 08:19 AM**

*David L. Lawrence Convention Center - 408*


Atmospheric models simulate mass balances of gaseous pollutants and particulate matter – or atmospheric aerosols – in the atmosphere. Typical methods to solve atmospheric inverse problems minimize the model error based on a simplified version of the model [1]. However, these methods may have issues estimating nonlinear aerosol dynamics. In this paper we propose a passivity-based input observer (PBIO) that estimates parameters based on nonlinear conservation laws that can be represented exactly. The method used here is similar to inventory control in which the model states and measurements are projected to a set of inventories. Inventories are positive, additive, and continuously-differentiable variables that follow general conservation laws (like mass or energy).

The PBIO estimates a set of parameters by forcing the model bias, in terms of the inventory variables, to zero. We formulate a quadratic storage function in terms of the model bias. Additionally, the storage function rate of change is constrained to a negative quadratic in terms of the model bias. The parameter update law is derived using these two constraints, leading to a proportional feedback and nonlinear feedforward decoupling on the model bias to update the parameters. This approach reduces the closed-loop system's sensitivity to model mismatch, which can be tuned by adjusting the feedback gain.

Atmospheric models include significant model mismatch due to uncertainty in model inputs (emission fields, meteorology, land type, etc.), model parameters (reaction rate constants, empirical parameters, etc.), and sometimes unknown atmospheric physics and chemistry. The Intergovernmental Panel on Climate Change has characterized how this model mismatch propagates to uncertainty in aerosol's radiative effects in the atmosphere and in clouds [2]. Predicted concentrations of cloud condensation nuclei (CCN), or particles that activate to form cloud droplets, is one of the significant drivers of uncertainty in recent climate change. The particle number size distribution and composition are necessary to predict CCN concentrations. Therefore, in this work, we focus on improving predictions of the number size distribution by estimating uncertain aerosol dynamics based on ground-site measurements.

We choose to estimate scaling factors for the rates of key uncertain processes in the model so that the system is input-affine and is easily inverted for the unknown scaling factors. Key uncertain aerosol processes that we consider are primary organic aerosol emissions, nucleation of new particles from condensable vapors, and condensation of volatile organic compounds (VOCs). Primary aerosol emissions are difficult to monitor and calculate due to the variability in emission fluxes and intensity levels within each type of source. The chemistry that drives nucleation of particles from molecular clusters of compounds possibly including sulfuric acid, water, amines, and organic compounds is still an active area of research. There are large uncertainties in the identity of VOCs and the yield of VOCs partitioning to the particle phase.

Here, we present the design of a PBIO to estimate aerosol dynamics and preliminary applications to a zero-dimensional or box model. We use a zero-dimensional version of the TwO-Moment Aerosol Sectional (TOMAS) microphysical code, which models size-resolved aerosols with the discretized general dynamic equation. We generate synthetic measurements by running TOMAS with a set of "true" rates for primary emissions, nucleation, and SOA production. Then, we start a TOMAS simulation with a set of process rates biased from their "true" value. We quantify the PBIO performance in terms of the bias in the estimated inventory variables, process rates, and the full number size distributions. The methodology developed here will facilitate future work that estimates primary aerosol emissions, nucleation, and VOC condensation rates in a global model based on a measurement network.



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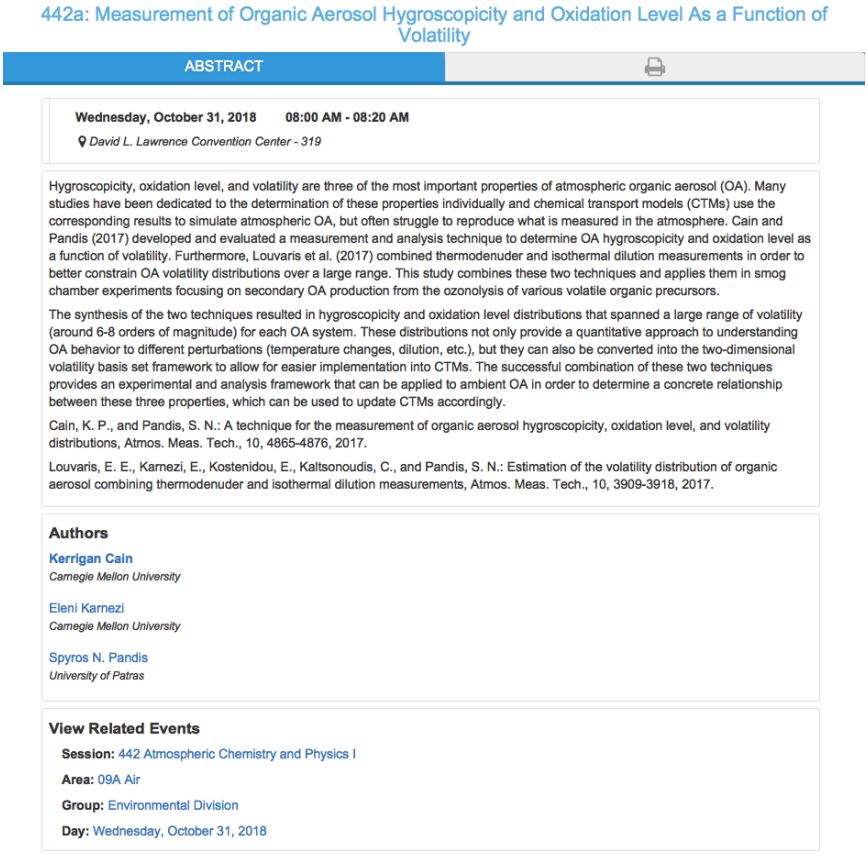
**Session:** 456 Estimation and Control of Uncertain Systems

**Area:** 10B Systems and Process Control

**Group:** Computing and Systems Technology Division

**Day:** Wednesday, October 31, 2018

## 4.2 442a: Measurement of Organic Aerosol Hygroscopicity and Oxidation Level As a Function of Volatility https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528041 @CAPS_CMU



442a: Measurement of Organic Aerosol Hygroscopicity and Oxidation Level As a Function of Volatility

ABSTRACT

**Wednesday, October 31, 2018 08:00 AM - 08:20 AM**

*David L. Lawrence Convention Center - 319*

Hygroscopicity, oxidation level, and volatility are three of the most important properties of atmospheric organic aerosol (OA). Many studies have been dedicated to the determination of these properties individually and chemical transport models (CTMs) use the corresponding results to simulate atmospheric OA, but often struggle to reproduce what is measured in the atmosphere. Cain and Pandis (2017) developed and evaluated a measurement and analysis technique to determine OA hygroscopicity and oxidation level as a function of volatility. Furthermore, Louvaris et al. (2017) combined thermodenuder and isothermal dilution measurements in order to better constrain OA volatility distributions over a large range. This study combines these two techniques and applies them in smog chamber experiments focusing on secondary OA production from the ozonolysis of various volatile organic precursors.

The synthesis of the two techniques resulted in hygroscopicity and oxidation level distributions that spanned a large range of volatility (around 6-8 orders of magnitude) for each OA system. These distributions not only provide a quantitative approach to understanding OA behavior to different perturbations (temperature changes, dilution, etc.), but they can also be converted into the two-dimensional volatility basis set framework to allow for easier implementation into CTMs. The successful combination of these two techniques provides an experimental and analysis framework that can be applied to ambient OA in order to determine a concrete relationship between these three properties, which can be used to update CTMs accordingly.

Cain, K. P., and Pandis, S. N.: A technique for the measurement of organic aerosol hygroscopicity, oxidation level, and volatility distributions, Atmos. Meas. Tech., 10, 4865-4876, 2017.

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**View Related Events**

**Session:** 442 Atmospheric Chemistry and Physics I

**Area:** 09A Air

**Group:** Environmental Division

**Day:** Wednesday, October 31, 2018

## 4.3 449b: Symbolic Regression of Alpha Functions for Cubic Equations of State https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530881

449b: Symbolic Regression of Alpha Functions for Cubic Equations of State

ABSTRACT

**Wednesday, October 31, 2018 08:15 AM - 08:30 AM**

*David L. Lawrence Convention Center - 307*

Since Van der Waal first proposed a formulation for a thermodynamic cubic equation of state, modifications to the thermodynamic cubic equations of state have focused on the alpha function of the attraction term to model the temperature dependence of attraction for real fluids [1]. The alpha function is dependent on reduced temperature and density and acentric factor. Modifications to Peng Robinson and Redlich-Kwong forms of the cubic equation have focused on the functional form of the alpha function fit prior to fitting the attraction and repulsion values for the cubic equation. Traditionally, cubic equations are derived from the critical point which leads to errors in liquid density and in the critical region. Alternatively, when fit to chemical-specific liquid density and vapor pressure data, the equations fail to represent the critical constraints [2].

To improve the development of cubic equations and their regression, we focus on a data-driven constrained symbolic regression to simultaneously determine the model form of the alpha function and fit parameters of the cubic equation. Symbolic regression learns both the model structure and parameters to model a data set, unlike traditional regression that limits the scope of the regression to a fixed functional form [4]. The regression only requires the specification of a set of operators and operands (+, -, *,÷, exp(·), log(·), $(\cdot)^2$, $(\cdot)^3$, $\sqrt{\cdot}$, etc.) to flexibly develop new functional forms that accurately represent the data. Symbolic regression has typically been performed with genetic programming [4], but recent developments in applying global deterministic approach have shown improved fitting metrics, such as sum of squared error and other information criteria [3]. We use this approach to apply symbolic regression to pure fluids to determine new alpha function forms for cubic equations of state and compare their accuracy to the data, specifically in liquid density and supercritical regions, and adherence to theoretical thermodynamic properties with other alpha modifications of cubic equations of state.

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**View Related Events**

**Session:** 449 Data-Driven Screening of Chemical and Materials Space

**Area:** 04 Education

**Area:** 21 Computational Molecular Science and Engineering Forum

**Area:** 01A Thermodynamics and Transport Properties

**Group:** Engineering Sciences and Fundamentals

**Day:** Wednesday, October 31, 2018

## 4.4 452c: Erythrocytes As Carriers of Immunoglobulin Based Therapeutic Drugs @DrAlanRussell https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/535691

452c: Erythrocytes As Carriers of Immunoglobulin Based Therapeutic Drugs

ABSTRACT

**Wednesday, October 31, 2018 08:36 AM - 08:54 AM**

*Westin Convention Center - Cambria*

**Erythrocytes as Carriers of Immunoglobulin Based Therapeutic Drugs**

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**ABSTRACT**

The spectacular success of immunoglobulin-based therapeutics in treating a wide range of diseases has heightened the need to further enhance their efficacy and lifetime, while diminishing any deleterious side effects. We have explored a range of chemistries that allow us to couple antibodies to red blood cells. The couplings can enable controlled delivery from the cell surface to plasma or the antibody can remain permanently attached to the membrane. For all methods we can attach on the order of $10^5$ antibodies per red blood cell (human and rat) and these cell-displayed antibody arrays were shown to be stable in physiological buffer for more than 60 days. In a rodent model, our preliminary results showed that erythrocytes with arrays of antibodies were stable and functional. We are currently investigating the impact of covalently attaching antibody arrays on the circulation lifetime and the antigen-scavenging efficiency of red blood cells in a rat model. In summary, we believe that using our bioconjugation strategies, we could build a potentially long-lived, biocompatible functional red blood cell that could efficiently remove antigens from circulation.

**Acknowledgments:**

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**View Related Events**

**Session:** 452 Drug Delivery I: Biologics

**Area:** 08B Biomaterials

**Area:** 15D Engineering Fundamentals in Life Science

**Group:** Food, Pharmaceutical & Bioengineering Division

**Day:** Wednesday, October 31, 2018

## 4.5 441c: On Solving Nonconvex Two-Stage Stochastic Programs with Generalized Benders Decomposition https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/524155 @AIChECAST10

441c: On Solving Nonconvex Two-Stage Stochastic Programs with Generalized Benders Decomposition

ABSTRACT

**Wednesday, October 31, 2018 08:38 AM - 08:57 AM**

*David L. Lawrence Convention Center - 409*

Stochastic programming is one of the major mathematical frameworks in the area of decision-making under uncertainty. A special case is two-stage stochastic programming in which stage-one decisions correspond to "here and-now" decisions that must be taken before the uncertainties are revealed; stage-two decisions correspond to "wait and see" decisions or recourse decisions that can be adjusted in the face of the realization of the uncertainties.. Solving the deterministic equivalent of the two-stage stochastic programs directly can be prohibitive when the number of scenarios is large because the computational time generally grows exponentially with the number of scenarios. Generalized Benders decomposition [1] can be applied to problems with continuous variables and convex constraints in the second stage. However, it cannot be applied directly to the problems with nonconvex second stage constraints.

In the area of nonconvex nonlinear stochastic programming, Li et al [2] has proposed a NGBD algorithm for problems with pure binary first stage variables. Cao and Zavala [3] propose a scalable algorithm that can globally optimize nonconvex two-stage stochastic programs. Ogbe and Li [4] propose a joint decomposition algorithm for general mixed-integer nonlinear nonconvex stochastic programs.

In this work, we further explore algorithms based on generalized Benders decomposition for solving two-stage stochastic programs with nonconvex constraints and mixed-integer variables in both first and second stage decisions. The validity and tightness of different types of cuts including Benders cuts, strengthened Benders cuts, and Lagrangean cuts are studied. Moreover, different convexification schemes for nonconvex problems are tested under this framework. In order for the algorithm to have finite convergence, a spatial branch and bound search is performed on the first stage decisions.

The algorithm is tested with several applications in process systems engineering, that includes the generalized pooling problem under uncertainty. The decomposition algorithm is benchmarked against the full-space solution with state-of-art global solvers, such as BARON, ANTIGONE.

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**View Related Events**

**Session:** 441 Advances in Optimization Under Uncertainty

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Wednesday, October 31, 2018

## 4.6 441d: New Developments in Flexibility Analysis in the Framework of Design Space Definition https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/529193 @LillyPad @AIChECAST10

441d: New Developments in Flexibility Analysis in the Framework of Design Space Definition

ABSTRACT

**Wednesday, October 31, 2018 08:57 AM - 09:16 AM**

*David L. Lawrence Convention Center - 409*


The importance of the Design Space (DS) definition lies in the assurance of quality, a key goal of Quality by Design, and in the broadening from an acceptable operating set point, to a collection of tolerable operating regions. Working within the DS is considered acceptable and not additional actions are needed (e.g., notification). Operating outside the DS represents a deviation and would normally initiate an investigation. Defining properly the limits of the design space is vital to provide manufacturing flexibility and quality assurance. Even though process DS has been only recently formally introduced in the pharmaceutical industry, it is not a completely new concept. Identifying design and operating constraints as well as the necessary control scheme for a process are concepts and ideas that have been extensively studied in other industries and literature (Boukouvala *et al.*, 2017). The use of concepts introduced by the process system literature, such as process feasibility and flexibility, have proven to be useful in the DS identification (García-Muñoz *et al.*, 2015).

Flexibility addresses the guaranteed feasibility of operation of a plant over a range of conditions, with the ultimate goal being on how to design a process for guaranteed feasible operation over a specified parameter range (Grossmann *et al.*, 2014). There are three main problems to address: the flexibility test problem, flexibility index calculation and the design under uncertainty with flexibility constraints. First, the flexibility test problem (Halemane and Grossmann, 1983) for a given design consists in determining whether by proper adjustment of the controls variables the process constraints hold for any realization of uncertain parameters. To indicate how much flexibility can actually be achieved in a given design (Swaney and Grossmann, 1985), the flexibility index is defined as the largest scaled factor that multiplies the deviation of the uncertain set for which the inequalities hold. As for design optimization problem, the selection of the design variables is involved so as to minimize cost and either satisfy the flexibility test, or maximize the flexibility index, where the latter problem gives rise to a multi-objective optimization problem.

However, these formulations are based implicitly on the assumption that control variables can compensate any variation in the uncertain parameter set and that during operation stage uncertain parameters can be determined with enough precision. Ostrovsky *et al.* (2004) and Rooney and Biegler (2004) extended the analysis by taking into account the level of parametric uncertainty in the mathematical models at the operation stage, by grouping the uncertain parameters, $\in \Theta$, into two types: unmeasured and measured. The flexibility constraint was then extended to account for unmeasured process variability, . The flexibility constraint was re-defined as a multi-level optimization problem.

In this work, we propose new formulations for the extended flexibility analysis. The basic idea relies on recursively reformulating the inner optimization problems by the Karush-Kuhn-Tucker conditions, and with a mixed-integer representation of the complementarity conditions to solve the resulting multilevel optimization problem. Three types of models with uncertain parameters are tackled and solved with the proposed strategy: 1) linear programming model, 2) nonlinear programming model with monotonic variation of unmeasured uncertain, and finally, 3) nonlinear programming model. Similar mixed-integer formulations to the traditional flexibility analysis are obtained for the first two cases, whereas a novel nonconvex MINLP is obtained for the latter case. Specifically, for the nonlinear programming model, we study a thermal de-protection reaction performed in a CSTR followed by an evaporation step. Numerical results are compared to the traditional flexibility analysis (Grossmann and Floudas, 1987), leading to larger values of constraint violation and lower values of flexibility indices due to the fact that the recourse action can only compensate for measured uncertain parameter variations.



Keywords

Flexibility Analysis, Quality by Design, Design Space


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**View Related Events**

**Session:** 441 Advances in Optimization Under Uncertainty

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

## 4.7 472f: Identification of Optimally Stable Nanoparticle Geometries Via Mathematical Optimization and Density-Functional Theory https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/537457 @mpourmpakis_lab @aiche_cre

472f: Identification of Optimally Stable Nanoparticle Geometries Via Mathematical Optimization and Density-Functional Theory

ABSTRACT

**Wednesday, October 31, 2018 09:40 AM - 10:00 AM**

*David L. Lawrence Convention Center - 415*

One of the key research objectives in the area of transition metal nanoparticles is to determine the most stable structure for particles of a given size (mass) and composition. In this work, we propose a mathematical optimization based modeling framework, coupled with density functional theory (DFT) calculations, for determining the proven minimum energy structures of three-dimensional, mono-metallic nanoclusters of a given size. More specifically, we apply this framework to mono-metallic copper (Cu) nanoclusters, as previous work has shown potential uses for mono- and bi-metallic Cu nanoparticles in $CO_2$ activation and reduction [1],[2],[3].

In our work, we consider the global minimum energy Cu nanocluster of a given size to be the one that attains the maximum average dimensionless cohesive energy, which is the average cohesive energy normalized to the bulk energy. The cohesive energy measures the strength of inter-atomic bonding between atoms and the overall stability. Thus, maximizing the average dimensionless cohesive energy leads to the most stable, ground-state nanocluster geometry. Our proposed model uses a cohesive energy function, first proposed by Tomanek et al., which depends only on the geometric descriptor of coordination number (CN) [4]. Because CN is a local geometric descriptor of the nanoparticle (NP), it can be encoded within a rigorous, "bottom-up" design optimization model. In fact, previous work has successfully utilized CN as a local descriptor to design transition metal surfaces via a mixed-integer linear programming (MILP) [5], a class of mathematical optimization problems that can be solved readily and reliably with appropriate numerical optimization techniques. The current work serves as a first step in extending such tools for targeted nanoparticle design in three dimensions. It should be highlighted that the proposed method deviates from the existing literature, inasmuch as existing methods for determining lowest energy small nanoclusters typically involves heuristic search of the design space and evaluation of incumbent solutions via inter-atomic potentials [6],[7],[8],[9]. However, global minimum energy nanoclusters reported in this way are only considered "putative," as the search methods utilized provide no mathematical proof of optimality. In contrast, the use of MILP based optimization techniques proposed in this work guarantees the optimality of the final designs.

However, the cohesive energy model originally proposed by Tomanek el al. also includes a repulsive potential energy term. Whereas previous work has shown that neglecting the repulsive contribution for sufficiently large nanoclusters compares favorably to the DFT calculated cohesive energy [10], doing so at small NP sizes is not appropriate due to surface contraction and quantum effects. In order to create an optimization model that is accurate for small NPs, we first assemble a large collection of nanoparticles of given size that feature a wide variety of coordination number distributions, and then use first-principles DFT calculations to evaluate their cohesive energy. We then perform linear regression to elucidate a correction term that can be represented in our MILP optimization framework. This way, optimally cohesive NPs at small sizes can be correctly and quickly predicted.

Following the above methodology, a pool of optimal and near-optimal Cu NPs of different small sizes (up to 50 atoms) was collected, and single-point DFT calculations (using the PBE exchange-correlation functional [11]) were performed to determine the expected cohesive energies for these structures. Using this data, a linear regression model was developed in order to identify corrections for the cohesive energy contribution of a single atom in a given Cu NP. These corrective factors were then incorporated into the mixed-integer optimization model, which determined highly cohesive nanoclusters for each given size, establishing morphological trends related to stability in NPs of any size. The results confirm various "magic-number" designs that are well-known to be highly stable, as well as reveal a number of unintuitive designs that were not previously reported in the literature, yet are confirmed to be highly stable after DFT geometry optimizations. These particles can serve as model particles for a multitude of follow-up studies within the realm of computational chemistry to include, among others, adsorption and catalytic activity studies. Finally, by collecting sequences of most cohesive designs for given NP sizes, our results can also be used to determine expected distributions of clusters that arise during experimental synthesis.

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## 4.8 452g: Lipid Nanoparticle Formulations for the Synergistic Co-Delivery of siRNA and mRNA #Invited https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/525835 @KWhiteheadLab

452g: Invited Speaker: Lipid Nanoparticle Formulations for the Synergistic Co-Delivery of siRNA and mRNA

ABSTRACT

**Wednesday, October 31, 2018 09:48 AM - 10:28 AM**

*Westin Convention Center - Cambria*

Despite the promise of RNA therapeutics, progress towards the clinic has been slowed by the difficulty of delivering RNA drugs, such as short interfering RNA (siRNA) and messenger RNA (mRNA), into cellular targets within the body. RNA therapeutics are large ($10^4$ – $10^6$ g/mol) and negatively charged; they do not have favorable biodistribution properties in vivo nor an ability to cross the cellular membrane of target cells. In response to these challenges, our lab has created biodegradable, ionizable lipid-like materials called 'lipidoids' that readily formulate into nanoparticles containing RNA. Lipidoids efficiently manipulate gene expression in a variety of biological systems, including hepatocytes, white blood cells and tumors.

This talk will focus on the simultaneous delivery of siRNA and mRNA. To facilitate the treatment of diseases associated with aberrant gene upregulation and downregulation, we sought to co-formulate siRNA and mRNA in a single lipidoid nanoparticle (LNP) formulation. We accommodated the distinct molecular characteristics of mRNA and siRNA in a formulation consisting of an ionizable biodegradable lipidoid, cholesterol, DSPC, DOPE, and PEG-lipid. Surprisingly, the co-formulation of siRNA and mRNA in the same LNP enhanced the efficacy of both drugs *in vitro* and *in vivo*. Compared to LNPs encapsulating siRNA only, co-formulated LNPs improved Factor VII gene silencing in mice from 44 to 87% at an siRNA dose of 0.03 mg/kg. Co-formulation also improved mRNA delivery, as a 0.5 mg/kg dose of mRNA co-formulated with siRNA induced three times the luciferase protein expression compared to when siRNA was not included.

As not all gene therapy applications require both RNA drugs, we sought to extend the synergy of co-formulated LNPs to formulations encapsulating only a single type of RNA. We accomplished this by substituting the "helper" RNA with a negatively charged polymer. LNPs containing negatively charged polymer mediated the same level of protein silencing or expression as standard LNPs using 2-3 fold less RNA. For example, LNPs formulated with and without polymer induced 50% Factor VII silencing at siRNA doses of 0.01 and 0.03 mg/kg, respectively. Together, these studies demonstrate potent co-delivery of siRNA and mRNA and show that inclusion of a negatively charged "helper polymer" enhances the efficacy of LNP delivery systems.

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**View Related Events**

**Session:** 452 Drug Delivery I: Biologics

**Area:** 08B Biomaterials

**Area:** 15D Engineering Fundamentals in Life Science

**Group:** Food, Pharmaceutical & Bioengineering Division

**Day:** Wednesday, October 31, 2018

## 4.9 530a: Scheduling with Preemption https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/533563 @ABBgroupnews @AIChECAST10

530a: Scheduling with Preemption

ABSTRACT

**Wednesday, October 31, 2018 12:30 PM - 12:49 PM**

*David L. Lawrence Convention Center - 409*

The most well-known scheduling formulations from the Process Systems Engineering (PSE) community do not account for the possibility of processing tasks to be interrupted. Yet, preemption is often encountered in daily practice, with the ISA-95 standard, which can act as a data-exchange platform for production scheduling [1], including planned break periods (e.g., due to preventive maintenance, weekends), number of stops allowed per task, minimum duration of each partial task, and possible penalties for interrupting a task.

Models allowing for preemption exist in the context of project scheduling. In [2], an activity with discrete duration *d*can be interrupted up to *d*times, with the time(s) of interruption being determined by a genetic algorithm. In this paper, we consider a more constrained form of preemption, where tasks can be interrupted at a given point in time, provided that they continue immediately after the break. Long tasks may be interrupted multiple times.

We will show how to handle preemption in discrete- and continuous-time formulations. On the one hand, we focus on the Resource-Task Network (RTN) discrete-time formulation [3], which is known to be very tight and better at handling events such as time-dependent electricity pricing [4]. The required changes occur at the level of the structural parameters and excess resource balance constraints, with the model variables remaining the same (it is straightforward to extend a State-Task Network based discrete-time formulation, such as the one in [5-6], to handle preemption in a similar way). On the other hand, we use Generalized Disjunctive Programming (GDP) [7-9] and a compact convex hull reformulation to derive the complex timing constraints of the continuous-time formulation. These are incorporated into a model for multistage batch plants that relies on the general precedence concept [10-11].

The performance of the proposed formulations is evaluated on an industrial problem featuring 24 orders and 4 stages (96 tasks) [4] considering the objective of makespan minimization. The results show a better performance for the discrete-time formulation (DT), which is able to solve the problem to optimality in less than one hour, whereas the continuous-time model (CT) can only handle 14 orders in the same computational time. This, despite DT having a worse relaxation and generating MILPs that are one order of magnitude larger in size. Preemption allows to reduce the makespan without bringing too much complexity into the model. In fact, CT can actually be solved faster compared to the non-preemptive case, probably due to the lower integrality gap.

Future work will deal with single stage batch plants with parallel units, to see if the ranking is altered when using a tighter continuous-time formulation, one with multiple time grids [9] and without super big-M constraints.

**Acknowledgments**: Financial support from Fundação para a Ciência e Tecnologia (FCT) through projects IF/00781/2013 andUID/MAT/04561/2013.

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**Session:** 530 Planning and Scheduling I

**Area:** 10B Systems and Process Control

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Wednesday, October 31, 2018

## 4.10 530f: Batch Scheduling with Quality-Based Changeovers @bbrunaud @DowChemical https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528580 @AIChECAST10

530f: Batch Scheduling with Quality-Based Changeovers

ABSTRACT

**Wednesday, October 31, 2018 02:05 PM - 02:24 PM**

*David L. Lawrence Convention Center - 409*

Batch scheduling is a frequent and complex operation performed in all process industries. The combinatorial nature of the problem difficults finding non intuitive solutions to capture additional value. Because of the high frequency of these operations, even small improvements can quickly translate into millions of dollars in savings. The planning horizon spans from two days to a few weeks. In the case of discrete-time models, the horizon is divided in hourly periods. The resulting models are difficult to solve, thereby motivating intensive research in this area[1]. One of the features that adds more complexity to the problem is the presence of sequence-dependent changeovers (SDC)[2], in which the length of the cleaning task performed between two successive processing tasks depends on the sequence in which the processing tasks are performed. For example, if in a given reactor, task B is performed after task A, then a cleaning of 2 hours might be required between the tasks. However, if task A is performed after task B, maybe the cleaning time could be longer, not required, or the transition from B to A might be even forbidden. The cleaning is frequently required to prevent cross contamination between products.

In many applications, like the production of chemicals of the same product family, the cleaning can be avoided by making sure enough batches of the second product are performed in a row, and any impurities from the previous product are diluted enough such that the resulting product meet quality standards. We refer to this type of transitions as quality-based changeovers (QBC). This kind of transitions have not been addressed in the Process Systems Engineering (PSE) literature. The required models need to consider the option of performing either a quality-based changeover or a traditional cleaning changeover. We implement and test quality-based changeovers in three of the major general purpose scheduling frameworks in PSE, namely State-Task-Network (STN)[3], Resource-Task-Network (RTN)[4], and Unit-Operation-Port-State-Superstructure (UOPSS)[5]. Implementing in the three frameworks allows to compare their computational efficiency, and main features, including extensibility to accommodate novel features such as quality based changeovers. The implementations are tested in a small process network, and in an industrial-sized network. For the second case, a richer objective function is proposed, including minimization of priority-dependent backlog, changeover time, and deviation from a previously determined plan.

The results obtained give valuable insights on the three frameworks. Some highlights of the comparisons performed are:

QBC can seamlessly integrated in the STN and UOPSS frameworks by extension of their SDC constraints. On the other hand, RTN requires the addition of many auxiliary tasks and resources.

QBC reduces computational burden to the problem by tightening the SDC constraints.

Since few SDC constraints are active at the optimum, they can be effectively handled with use of lazy constraints.

UOPSS requires the addition of less SDC constraints, which makes it more efficient for large problems.

The modeling extensions allow to better represent industrial problems, while the insights obtained yield guidelines for model selection. These two aspects combined represent an important contribution to the adoption optimization of scheduling operations.

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**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Wednesday, October 31, 2018

## 4.11 544at Prediction of Surface Energies for Complex Pt Structures from Coordination Number and Generalized Coordination Number https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/537389 @UlissiGroup @aiche_cre

544at: Prediction of Surface Energies for Complex Pt Structures from Coordination Number and Generalized Coordination Number

ABSTRACT

**Wednesday, October 31, 2018 03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

**Prediction of Surface Energies for Complex Pt Structures from Coordination Number and Generalized Coordination Number**

*Wen Zhong, Christopher L. Hanselman, Kevin Tran, Chrysanthos E. Gounaris, Zachary W. Ulissi*

Transition metal nanoparticles can show enhanced catalytic activity over single surfaces. Nanoparticle site activity has been shown to be correlated with simple descriptors like the generalized coordination number [1] . Textured or patterned surfaces can be formulated as an optimization problem and can show even higher activities but the stability of these surfaces needs to be considered [2]. We use simple regressions to directly predict the surface stability from the same generalized coordination number descriptor, allowing stability to be added to the optimization problem. We compare with 106 explicit DFT surface energy calculations for patterned surfaces, achieving an RMSE of 0.0023 eV/A^2 and 0.0020 eV/A^2 for coordination number and generalized coordination number descriptors respectively. We show that this model can also apply to larger and more complicated surfaces not in the training set. We also find that 80% of the patterned structures have surface energies that are greater than typical high-energy nanoparticle facets. We use these predictions to identify a number of pareto-optimal surfaces with both high activity and stability.

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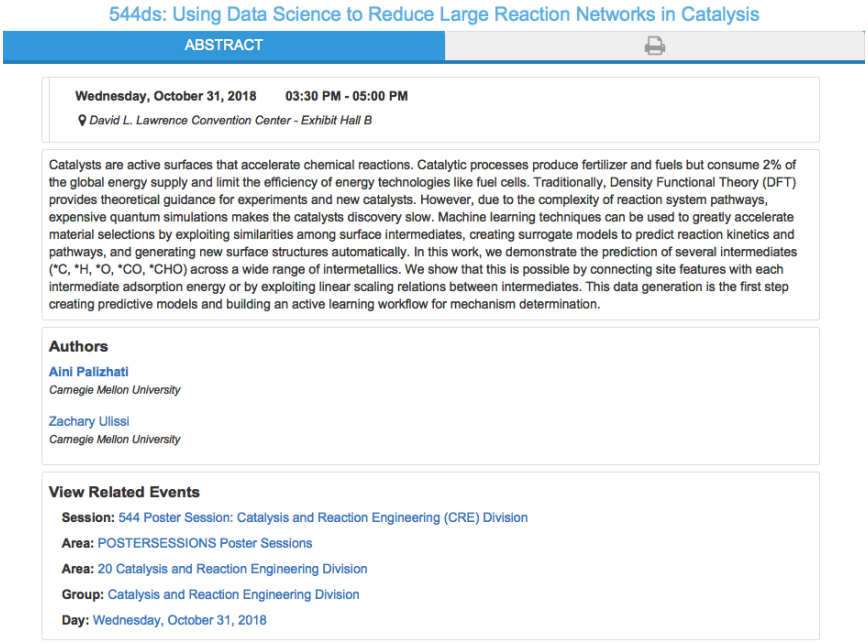
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**Day:** Wednesday, October 31, 2018

## 4.12 544ds Using Data Science to Reduce Large Reaction Networks in Catalysis https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/531716 @UlissiGroup @aiche_cre



544ds: Using Data Science to Reduce Large Reaction Networks in Catalysis

ABSTRACT

**Wednesday, October 31, 2018** **03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

Catalysts are active surfaces that accelerate chemical reactions. Catalytic processes produce fertilizer and fuels but consume 2% of the global energy supply and limit the efficiency of energy technologies like fuel cells. Traditionally, Density Functional Theory (DFT) provides theoretical guidance for experiments and new catalysts. However, due to the complexity of reaction system pathways, expensive quantum simulations makes the catalysts discovery slow. Machine learning techniques can be used to greatly accelerate material selections by exploiting similarities among surface intermediates, creating surrogate models to predict reaction kinetics and pathways, and generating new surface structures automatically. In this work, we demonstrate the prediction of several intermediates (*C, *H, *O, *CO, *CHO) across a wide range of intermetallics. We show that this is possible by connecting site features with each intermediate adsorption energy or by exploiting linear scaling relations between intermediates. This data generation is the first step creating predictive models and building an active learning workflow for mechanism determination.

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**Day:** Wednesday, October 31, 2018

## 4.13 544eu: Impact of Polymer-Based Protein Engineered -Chymotrypsin on Enantioselective Transesterification in Organic Media @DrAlanRussell @MatyPolymerLab https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/535932 @aiche_cre

544eu: Impact of Polymer-Based Protein Engineered α-Chymotrypsin on Enantioselective Transesterification in Organic Media

ABSTRACT

**Wednesday, October 31, 2018 03:30 PM - 05:00 PM**

*David L. Lawrence Convention Center - Exhibit Hall B*

Enzymes are able to catalyze reactions, such as transesterification or aminolysis with impressive stereo- or enantio- selectivity, in essentially organic media and ionic liquids. However, enzyme activity in organic media is often several orders of magnitude below that in aqueous conditions due mainly to lack of solubility and inefficient substrate binding. Many different strategies have been used to increase solubility and enzymatic activity in organic solvents. Specifically, conjugation of a variety of enzymes with synthetic polymers resulted in better solubility and activity increases. However, conjugates often exist as aggregates in organic media.

We have been developing next generation conjugates using polymer-based protein engineering to nano-armor proteins with densely conjugated polymers on the surface of proteins. We have previously examined the activity and stability response of nano-armor enzymes to a variety of stressors (pH and temperature) in aqueous media. Specifically, chymotrypsin-poly(2-(dimethylamino)ethyl methacrylate) (CT-pDMAEMA) prepared by atom transfer radical polymerization, showed pH-dependent activity and increased thermal stability.

Due to the amphiphilic quality of pDMAEMA and the "nano-armor" around the enzymatic core, CT-pDMAEMA conjugates can be molecularly dissolved and active in organic solvents. We first determined the effect of polymer conjugation on solubility of the conjugates in organic solvents. The hydrodynamic diameter of CT-pDMAEMA ($D_h$ ~ 29 nm) was measured using dynamic light scattering in acetonitrile, which was equivalent to that in aqueous conditions ($D_h$ ~ 34 nm), indicating that CT-pDMAEMA readily dissolves at the molecular scale in acetonitrile.

We have examined the kinetics of chymotrypsin-catalyzed transesterification and hydrolysis of *N*-acetyl L-phenylalanine thiophenyl ester (L-APTE). Specifically, we have investigated the effect of water content on the rate of reaction ($k_{cat}$) and substrate affinity ($K_M$). Since thiophenol is a byproduct of both the transesterification and hydrolysis reactions, we used 4,4'-dithiopyridine as a colorimetric indicator to quantify product formation. Thus, the resulting apparent $k_{cat}$ and $K_M$ values corresponded to the total rate of substrate removal. The conjugate exhibited good substrate binding with L-APTE ($K_M$ ~ 17 mM). The activity ($k_{cat}$) of CT-pDMAEMA increased with increasing water concentration, while the consistent $K_M$ values indicate that L-APTE affinity was not affected by water. The conjugate had an activity (peak activity 330 μM/min/mg enzyme) many orders of magnitude higher than that of the insoluble native chymotrypsin. In this presentation we extend our analysis to discuss the impact of molecular dissolution of nano-armored enzymes on enantioselectivity.

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## 4.14 559c: Strawberry Polyphenols As Intestinal Permeation Enhancers for Oral Drug Delivery https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/534763 @Nick_-Lamson @beccaleeball @KWhiteheadLab

559c: Strawberry Polyphenols As Intestinal Permeation Enhancers for Oral Drug Delivery

ABSTRACT

**Wednesday, October 31, 2018 04:06 PM - 04:24 PM**

*Westin Convention Center - Cambria*


**Introduction:** Oral delivery of biologic drugs, especially therapeutic proteins and nucleic acids, is highly desired due to its ease of use and improved patient compliance when compared with injection[1]. However, successful oral formulations must overcome poor transport of large therapeutic molecules from the intestines to the bloodstream. Many studies have examined the use of chemical permeation enhancers to improve the absorption of biologics through the intestinal epithelium[2]. However, effective species tend to induce cytotoxicity or damage the intestinal mucosa. Given the lingering need for efficacious permeation enhancers that are well-tolerated by the intestines, this work presents the results of screening an extensive, food-base library for natural chemicals that improve intestinal absorption of large molecules. Of these, refined strawberry extracts are identified as the most promising, achieving successful oral macromolecule delivery in treated mice.

**Methods:** A library of crude extracts prepared from over 100 fruits, vegetables, herbs, and fungi was screened for cytotoxicity and absorption enhancing effects on Caco-2 intestinal barrier models. The most effective and well-tolerated permeation enhancing extract (strawberry) was chosen for separation via sequential adsorption/desorption on an acrylic ester resin, yielding active polyphenol fractions. To test *in vivo* efficacy of the materials, mice received oral gavages of strawberry polyphenols, then gavages of 4 kDa FITC-labelled dextran (FITC-DX4) or 40 kDa FITC-dextran (FITC-DX40). Blood samples were collected and screened for FITC fluorescence. For mechanistic studies, Caco-2 monolayers were used in conjunction with fluorescent confocal microscopy to investigate tight junction rearrangement.

**Results:** Of the crude food extracts examined, the vast majority did not induce cytotoxicity in intestinal cells. Those that did significantly reduce cell viability mostly contained portions of the foods not typically consumed in large quantities (e.g. citrus rinds). Among the non-toxic extracts, most did not significantly change the barrier properties of Caco-2 monolayers, though some did significantly increase permeability. Of these, strawberry was the most effective and was chosen for further separation and investigation. Isolation of the polyphenol contents from the strawberry extract via Amberlite resin yielded fractions that were highly potent permeabilizers of Caco-2 barriers. When combined and orally administered to mice, these fractions boosted the intestines-to-bloodstream transport of both FITC-DX4 and FITC-DX40 by over 100%. Further, *in vitro* mechanistic studies revealed that the nanoparticle treatment does cause a rearrangement of the cytoskeletal protein actin, as well as the tight junction protein ZO-1, an integral component to the barrier function of the intestinal epithelium.

**Conclusions and Implications:** This is the first study demonstrating the use of nontoxic, food-derived chemicals for improving adsorption of orally administered protein drugs. The extensive library of crude extracts examined demonstrated that most foods do not appreciably change intestinal permeability. Of the few that do, strawberry was the most potent permeation enhancer, and demonstrated increased potency as extracts were refined and screened for oral drug delivery capabilities in mice. These conclusive *in vivo* results demonstrate the ability of strawberry polyphenols to effectively increase epithelial permeability, enabling the oral delivery of macromolecular drugs. Future investigations will include delivery of active protein drugs and comprehensive safety studies in mice and higher animal models.



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**Group:** Food, Pharmaceutical & Bioengineering Division

**Day:** Wednesday, October 31, 2018

## 4.15 583g: Optimal Mass Exchanger Network Synthesis Using a 2-Step Hybrid Algorithm Including Packed Column Design https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/535724 @mchlshort @UCT_news @AIChECAST10

583g: Optimal Mass Exchanger Network Synthesis Using a 2-Step Hybrid Algorithm Including Packed Column Design

ABSTRACT

**Wednesday, October 31, 2018 05:24 PM - 05:43 PM**

*David L. Lawrence Convention Center - 409*

A method for the synthesis of mass exchanger networks (MENs) is presented that focuses on including detailed packed column design considerations at the network optimization level to deliver designs that are optimal and more realistic. Most approaches to simultaneous optimization of MENs have been based upon a stage-wise superstructure with simplified representations of the individual mass exchangers. Even with these simplified representations, the MINLP is still non-convex and computationally difficult to solve with modern deterministic solvers. Additionally, it is not possible to guarantee that when a solution is found that it is globally optimal.

The method developed in this study, uses a version of this aforementioned stage-wise MINLP formulation in the first step. Once the solution to this is found, a secondary nonlinear programming (NLP) step, based on orthogonal collocation on finite elements, is performed to find the optimal column dimensions, packing sizes, fluid velocities, and pressure drops, based on the flows and stream matches from the network optimization. Once the new individual column designs are found, the MINLP is then updated with a series of correction factors based on the solution of the detailed designs. This procedure repeats between the MINLP and NLP sub-optimization until the differences between the solutions converge or until a maximum number of iterations is reached.

The methodology, presented in detail in the work of Short, et al. (2018), is moved from the GAMS environment into PYOMO, allowing for powerful new ways to enhance the algorithm. The previous methodology's correction factor determination is improved upon in order to speed up and increase the chances of convergence. Additionally, through the application of different MINLP solvers, larger examples that were previously not able to be solved are included.

Reference: Short, M., Isafiade, A.J., Biegler, L.T.,Kravanja, Z., 2018, Synthesis of mass exchanger networks in a two-step hybrid optimization strategy, Chemical Engineering Science, 178, 118 - 115.

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**Area:** 10A Systems and Process Design

**Group:** Computing and Systems Technology Division

**Day:** Wednesday, October 31, 2018

# 5 #AICHeAnnual Here are a few talks from @CMU_Cheme on Thursday @ChEnected :tweet:thread

## 5.1 611g Practical Applications of Machine Learning to Catalyst Design and Discovery https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/536780 @UlissiGroup @aichecomsef



611g: Practical Applications of Machine Learning to Catalyst Design and Discovery

ABSTRACT

**Thursday, November 01, 2018** **09:30 AM - 09:45 AM**

*David L. Lawrence Convention Center - 308*

Heterogeneous catalysis is fundamental to chemical industry and consumes several percent of the entire global energy supply. Reducing this usage and enabling next-generation energy solutions such as direct conversion of CO2 to fuels requires the design of new catalysts with optimal activity, selectivity, and stability. Scientific computing advances have enabled electronic structure codes to aid in this design process but fundamental limitations make it unlikely that direct simulation of macroscopic catalysts will be possible. The huge design space can be reduced by recognizing similarities in materials (developing structural fingerprints) and adopting regression tools from the systems engineering or machine learning communities to provide useful surrogate models as a guide for full-accuracy calculations. I will discuss how we approach the coupled problems of database generation, calculation automation, and active learning to search for intermetallic catalysts with near-optimal properties. We distinguish these approaches from typical high throughput screens because the final calculations are functionally identical to the standard computational chemistry workflow that would be used for individual surfaces. We demonstrate the successful identification of more than 100 new candidates for the CO2 reduction reaction, and show that the final predictive models can aid in the determination of the active facet in polycrystalline catalysts.

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**Area:** 10 Computing Systems and Technology Division

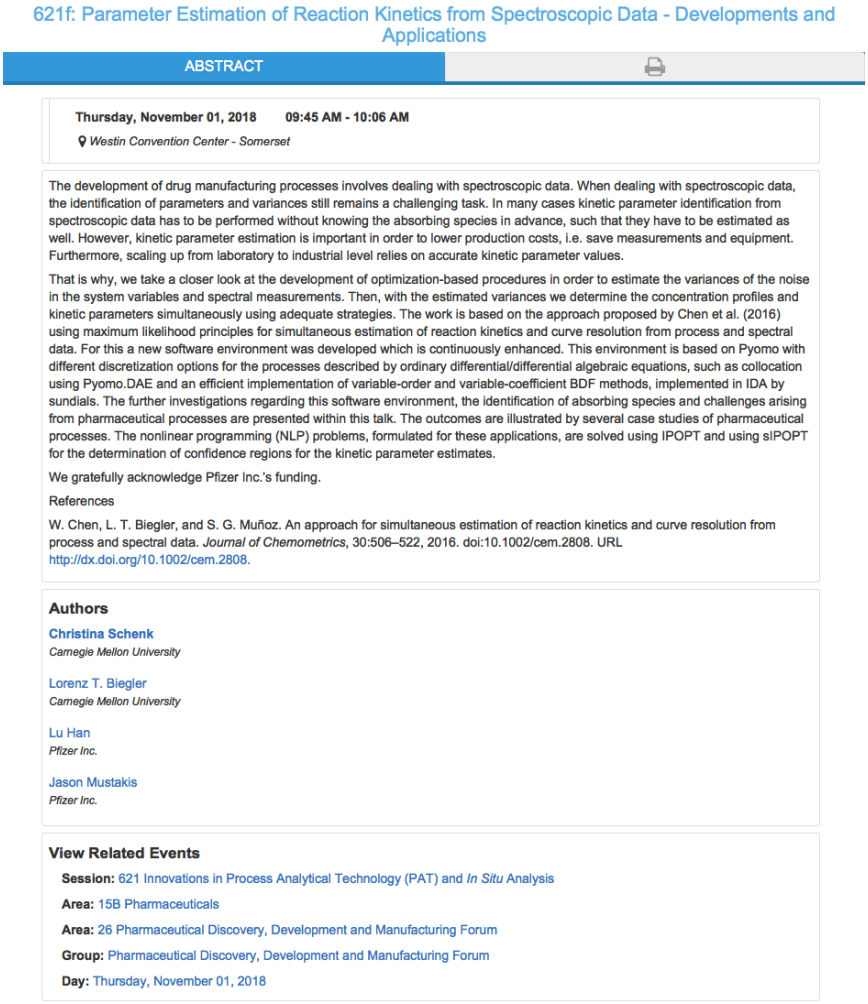
**Area:** 10E Data and Information Systems

**Area:** 21 Computational Molecular Science and Engineering Forum

**Group:** Computational Molecular Science and Engineering Forum

**Day:** Thursday, November 01, 2018

## 5.2 621f: Parameter Estimation of Reaction Kinetics from Spectroscopic Data - Developments and Applications https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530489 @pfizer



621f: Parameter Estimation of Reaction Kinetics from Spectroscopic Data - Developments and Applications

ABSTRACT

**Thursday, November 01, 2018** **09:45 AM - 10:06 AM**

*Westin Convention Center - Somerset*

The development of drug manufacturing processes involves dealing with spectroscopic data. When dealing with spectroscopic data, the identification of parameters and variances still remains a challenging task. In many cases kinetic parameter identification from spectroscopic data has to be performed without knowing the absorbing species in advance, such that they have to be estimated as well. However, kinetic parameter estimation is important in order to lower production costs, i.e. save measurements and equipment. Furthermore, scaling up from laboratory to industrial level relies on accurate kinetic parameter values.

That is why, we take a closer look at the development of optimization-based procedures in order to estimate the variances of the noise in the system variables and spectral measurements. Then, with the estimated variances we determine the concentration profiles and kinetic parameters simultaneously using adequate strategies. The work is based on the approach proposed by Chen et al. (2016) using maximum likelihood principles for simultaneous estimation of reaction kinetics and curve resolution from process and spectral data. For this a new software environment was developed which is continuously enhanced. This environment is based on Pyomo with different discretization options for the processes described by ordinary differential/differential algebraic equations, such as collocation using Pyomo.DAE and an efficient implementation of variable-order and variable-coefficient BDF methods, implemented in IDA by sundials. The further investigations regarding this software environment, the identification of absorbing species and challenges arising from pharmaceutical processes are presented within this talk. The outcomes are illustrated by several case studies of pharmaceutical processes. The nonlinear programming (NLP) problems, formulated for these applications, are solved using IPOPT and using sIPOPT for the determination of confidence regions for the kinetic parameter estimates.

We gratefully acknowledge Pfizer Inc.'s funding.

References

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**Area:** 15B Pharmaceuticals

**Area:** 26 Pharmaceutical Discovery, Development and Manufacturing Forum

**Group:** Pharmaceutical Discovery, Development and Manufacturing Forum

**Day:** Thursday, November 01, 2018

## 5.3 598h: Quadratic Cut Decomposition Method for Convex Mixed-Integer Nonlinear Programs https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/527215 @bernalde @AIChECAST10

598h: Quadratic Cut Decomposition Method for Convex Mixed-Integer Nonlinear Programs

ABSTRACT

**Thursday, November 01, 2018 10:13 AM - 10:32 AM**

*David L. Lawrence Convention Center - 409*

**Quadratic cut decomposition method for convex mixed-integer nonlinear programs**

David E. Bernal*, Lijie Su**, Lixin Tang**, Ignacio E. Grossmann*

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In this work, we address the incorporation of 2nd order derivative information to the Outer-Approximation (OA) method to solve convex mixed-integer nonlinear programming (MINLP) problems. An MINLP problem arises when we represent an optimization problem using nonlinear algebraic equations involving continuous and discrete variables. A method to solve this problem is OA [1], [2], where the original problem is decomposed into two subproblems. The first subproblem, a master problem, considers the discrete nature of the integer-constrained variables and approximates the nonlinear functions with linear equations through a Mixed-Integer Linear Program (MILP). The second subproblem optimizes the nonlinear objective and constraints for fixed discrete variables giving rise to a Nonlinear Program (NLP). The linearization point of the first subproblem is provided by the solution of the second one, while the values for the discrete variables fixed in the second subproblem are provided by the solution of the first one. The solution of the master problem yields a lower bound (minimization) while the NLP subproblem provides an upper bound for the objective of the original MINLP. Solving these problems iteratively results in the reduction in the difference between these bounds. The difference between the bound given by the linear approximated subproblem and the optimal solution of the MINLP relies heavily on how well the linearizations approximate the nonlinear equations of the original problem.

Several previous contributions have proposed using quadratic approximations for MINLP decomposition methods. Grossmann[3] and Fletcher and Leyffer[2] discussed including 2nd order derivative information in the master problem of the OA method. Specifically, considering the Hessian of the Lagrange function in the objective of the master problem gives rise to an Mixed –Integer Quadratic Problem (MIQP) problem. Also, Ding-Mei and Sargent[4] showed that the quadratic approximation helps to define NLP subproblems that have better initial guesses of the continuous variables from the MILP.

In this work, we propose a new approach for using quadratic accurately. In particular, we propose to use a scaled quadratic approximation of the nonlinear functions instead of a linear approximation and incorporate it into the OA method. This scaled quadratic approximation uses 2nd order derivative information and therefore approximates the nonlinear functions more accurately than the linearization, yielding stronger lower bounds while ensuring the validity of the relaxation. The scaled quadratic cut is proved to be a stricter and tighter underestimation for convex nonlinear functions than classical supporting hyperplanes, which results in the improvement of the OA solution method. The resulting master problem becomes a Mixed-Integer Quadratically Constrained Program (MIQCP).

Having a quadratic approximation of the nonlinear constraints allows the master problem to provide a closer approximation of the continuous feasible region, and therefore, improve the lower bound provided by the relaxation[3]. Recently, MILP solvers like CPLEX[5] and Gurobi[6] have expanded their capabilities to efficiently solve MIQCP problems[7]. The crucial factor of MIQCP solvability is the Hessian matrix, the second-order partial derivatives matrix. Convex MIQCP problems, which have a positive semi-definite Hessian matrix, can be solved to optimality as one special convex MINLP with state-of-the-art MILP solvers. These developments allow us to consider quadratic objectives and constraints in the master problem. The Hessian of the nonlinear functions is available thanks to the NLP solvers, and therefore can be included in the problem efficiently.

The implementation of this method in the Mixed-Integer Nonlinear Decomposition Toolbox in Pyomo MindtPy is presented together with numerical results of benchmark MINLP problems demonstrate the effectiveness of the proposed MINLP solution methods with scaled quadratic cuts compared to the traditional OA method.

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**Session:** 598 Advances in Optimization with Surrogate and Mixed-Integer Models

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Thursday, November 01, 2018

## 5.4 668b: Role of Stefan-Maxwell Fluxes in the Dynamics of Concentrated Electrolytes @AdityaSKhair1 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/524671

668b: Role of Stefan-Maxwell Fluxes in the Dynamics of Concentrated Electrolytes

ABSTRACT

**Thursday, November 01, 2018 12:48 PM - 01:06 PM**

*Omni William Penn Hotel - Conference Center B*

This theoretical analysis investigates the effect of ambi-polar diffusion and coupled ionic fluxes on the dynamics of the charging of an electrochemical cell. We consider a model cell consisting of a concentrated, binary electrolyte between parallel, blocking electrodes, under a suddenly applied DC voltage. It is assumed that the applied voltage is small compared to the thermal voltage, $RT/F$, where $R$ is the universal gas constant, $T$ is the temperature and $F$ is the Faraday's constant. We employ the Stefan-Maxwell equations to describe the hydrodynamic coupling of ionic fluxes that arise in concentrated electrolytes. These equations inherently account for asymmetry in the mobilities of the ions in the electrolyte. A modified set of linearized Poisson-Nernst-Planck equations, obtained by incorporating Stefan-Maxwell fluxes into the species balances, are formulated and solved analytically. A long-time asymptotic analysis reveals that the electrolyte dynamics occur on two separate time scales. A faster, RC time, $\tau_{RC} \sim \kappa^{-1} L/D_e$, where $D_e$ is an effective diffusivity, characterizes the evolution of charge density at the electrode. The effective diffusivity, $D_e$, is a function of the ambi-polar diffusivity of the salt, $D_a$, as well as a cross-diffusivity, $D_{+-}$, of the ions. This time scale also dictates the initial exponential decay of current in the external circuit. At times longer than $\tau_{RC}$, the external current again decays exponentially on a slower, diffusive time scale, $\tau_D \sim L^2/D_a$. This diffusive time scale is due to the unequal ion mobilities that result in a non-uniform bulk concentration of the salt during the charging process. Finally, we demonstrate how these dynamics can be incorporated into an equivalent circuit model.

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**Session:** 668 Interfacial Phenomena in Electrochemical Systems

**Area:** 01C Interfacial Phenomena

**Group:** Engineering Sciences and Fundamentals

**Day:** Thursday, November 01, 2018

## 5.5 653d: Dehydroaromatization of Ethylene over Bifunctional Lewis-Brønsted Acid Pairs in Ag-ZSM-5 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/526494 #MSAlum @sudochemeng @TheGrabowGroup @aiche_cre

653d: Dehydroaromatization of Ethylene over Bifunctional Lewis-Brønsted Acid Pairs in Ag-ZSM-5

ABSTRACT

**Thursday, November 01, 2018 01:30 PM - 01:50 PM**

*David L. Lawrence Convention Center - 404*

Aromatic hydrocarbons such as benzene, toluene and xylene are commercially used as fuel additives and as raw materials in the synthesis of desirable aromatics-derived products, amongst other uses.[1] Synthesis of these cyclic compounds are primarily dependent on catalytic reforming processes using catalysts with limited scope in alternate production routes. The surge in natural gas production across the world has incentivized the search for processes that can utilize methane and light olefin derivatives in the manufacture of aromatic products. The conversion methanol to hydrocarbons over zeolite catalysts such as H-ZSM-5 has evolved into a mature field of research occupying the minds of many researchers across the world. Another such process that has garnered only little attention is the conversion of ethylene to aromatics.[2]

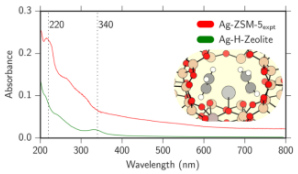



**Figure 1:** Comparison of the UV-vis spectra obtain from simulations (in green) with that obtained experimentally (in red). The inset figure shows the isolated $Ag^+$ Lewis acid site coordinated to two ethylene molecules.

In this work, we use a combination of density functional theory (DFT) calculations, flow catalytic experiments and the transient temporal analysis of products (TAP) technique to shed light on the enhanced aromatic production exhibited by Ag exchanged ZSM-5 from ethylene.[3] The identity of the Ag exchanged sites was confirmed through the simulation of spectroscopic techniques such as UV-vis spectroscopy and EXAFS. As seen in Figure 1, our UV-vis simulations on representative Ag-zeolite models show good agreement with the experimental spectra consistent with the presence of isolated $Ag^+$ as the Lewis acid site. DFT calculations and TAP experiments were used to decouple to role of the Brønsted ($H^+$) and Lewis ($Ag^+$) acid sites in the commonly accepted hydrocarbon pool mechanism.[4] The Brønsted acid sites were found to facilitate the alkene cycle i.e. the initiation of ethylene oligomerization forming longer hydrocarbon chains and other cracking reactions. Ring closing mechanisms for unsaturated $C_5$ and $C_6$ species adsorbed to $Ag^+$ Lewis acid site in the zeolite showed exceptionally low activation barriers, thus highlighting the role of Ag in the arene cycle of the hydrocarbon pool mechanism. The distinct functions of the Brønsted and Lewis acid sites in the ethylene conversion process highlight the bifunctional nature of metal exchanged zeolites and their immense potential in hydrocarbon upgrade reactions.

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4. Pappas, D. K. et al. *J. Am. Chem. Soc.* **(2017)** *139* (42) 14961–14975.


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**Session:** 653 Catalysis with Microporous and Mesoporous Materials IV: Conversion of Renewables, Natural Gas, and Petroleum

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Thursday, November 01, 2018

## 5.6 699a: Catkit: Symmetry Methods for Automated Generation of Catalytic Structures https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/533449 #PhDAlum @jacob_boes @SUNCAT_Center @aiche_cre

699a: Catkit: Symmetry Methods for Automated Generation of Catalytic Structures

ABSTRACT

**Thursday, November 01, 2018 03:30 PM - 03:50 PM**

*David L. Lawrence Convention Center - 402*

Cheminformatics have dramatically accelerated the rate at which we discover useful properties of chemical species, originally for the purpose of drug discovery. This acceleration is made possible by taking advantage of bond symmetries present in organic molecules, allowing computers to systematically enumerate through an otherwise infinite search space. These techniques are also easily extend to other forms of symmetry, such as the space group symmetries present in bulk systems.

Since the systems in heterogeneous catalysis contain the complexity of both of these chemical domains, high-throughput techniques remain difficult to automate effectively for catalyst design. In this work, we propose a methodology which combines space group and bond symmetry techniques general enough to enumerate a large number of possible catalytic systems. This includes surfaces of any miller index produced from a provided bulk system, unique adsorption site identification, and creation of 3D structures bonded by one or multiple sites. This technique also produces a graph structure which provides the basis for all machine-learning fingerprints which require connectivity information; ideal for accurate descriptors which are not dependent on information gathered by expensive techniques such as Density Functional Theory.

We demonstrate these techniques for the enumeration of unique adsorption configurations across a wide range of alloy systems and catalytically relevant adsorbates. Utilizing these enumerated structures we show how the graph of each can be easily utilized to construct a variety of fingerprints as a basis for machine-learning accelerated predictions of adsorption energies. Finally, we discuss our Python implementation of these algorithms in the open source software package known as CatKit, intended to make high-throughput screening techniques easier to implement for the community.

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**Session:** 699 Data Science in Catalysis II

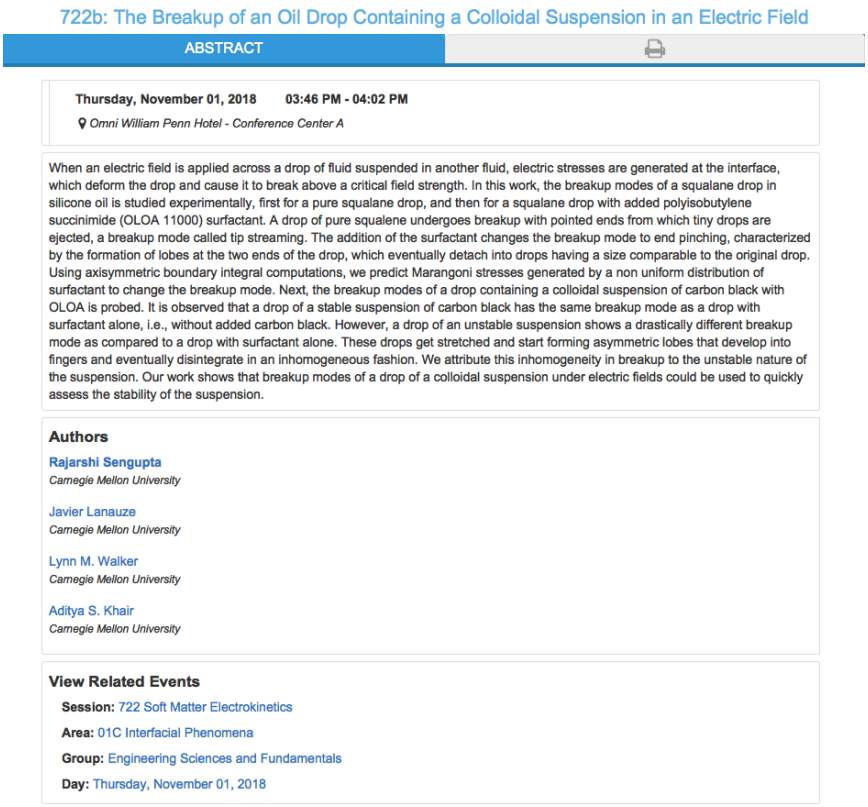
**Area:** 21 Computational Molecular Science and Engineering Forum

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Thursday, November 01, 2018

## 5.7 722b: The Breakup of an Oil Drop Containing a Colloidal Suspension in an Electric Field @AdityaSKhair1 @LwalkerCMU https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/527942



722b: The Breakup of an Oil Drop Containing a Colloidal Suspension in an Electric Field

ABSTRACT

**Thursday, November 01, 2018 03:46 PM - 04:02 PM**

*Omni William Penn Hotel - Conference Center A*

When an electric field is applied across a drop of fluid suspended in another fluid, electric stresses are generated at the interface, which deform the drop and cause it to break above a critical field strength. In this work, the breakup modes of a squalane drop in silicone oil is studied experimentally, first for a pure squalane drop, and then for a squalane drop with added polyisobutylene succinimide (OLOA 11000) surfactant. A drop of pure squalene undergoes breakup with pointed ends from which tiny drops are ejected, a breakup mode called tip streaming. The addition of the surfactant changes the breakup mode to end pinching, characterized by the formation of lobes at the two ends of the drop, which eventually detach into drops having a size comparable to the original drop. Using axisymmetric boundary integral computations, we predict Marangoni stresses generated by a non uniform distribution of surfactant to change the breakup mode. Next, the breakup modes of a drop containing a colloidal suspension of carbon black with OLOA is probed. It is observed that a drop of a stable suspension of carbon black has the same breakup mode as a drop with surfactant alone, i.e., without added carbon black. However, a drop of an unstable suspension shows a drastically different breakup mode as compared to a drop with surfactant alone. These drops get stretched and start forming asymmetric lobes that develop into fingers and eventually disintegrate in an inhomogeneous fashion. We attribute this inhomogeneity in breakup to the unstable nature of the suspension. Our work shows that breakup modes of a drop of a colloidal suspension under electric fields could be used to quickly assess the stability of the suspension.

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**Area:** 01C Interfacial Phenomena

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**Day:** Thursday, November 01, 2018

## 5.8 715d: Novel Formulation for Optimal Schedule with Demand Side Management in Multi-Product Air Separation Processes https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/525924 @airliquidegroup @AIChECAST10

715d: Novel Formulation for Optimal Schedule with Demand Side Management in Multi-Product Air Separation Processes

ABSTRACT

**Thursday, November 01, 2018 04:27 PM - 04:46 PM**

*David L. Lawrence Convention Center - 409*

Since the deregulation of electricity markets and the integration of renewable energy systems into the electricity grid, the electricity market has more flexible clearing pricing strategies on customers, which significantly influence the profitability of energy-intensive industries, like air separation plants. In this context, it is critical for industries to optimally adjust their production in order to benefit from the variable energy cost.

In this work, we focus on the optimal scheduling of air separation in industrial gases processes with electricity purchased from spot market. We consider the electricity price forecast in the scheduling horizon as a model input, based on fixed time intervals. As for the continuous process, the variable production levels are achieved by properly adjusting the units to meet various production constraints, such as different product demands. This could be interpreted as the transition of operation states of the production system. In this work, the continuous space of plant operation is discretized into several operation states. Therefore, the main goal of the scheduling problem is to minimize the total production costs by selecting the optimal operation state for each unit in the production system for each time period. Besides the energy consumption, we also consider the setup costs and other related cost item.

To represent the production states and the transition between them, an advanced process state transition network (APSTN) is proposed. The formulation relies on a basic concept of previous work [1], where states with minimum processing time are disaggregated into sub states. However, the APSTN representation is more compact, especially when dealing with more complex production systems. Specifically, for multi-product air separation processes, a discrete-time mixed integer linear (MILP) model is developed based on the APSTN, which has proven to provide a tighter LP relaxation to handling industrial-scale instances.

Three main features characterize the air separation process: 1) minimum consecutive operating time for specific states; 2) setup costs of transition of states and finally, 3) pre-defined direction for some transitions. The proposed MILP model involves the constraints with the previously mentioned features, as well as other considerations, such as inventory level and product demand. The computational efficiency of the model is demonstrated with data provided by the company. Numerical experiments are performed in order to make comparisons between our model and previous ones [1-3]. The response of the scheduling and production level is also tested regarding to various interval's length for the electricity pricing and the time horizon for the scheduling.

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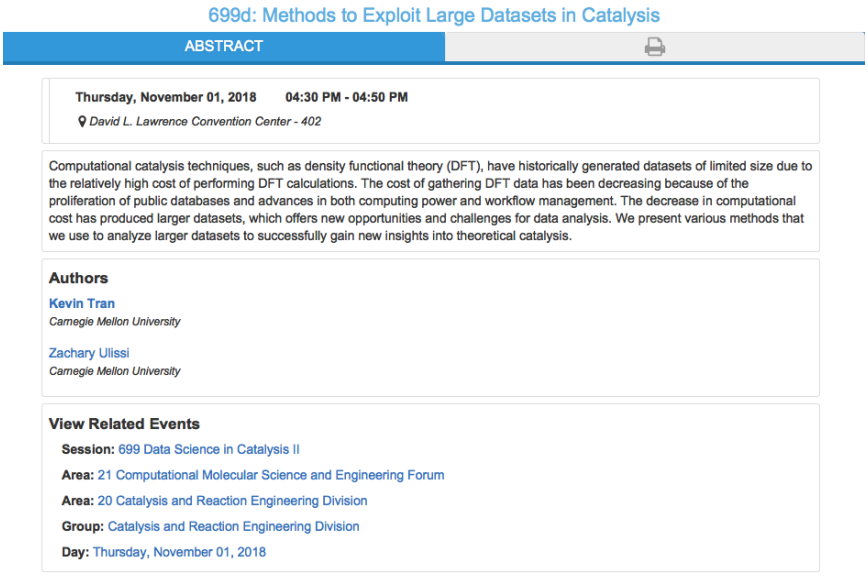
**Area:** 10B Systems and Process Control

**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Thursday, November 01, 2018

## 5.9 699d Methods to Exploit Large Datasets in Catalysis https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/528022 @UlissiGroup @aiche_cre



699d: Methods to Exploit Large Datasets in Catalysis

ABSTRACT

**Thursday, November 01, 2018** **04:30 PM - 04:50 PM**

*David L. Lawrence Convention Center - 402*

Computational catalysis techniques, such as density functional theory (DFT), have historically generated datasets of limited size due to the relatively high cost of performing DFT calculations. The cost of gathering DFT data has been decreasing because of the proliferation of public databases and advances in both computing power and workflow management. The decrease in computational cost has produced larger datasets, which offers new opportunities and challenges for data analysis. We present various methods that we use to analyze larger datasets to successfully gain new insights into theoretical catalysis.

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**Area:** 21 Computational Molecular Science and Engineering Forum

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Thursday, November 01, 2018

## 5.10 715e: Multi-Operational Development Planning for Multi-System Shale Gas Production https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/524663 @EQTCorp @AIChECAST10

715e: Multi-Operational Development Planning for Multi-System Shale Gas Production

ABSTRACT

**Thursday, November 01, 2018 04:46 PM - 05:05 PM**

*David L. Lawrence Convention Center - 409*


Shale gas production is expected to increase by more than 50% over the next 34 years, from 14 trillion cubic feet (TCF) in 2016 to approximately 29 TCF in 2040 [1]. This increase in production will make shale gas a major player in the natural gas field, with shale gas production fulfilling 69% of the expected total dry gas natural gas demand [1]. With this expected growth in shale gas, informed decisions, as to where, when, and how to develop assets, must be made to keep the development and production economical. This includes development scheduling, such as when and where to drill, and strategic planning, such as when and which contracts should be made for gas transportation and resource utilization.

Due to the increased impact of shale gas on the energy sector, the optimization of the development and transportation of shale gas has grown in popularity during the past decade. This research has concentrated on adding breadth to the supply chain optimization problem [2,3], and depth to the development and production scheduling [2,4]. However, previous models lumped all pre-production operations (i.e., top-setting, horizontal drilling, fracturing, and turning-in-line) into the generic group of development operations. While this does provide a method of tracking well production as a function of development, it does not allow for the consideration of drilling rig and fracturing crew contracts or mobilization costs, which may significantly impact the development of a shale gas region.

Motivated by the above, in this paper, we present the planning of the multi-operational asset development and shale gas production for a set of prospective capacity systems in an expansive shale gas region. By explicitly accounting for the sequencing of development operations at the pad level, the planning considers the time to perform individual operations, the length of contracts and the number of resources (rigs, crews) needed for development, and costs for moving rigs from one pad to another. By initially formulating the multi-system development planning using General Disjunctive Programming (GDP), we can systematically derive mixed-integer inequalities for all disjunctions and logic propositions to obtain a discrete-time mixed integer programming (MILP) formulation [5,6]. The MILP can be used to optimize the planning of asset development and infrastructure commitments, considering the long-term impacts of gas production and transportation for multiple shale gas systems. To demonstrate the validity of this formulation and the impact of rig considerations, we apply the planning to an industrial case study based on real data, quantifying the value of optimizing shale gas development and production.


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**Area:** 10C Computers in Operations and Information Processing

**Group:** Computing and Systems Technology Division

**Day:** Thursday, November 01, 2018

# 6 #AICHeAnnual Here are a few talks from @CMU_-Cheme on Friday @ChEnected :tweet:thread

## 6.1 732a: Accurate Adsorbate Free Energies from First-Principles https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/529098 #MSAlum @prtk_m @profwschneider @aiche_-cre

732a: Accurate Adsorbate Free Energies from First-Principles

ABSTRACT

**Friday, November 02, 2018 08:00 AM - 08:18 AM**

*David L. Lawrence Convention Center - 318*

In first-principles-based microkinetic models used in catalysis, free energies of adsorption and reaction are typically obtained by combining density functional theory (DFT) energies with standard approximate models, such as the harmonic oscillator, the hindered translator, or the two-dimensional ideal gas. While there has been an increasing focus on developing approaches to benchmark the reliability of the 0 K DFT energies, the performance of models used to treat the finite temperature contributions to the free energies has been analyzed to a lesser extent. I will discuss our approach (illustrated in Figure 1) to calculate accurate free energies using quantum mechanical solutions for adsorbate translational energy states extracted directly from first-principles potential energy surfaces. Through a series of case studies of adsorbates on metal surfaces, we show that the no one free energy model performs satisfactorily in all cases. Moreover, even combinations of different approximations sometimes deviate significantly from the free energies calculated by our first principles approach. Using observations from these case studies, we discuss how a full quantum mechanical approach may be extended to calculate accurate free energies for arbitrary adsorbate potential energy surfaces at computational cost similar to standard models.

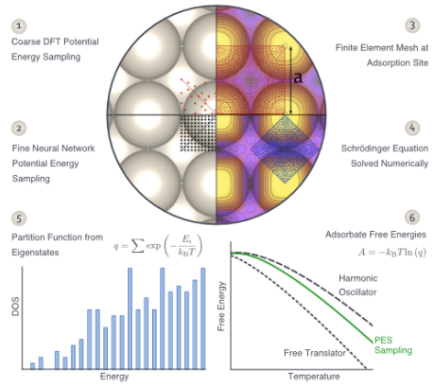



Figure 1: Protocol for calculation of adsorbate free energy

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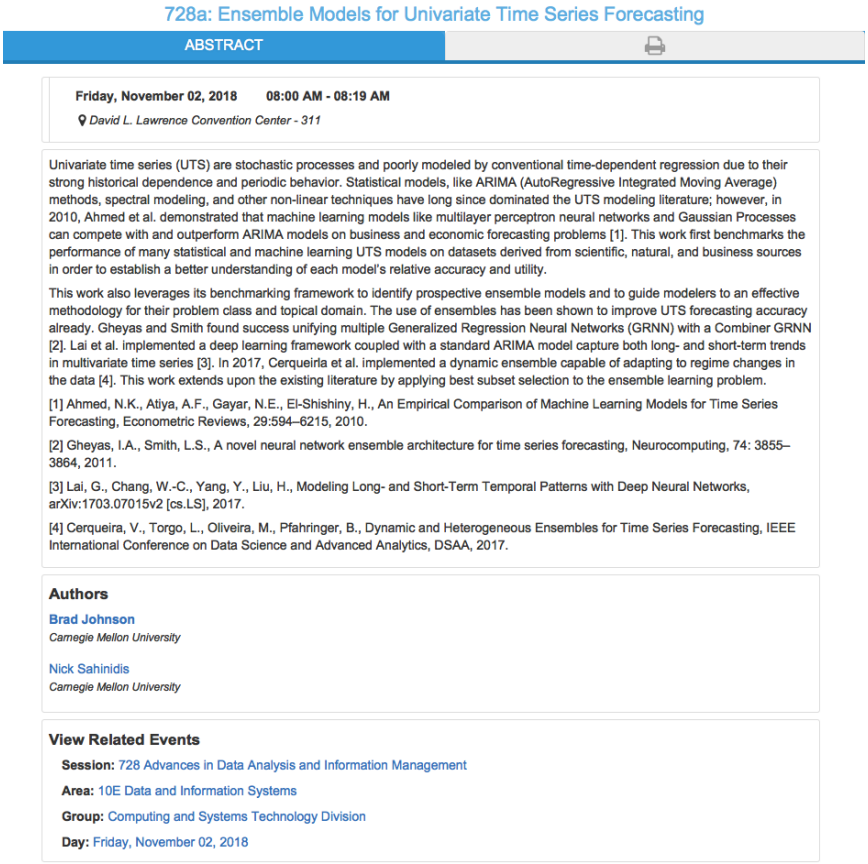
**Session:** 732 Fundamentals of Catalysis IV: Surface Reactivity

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Friday, November 02, 2018

## 6.2 728a: Ensemble Models for Univariate Time Series Forecasting https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/532401 @AIChECAST10



728a: Ensemble Models for Univariate Time Series Forecasting

ABSTRACT

**Friday, November 02, 2018 08:00 AM - 08:19 AM**

*David L. Lawrence Convention Center - 311*

Univariate time series (UTS) are stochastic processes and poorly modeled by conventional time-dependent regression due to their strong historical dependence and periodic behavior. Statistical models, like ARIMA (AutoRegressive Integrated Moving Average) methods, spectral modeling, and other non-linear techniques have long since dominated the UTS modeling literature; however, in 2010, Ahmed et al. demonstrated that machine learning models like multilayer perceptron neural networks and Gaussian Processes can compete with and outperform ARIMA models on business and economic forecasting problems [1]. This work first benchmarks the performance of many statistical and machine learning UTS models on datasets derived from scientific, natural, and business sources in order to establish a better understanding of each model's relative accuracy and utility.

This work also leverages its benchmarking framework to identify prospective ensemble models and to guide modelers to an effective methodology for their problem class and topical domain. The use of ensembles has been shown to improve UTS forecasting accuracy already. Gheyas and Smith found success unifying multiple Generalized Regression Neural Networks (GRNN) with a Combiner GRNN [2]. Lai et al. implemented a deep learning framework coupled with a standard ARIMA model capture both long- and short-term trends in multivariate time series [3]. In 2017, Cerqueirla et al. implemented a dynamic ensemble capable of adapting to regime changes in the data [4]. This work extends upon the existing literature by applying best subset selection to the ensemble learning problem.

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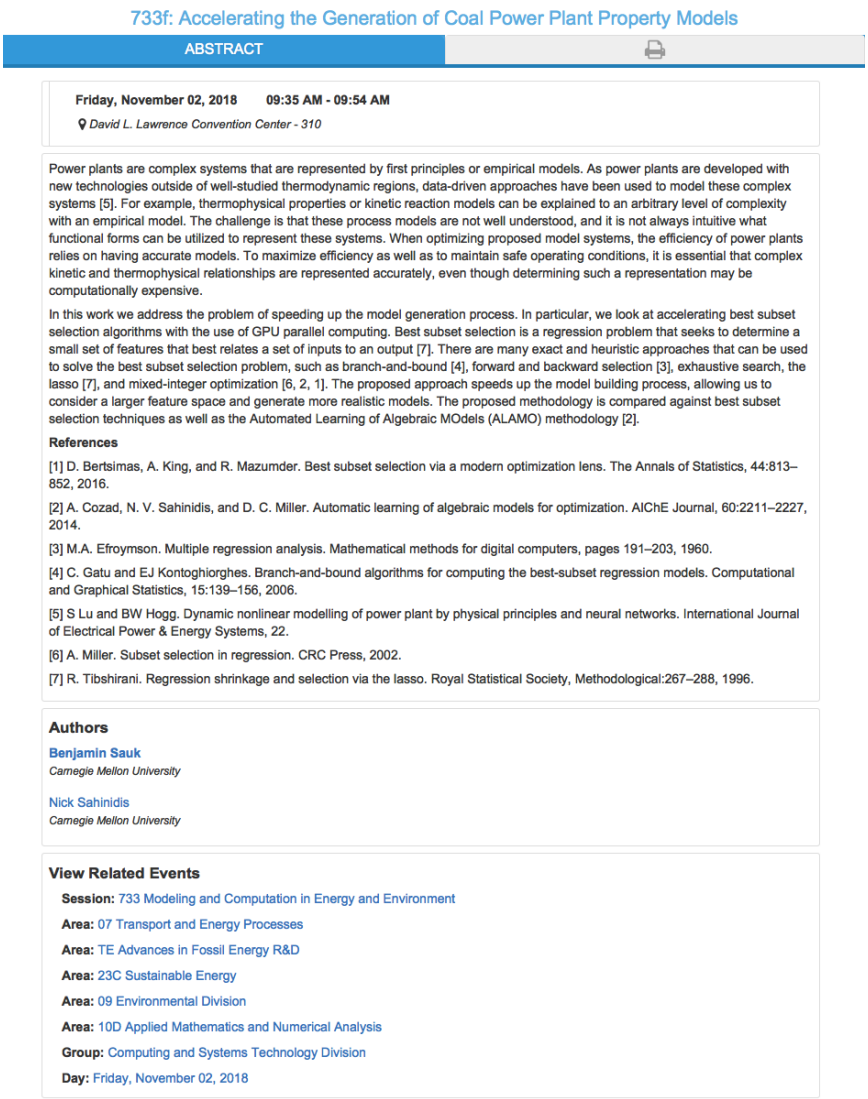
**Session:** 728 Advances in Data Analysis and Information Management

**Area:** 10E Data and Information Systems

**Group:** Computing and Systems Technology Division

**Day:** Friday, November 02, 2018

## 6.3 733f: Accelerating the Generation of Coal Power Plant Property Models @bsauk2 https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/530255 @AIChECAST10



733f: Accelerating the Generation of Coal Power Plant Property Models

ABSTRACT

**Friday, November 02, 2018 09:35 AM - 09:54 AM**

*David L. Lawrence Convention Center - 310*

Power plants are complex systems that are represented by first principles or empirical models. As power plants are developed with new technologies outside of well-studied thermodynamic regions, data-driven approaches have been used to model these complex systems [5]. For example, thermophysical properties or kinetic reaction models can be explained to an arbitrary level of complexity with an empirical model. The challenge is that these process models are not well understood, and it is not always intuitive what functional forms can be utilized to represent these systems. When optimizing proposed model systems, the efficiency of power plants relies on having accurate models. To maximize efficiency as well as to maintain safe operating conditions, it is essential that complex kinetic and thermophysical relationships are represented accurately, even though determining such a representation may be computationally expensive.

In this work we address the problem of speeding up the model generation process. In particular, we look at accelerating best subset selection algorithms with the use of GPU parallel computing. Best subset selection is a regression problem that seeks to determine a small set of features that best relates a set of inputs to an output [7]. There are many exact and heuristic approaches that can be used to solve the best subset selection problem, such as branch-and-bound [4], forward and backward selection [3], exhaustive search, the lasso [7], and mixed-integer optimization [6, 2, 1]. The proposed approach speeds up the model building process, allowing us to consider a larger feature space and generate more realistic models. The proposed methodology is compared against best subset selection techniques as well as the Automated Learning of Algebraic MOdels (ALAMO) methodology [2].

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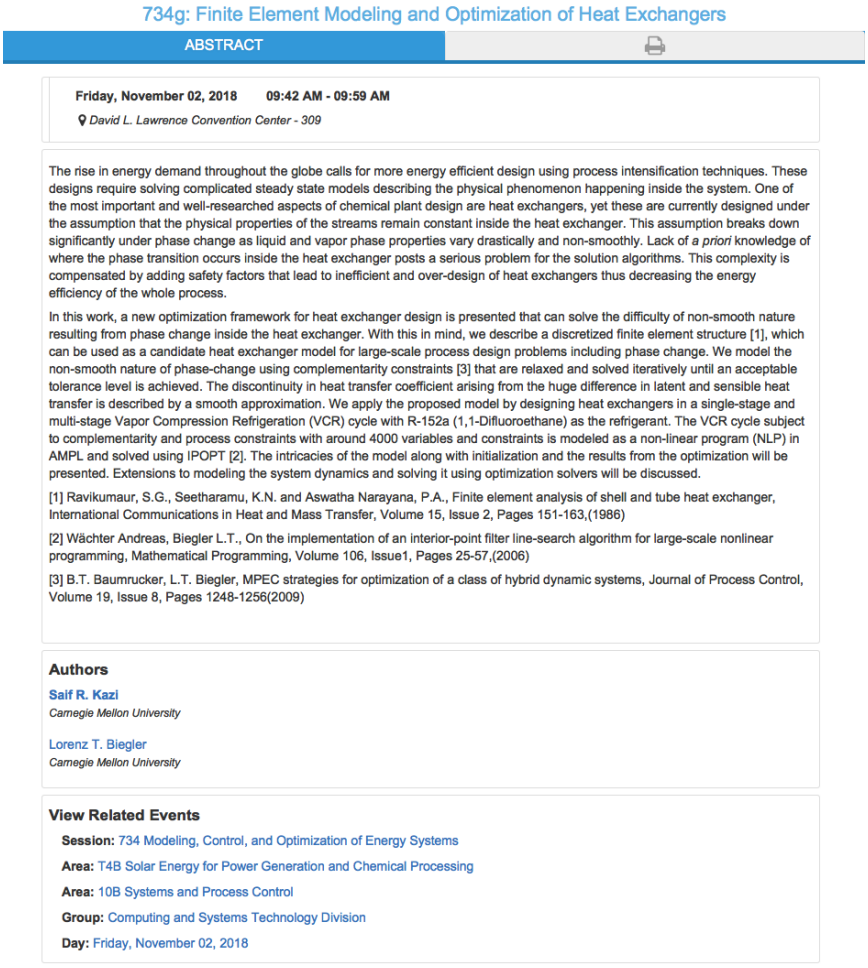
**Area:** 09 Environmental Division

**Area:** 10D Applied Mathematics and Numerical Analysis

**Group:** Computing and Systems Technology Division

**Day:** Friday, November 02, 2018

## 6.4 734g: Finite Element Modeling and Optimization of Heat Exchangers https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/521550 @AIChECAST10



734g: Finite Element Modeling and Optimization of Heat Exchangers

ABSTRACT

**Friday, November 02, 2018 09:42 AM - 09:59 AM**

*David L. Lawrence Convention Center - 309*

The rise in energy demand throughout the globe calls for more energy efficient design using process intensification techniques. These designs require solving complicated steady state models describing the physical phenomenon happening inside the system. One of the most important and well-researched aspects of chemical plant design are heat exchangers, yet these are currently designed under the assumption that the physical properties of the streams remain constant inside the heat exchanger. This assumption breaks down significantly under phase change as liquid and vapor phase properties vary drastically and non-smoothly. Lack of *a priori* knowledge of where the phase transition occurs inside the heat exchanger posts a serious problem for the solution algorithms. This complexity is compensated by adding safety factors that lead to inefficient and over-design of heat exchangers thus decreasing the energy efficiency of the whole process.

In this work, a new optimization framework for heat exchanger design is presented that can solve the difficulty of non-smooth nature resulting from phase change inside the heat exchanger. With this in mind, we describe a discretized finite element structure [1], which can be used as a candidate heat exchanger model for large-scale process design problems including phase change. We model the non-smooth nature of phase-change using complementarity constraints [3] that are relaxed and solved iteratively until an acceptable tolerance level is achieved. The discontinuity in heat transfer coefficient arising from the huge difference in latent and sensible heat transfer is described by a smooth approximation. We apply the proposed model by designing heat exchangers in a single-stage and multi-stage Vapor Compression Refrigeration (VCR) cycle with R-152a (1,1-Difluoroethane) as the refrigerant. The VCR cycle subject to complementarity and process constraints with around 4000 variables and constraints is modeled as a non-linear program (NLP) in AMPL and solved using IPOPT [2]. The intricacies of the model along with initialization and the results from the optimization will be presented. Extensions to modeling the system dynamics and solving it using optimization solvers will be discussed.

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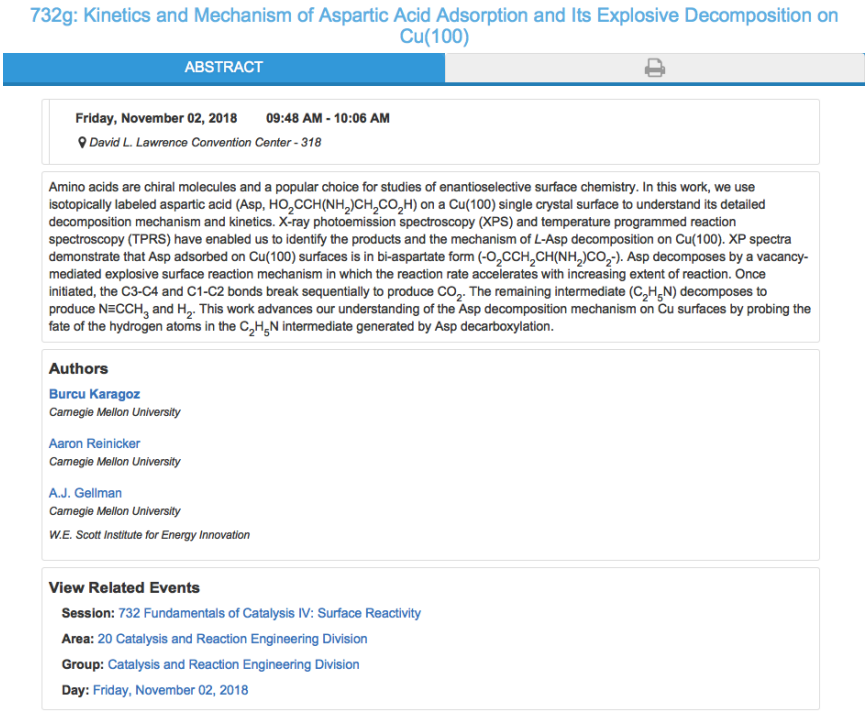
**Area:** T4B Solar Energy for Power Generation and Chemical Processing

**Area:** 10B Systems and Process Control

**Group:** Computing and Systems Technology Division

**Day:** Friday, November 02, 2018

## 6.5 732g: Kinetics and Mechanism of Aspartic Acid Adsorption and Its Explosive Decomposition on Cu(100) @BrcKaragoz https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/523551 @aiche_cre



732g: Kinetics and Mechanism of Aspartic Acid Adsorption and Its Explosive Decomposition on Cu(100)

ABSTRACT

**Friday, November 02, 2018** **09:48 AM - 10:06 AM**

*David L. Lawrence Convention Center - 318*

Amino acids are chiral molecules and a popular choice for studies of enantioselective surface chemistry. In this work, we use isotopically labeled aspartic acid (Asp, $HO_2CCH(NH_2)CH_2CO_2H$) on a Cu(100) single crystal surface to understand its detailed decomposition mechanism and kinetics. X-ray photoemission spectroscopy (XPS) and temperature programmed reaction spectroscopy (TPRS) have enabled us to identify the products and the mechanism of *L*-Asp decomposition on Cu(100). XP spectra demonstrate that Asp adsorbed on Cu(100) surfaces is in bi-aspartate form ($-O_2CCH_2CH(NH_2)CO_2-$). Asp decomposes by a vacancy-mediated explosive surface reaction mechanism in which the reaction rate accelerates with increasing extent of reaction. Once initiated, the C3-C4 and C1-C2 bonds break sequentially to produce $CO_2$. The remaining intermediate ($C_2H_5N$) decomposes to produce $N{\equiv}CCH_3$ and $H_2$. This work advances our understanding of the Asp decomposition mechanism on Cu surfaces by probing the fate of the hydrogen atoms in the $C_2H_5N$ intermediate generated by Asp decarboxylation.

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**Session:** 732 Fundamentals of Catalysis IV: Surface Reactivity

**Area:** 20 Catalysis and Reaction Engineering Division

**Group:** Catalysis and Reaction Engineering Division

**Day:** Friday, November 02, 2018

## 6.6 748e: Adaptive Control of System with Unknown Inputs with Application to Chemical Reaction Control https://aiche.confex.com/aiche/2018/meetingapp.cgi/Paper/535412 @AIChECAST10

748e: Adaptive Control of System with Unknown Inputs with Application to Chemical Reaction Control

ABSTRACT

**Friday, November 02, 2018 01:46 PM - 02:05 PM**

*David L. Lawrence Convention Center - 309*

Adaptive Control of System with Unknown Inputs with Application to Chemical Reaction Control

We present an approach to control a system composed of a deterministic nonlinear system plus an uncertain parts modeled as a time-varying parameter vector, also called unknown input or disturbance. These inputs may represent unknown reaction rates, heat or mass transfer rates. The adaptive controller is composed of two parts: a passivity-based input observer to estimate the uncertain, time-varying parameter and a passivity-based controller that updates its control decision based on the observer estimates and on-line measurements.

The main assumptions and the measurement vector has at least the same dimension as the unknown parameter vector, along with an observability condition [1], and that the input or disturbance is uniformly continuous [3].

The unknown input observer treats the estimation problem as a control problem. In this approach, the observer constructs the error between the observations and the output produced from the observer, and uses Lyapunov method to construct parameter adaptation law that will ensure both measurement and parameter tracking. The estimated parameters are used in a passivity-based back-stepping controller [2]. The back-stepping was necessary since we need to augment original dynamics with a first order filter to ensure the existence of a solution to both control and parameter estimation problems. Asymptotic control stability and tuning intuition of the observer and controller are presented based on control stability and trade-off between estimation convergence rate and amplification of measurements noise [3]. We show that the controller can be written in a PID with a filter.

The practical problem is motivated by the fact that reaction kinetics and other dynamic mechanisms usually have large uncertainty, and it can affect the effectiveness of model-based controller. The unknown input observer can be used to estimate the reaction rates, heat and mass transfer rates if their mechanisms are not clear. The passivity-based control follows naturally and uses estimates in the feedforward term to update control decisions.

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